FEL-1 Design and Techno-Economic Analysis for a Proposed Addition of a Hydrodealkylation Process to a Chemical Plant

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Executive Summary

This report details our team's FEL-1 design of a proposed hydrodealkylation of toluene process to produce benzene (sold for $1000 per MT) as an addition to a chemical plant. For this process, toluene (purchased at $300 per MT) and a combined methane and hydrogen gas feed (purchased at $1400 per MT) are reacted in an adiabatic plug-flow reactor to produce benzene, alongside co-product methane and by-product diphenyl.

This design aims to maximize the Net Present Value (NPV) by optimization of the key variables: reactor conversion, temperature, hydrogen purge fraction, inlet toluene-to-hydrogen molar ratio, distillation stages, and our heat exchange network. The reactor is operated at a temperature of 599 °C and a pressure of 34 bar, with a conversion of 82% and selectivity of 93%. Our FEL-1 analysis determined that this design has potential profitability, resulting in an NPV of $104.9 MM and IRR of 70%. This design also has a TCI of $31.5 MM. To produce 1 kg of C_6H_6, 1.1 kg CO2 are produced, and 8 MJ are used.

Ultimately, we recommend that APL moves forward to a FEL-2 design of this plant due to favorable profitability metrics.

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Section 1: Introduction

1.1: Background & Market Analysis

Benzene is an essential petrochemical manufactured globally that acts as an intermediate for many key commodity chemicals, including cumene, styrene and cyclohexane. The chemical itself is used across many different industries in products such as lubricants, dyes, drugs and pesticides. The benzene market size was valued at $48.9 billion USD in 2024 and is projected to grow to $88.5 billion by 2034[1]. Increasing demand for benzene is mainly due to its desirability as an intermediate for styrene, which is used in electric vehicle manufacturing and development. Styrene is a key component in the polymers SBR and ABS, both widely used in electric vehicles and in 2023, hybrid and electric vehicles composed more than 16% of all American vehicle sales in comparison to just 2% of sales in 2000[2].

As the demand for benzene grows, the costliness of its synthesis must be evaluated. Currently, the primary mode of production for benzene is as a by-product of naphtha reforming refinery operations. This process requires the use of a catalyst to facilitate the conversion, which is usually comprised of an expensive precious metal such as platinum[3]. Here, the use of hydrodealkylation of toluene may provide a more profitable production alternative for benzene, as this process does not need a catalyst.

Our report evaluates the prospective profitability of a hydrodealkylation of toluene process addition to a chemical plant in order to produce 100 kta of benzene product. Through ASPEN HYSYS and MATLAB simulations, we created a FEL-1 plant design, optimized key design variables including temperature, pressure, reactor conversion, and the hydrogen-to-toluene molar ratio to determine the Net Present Value (NPV) and Internal Revenue Rate (IRR) of this potential process.

1.2: Reaction Chemistry

Benzene is produced through the hydrodealkylation of toluene. This process proceeds as a high-temperature, high-pressure, exothermic reaction where toluene is reacted in the presence of hydrogen gas to form the simpler aromatic hydrocarbon benzene. An additional reaction forming diphenyl from benzene also occurs during this process.

$$\begin{aligned} \text{Primary Reaction: } & \mathrm{C_7H_8 + H_2 \longrightarrow C_6H_6 + CH_4} \quad \Delta H_1^\circ = -41.8\ \mathrm{kJ/mol} \\ \text{Side Reaction: } & \mathrm{2C_6H_6 \rightleftharpoons C_{12}H_{10} + H_2} \quad \Delta H_2^\circ = 15.9\ \mathrm{kJ/mol} \end{aligned} \tag{1.1}$$

The primary reaction proceeds faster than the secondary reaction due to its kinetic properties (see Appendix A.3), so the overall process is exothermic. This is contingent on the use of the derived kinetics within a range of temperatures (550°C < T < 650°C), pressures (30 bar < P < 35 bar) and the molar ratio of toluene to hydrogen in the reaction feed (4.5 < MR < 5.5).

For this chemistry, there must exist a coupled relationship between the conversion, X, of the limiting reactant toluene and that of the selectivity, S_B, of benzene over all other byproducts – which in this chemistry is represented by diphenyl. Figure 1.1 represents this relationship as the yield of the benzene product increasing with conversion, but at the expense of selectivity.

This must occur due to the side reaction preferring benzene conversion to diphenyl at the higher benzene concentrations. The kinetics of the side reaction is generally slower than the primary reaction and hence selectivity has a absolute lower bound of around 80%. This enables our process to optimize for higher conversions while maintaining relatively high selectivity.

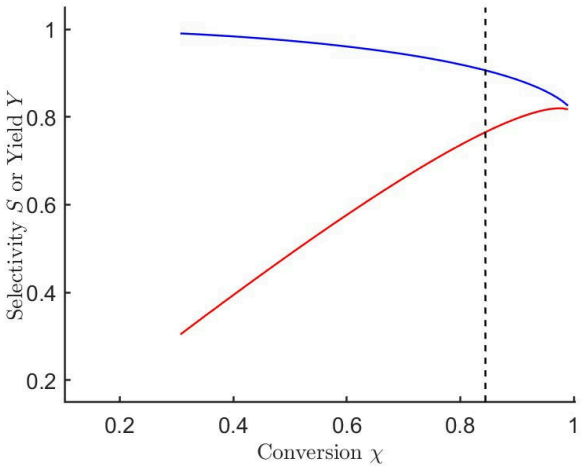


Figure 1.1 : Selectivity as a function of conversion for our HDA chemistry. Only profitable conversions for a single-reactor HDA process are shown. The optimized conversion 82% with a corresponding selectivity of 93%.

Section 2: Conceptual Design

Our plant design follows Douglas's Hierarchy up to Level 5, as part of a standard FEL-1 (Front-End Loading) analysis conducted with APL's methodology. Each level—abbreviated as ID, IO, PC, RR, SS, and HEN—represents a key step in developing the process.

First, we established foundational data: balanced chemical reactions, target production rates and purities, and process constraints. We also assessed raw material costs, properties, and plant utility requirements to estimate economic feasibility.

Second, we performed an overall mole balance to determine plant input/output flow rates and calculated the desired benzene production based on selectivity. These values were used to estimate the plant's maximum economic potential (Appendix A.1).

Third, we designed the reaction and recycle system. Using material balances and reaction mechanisms, we related reactor conversion, recycle flow, and selectivity to operating parameters—temperature, pressure, and the hydrogen to toluene. We evaluated a range of conditions to determine the required reactor volume and heat duty, then used cost correlations to estimate the capital and operating costs for the reactor and heater.

Fourth, we built the separation system based on the reactor's outlet composition and literature data. A flash drum removes methane and hydrogen for recycling or purge, and remaining separations were arranged by relative volatility. We used the full set of Underwood equations to estimate distillation stages and column sizing.

Fifth, we developed a heat exchanger network (HEN) using a comprehensive stream table to reduce utility costs identified in Level 4.

Finally, we used MATLAB to optimize reactor conversion, temperature, pressure, and feed ratio for maximum NPV. These values were input into a HYSYS simulation to determine detailed equipment sizes and finalize the process economic analysis.

2.1: Process Overview

The overall process flow diagram shown in Figure 2.2 is based on the HYSYS simulation and is consistent with industry standards for hydrodealkylation of toluene[4]. The process design based on MATLAB is located in Appendix D. Fresh and recycled toluene, as well as a fresh and recycled gas feed of methane and hydrogen enter separate inlet mixers and feed into a furnace, where the components are heated to 310°C. 2 heat exchangers in series are implemented to heat each stream to the desired reactor inlet of 598.9°C. The reactor outlet is followed by a series of heat exchangers to cool the outlet stream before entering 3 separation units to maximize purity and component recovery exiting the plant. A flash drum is used to remove methane and hydrogen from the system first to recycle those components, followed by a distillation column separating the ternary mixture of benzene, toluene, and diphenyl for high-purity benzene. A secondary distillation column is then used to further separate toluene and diphenyl to produce high-purity toluene to recycle back to the reactor and sell diphenyl as fuel.

2.2: Conceptual Design vs HYSYS Simulation

The complete process analysis given by the HYSYS simulation is slightly different than that given by the MATLAB conceptual design (Appendix C). These values differ due to the assumptions of each software system. Those include:

1. MATLAB treats all gasses as ideal and assumes that the heats of reaction are constant. Both systems use the Aly & Lee, or DIPPR thermodynamic equation to calculate the heat capacity[5]. The HYSYS simulation also uses the UNIQUAC thermodynamic package and is able to more accurately calculate and model thermodynamic behavior.
2. The mole balances for the flash were modeled after a perfect flash and feature complete separation of benzene, toluene, and diphenyl split from the light gases of hydrogen and methane. This assumption alters the amount of benzene theoretically recovered, but our designed flash conditions approximate this assumption quite well with less than 1% of the benzene feed lost to the vapor. Recovery of said benzene is provided by the scrubber unit before the recycle.
3. HYSYS provides a rigorous modelling of column design, but features no method to obtain equilibrium trays, which are provided only through estimation methods such as FUG or Underwood's equations. Difficulties in convergence of Underwood's equations meant that estimates of the number of stages employed FUG method, which are non-optimal and provides inaccurate reflux and reboiler ratios.
4. We make the assumption that the fresh feed of toluene is pure, but it will typically contain trace amounts of other hydrocarbons such as xylene or styrene[6]. This assumption allowed the toluene coming from the separator to be recycled back without the need for a purge. In reality, due to the trace components coming in with the toluene, a purge would be required to ensure there is no buildup of the trace components.
5. In MATLAB, distillation column mixtures were assumed to have a constant relative volatility, which retains most of its validity when comparing the HYSYS column parameters to their MATLAB counterparts. The model also assumed constant molal overflow due to minimal heat of vaporization changes when comparing the distillate and bottoms compositions of both columns. The MATLAB design did not include flash drums, as this model produced extremely high separation between components without the use of a flash drum, and so we exclusively used distillation columns, which still provided the high purity feeds desired for the outlet and recycle streams.

2.3: Process Flow Diagram

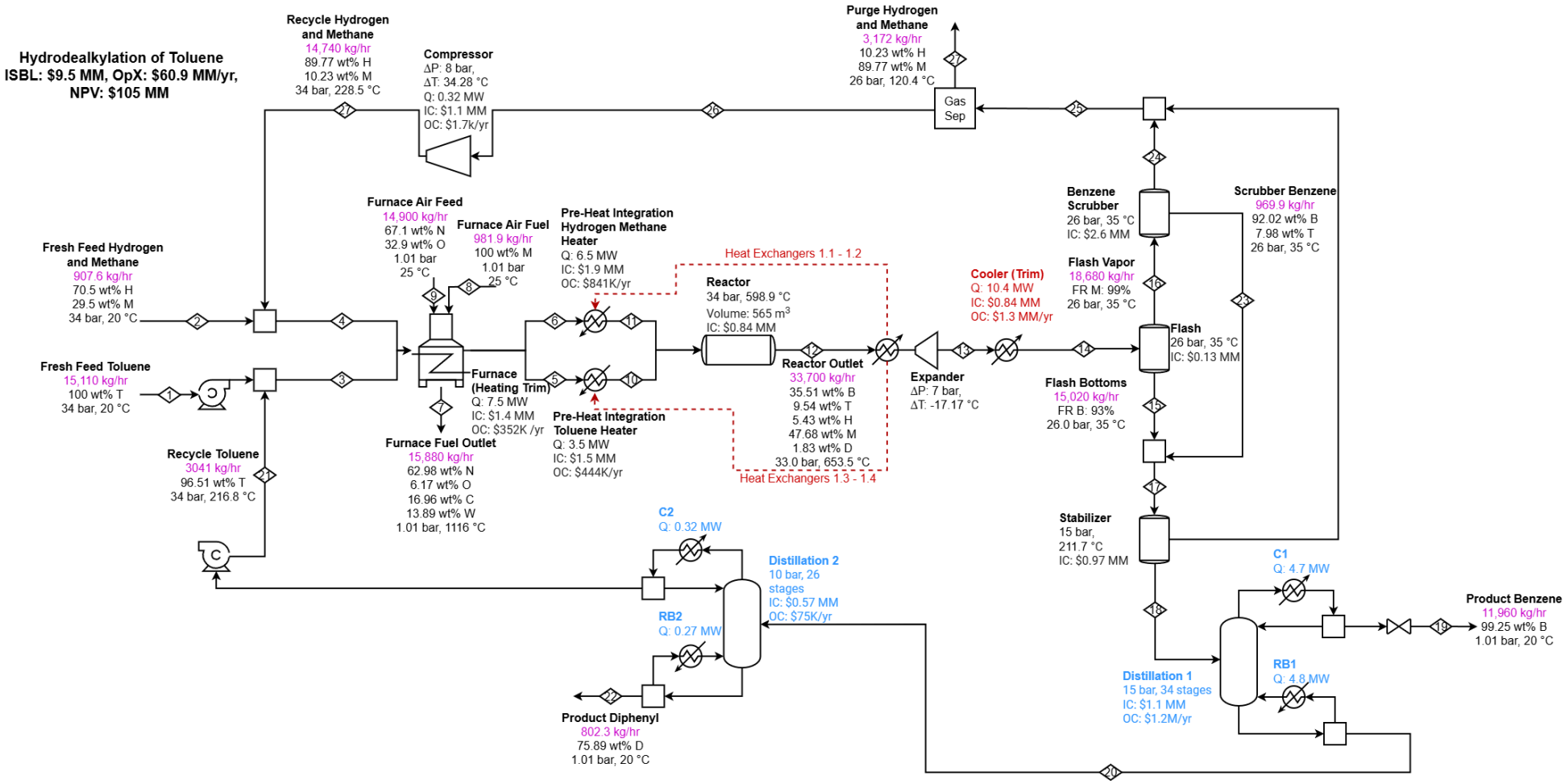


Figure 2.2 : Process flow diagram of the optimized HDA plant

For a detailed overview of the mass flow rates, compositions, temperature and pressure of each stream, see the stream summary in Table 2.1.

Table 2.1 : Components streams as modeled by HYSYS under the assumption that the process operates using plant years (8400 hr/yr).

Stream	Name	Temperature (°C)	Pressure (bar)	Mass Composition	Mass Flow Rate (kg/hr)	Mass Flow Rate (kta)
1	Toluene Feed	20.00	34.0	100% T	15,110	127
2	Gas Feed	20.00	34.0	70.5% H, 30.5% M	907.6	7.6
3	Toluene Mixer Outlet	58.77	34.0	100% T	18,070	152
4	Gas Mixer Outlet	134.6	34.0	86.3% M, 13.7% H	15,610	131
5	Toluene Furnace Outlet	310	34.0	100% T	18,070	152
6	Gas Furnace Outlet	310	34.0	86.3% M, 13.7% H	15,610	131
7	Furnace Flue Outlet	1116	1.01	62.98%, N 6.17% O, 16.96% C, 13.89% W	15,880	133
8	Furnace Fuel Gas	25.00	1.01	100% M	981.9	8.2
9	Furnace Air Feed	25.00	1.01	67.1% N, 32.9% O	14,900	125
10	Toluene HX Outlet	598.9	34.0	100% T	18,070	158
11	Gas HX Outlet	589.9	34.0	86.3% M, 13.7% H	15,610	137
12	Reactor Products	653.5	33.0	47.68% M, 5.43% H, 9.54% T, 35.51% B, 1.83% D	33,700	283
13	Reactor Cooler System Outlet	356.2	26.0	47.68% M, 5.43% H, 9.54% T, 35.51% B, 1.83% D	33,700	283
14	Flash Drum Inlet	35.00	26.0	47.60% M, 43.20% H, 1.66% T, 7.29% B, 0.19% D	33,700	283
15	Flash Bottoms	35.00	26.0	1.25% M, 0.01% H, 20.90% T, 73.77% B, 2.06% D	15,020	126
16	Flash Vapor	35.00	26.0	85.02% M, 9.79% H, 0.041% T, 4.78% B	18,680	157

Stream	Name	Temperature (°C)	Pressure (bar)	Mass Composition	Mass Flow Rate (kg/hr)	Mass Flow Rate (kta)
17	Stabilizer Feed	35.00	26.0	1.18% M, 0.01% H, 20.11% T, 74.88% B, 3.82% D	15,990	134
18	Distillation Column A Feed	211.7	15.0	20.36% T, 75.78% B, 3.87% D	15,800	133
19	Benzene Outlet	20	1.01	0.75% T, 99.25% B	11,960	100.5
20	Distillation Column A Bottoms	247.7	15	81.36% T, 2.73% B, 15.90% D	3,842	32.3
21	Toluene Pump Inlet	216.8	10.0	96.51% T, 3.45% B, 0.04% D	3,041	25.5
22	Diphenyl Outlet	20	1.01	24.11% T, 75.89% D	802.3	6.7
23	Scrubbed Volatiles	35.00	26.0	92.02% B, 7.98% T	969.9	8.1
24	Methane & Hydrogen Recycle	35.00	26.0	89.67% M, 10.33% H	17,720	149
25	Purge Inlet	120.4	26.0	89.77% M, 10.23% H	17,920	151
26	Purge Recycle	120.4	26.0	89.77% M, 10.23% H	14,740	124
27	Purge Outlet	120.4	26.0	89.77% M, 10.23% H	3,172	27
28	Compressed Gas Recycle	228.5	34.0	89.77% M, 10.23% H	14,740	124

2.4: Optimization of Key Design Variables

For complete optimization of this process, the variables that affect selectivity and conversion must be evaluated to produce the maximum NPV. The four design variables are the conversion, temperature, hydrogen purge mole fraction, and molar ratio. Those were optimized though our MATLAB optimization. These parameter effects are shown in the graphs of Figure 2.3, where the maximized net present value is bolded.

In general, hydrogen purge fraction represents the most influential variable of interest followed by the molar ratio (which is expected to remain at 5 due to chemical kinetics constraints). Interestedly, the initial reactor temperature and pressure play little to no role in the overall profitability of the plant.

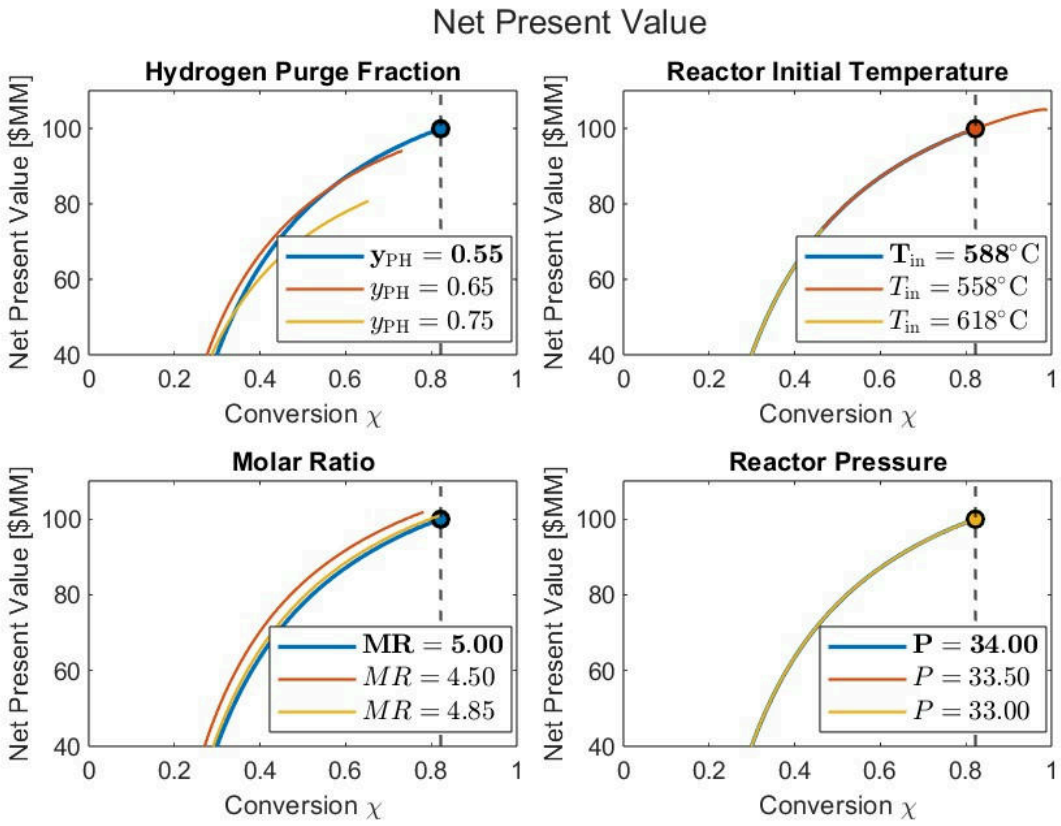


Figure 2.3 : MATLAB calculations of NPV for the process with varying design variables. Bolded values are the optimized values according to our MATLAB algorithm. Those may not exactly match our actual HYSYS design.

As outlined in Figure 1.1, selectivity of benzene over diphenyl is bounded from 80%-100% due to the favorable kinetics of the primary reaction over that of the side reaction – resulting in optimization of the plant to prefer higher conversions. The selectivity of this process is 93%. However, a single reactor cannot achieve past 82% conversion with the additional downside being a lower selectivity of our benzene product. Our analysis of our optimization concludes that this 82% conversion provides sufficient balance between a reasonable reactor size without the overhead capital cost of an additional reactor and good selectivity.

2.5: Reactor Design

In our conceptual design, we found that limiting the reactor configuration to a single large reactor at the barge limit (565 m^3) was both economical and simple. Shown in Figure 2.4 is the effect of theoretical

volumes plotted against conversions and select values of design variables. In general, the volume is a strong function of the hydrogen purge fraction, which is directly related to the flowrate of methane gas in the reactor recycle-feed loop. While the presence of methane increases the reactor size, it also inhibits the formation of diphenyl through equilibrium which is thought to be a larger contribution to the economics of the plant.

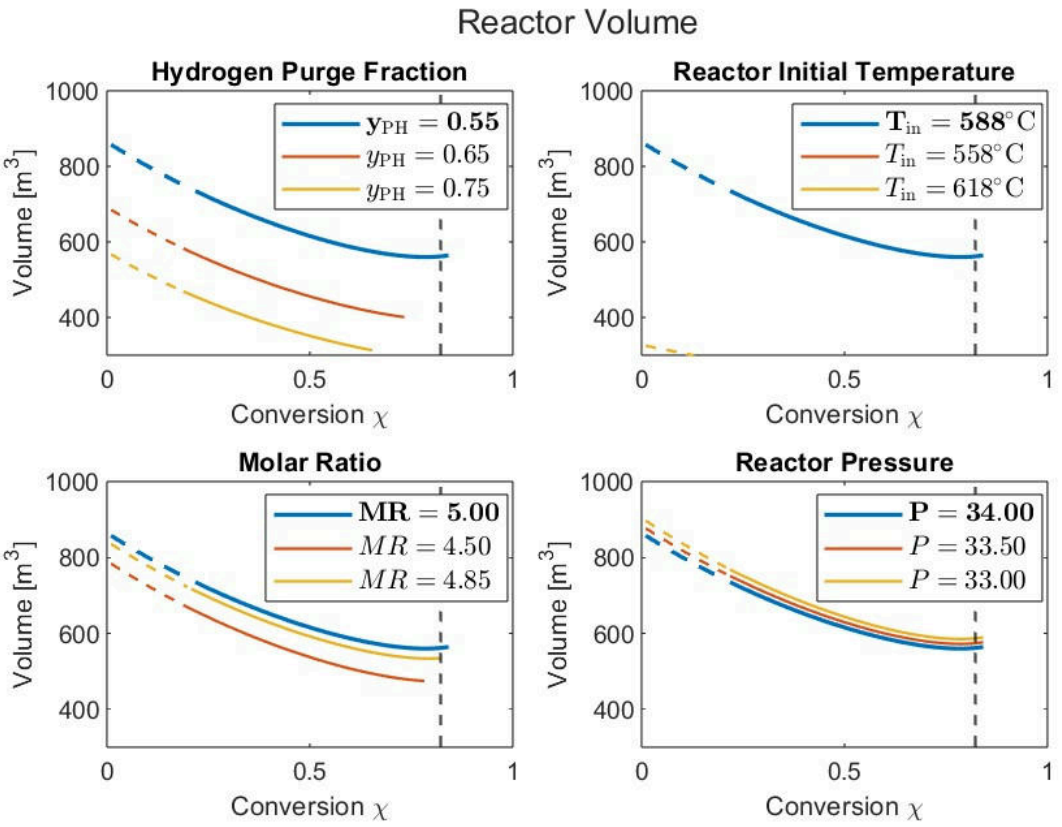


Figure 2.4 : Trends in theoretical reactor volume as a function of the four design variables from our MATLAB optimization algorithm. Bolded values represent the optimized values used in our HYSYS design. Dashed lines represent theoretical configurations that are not profitable. Note that the single reactor is limited to 565 m^3

In our conceptual design process, we considered both parallel reactors and a cold-shot scheme, but these featured either worse NPV after 15 years or only marginally better. A cold shot process has the potential to feature the most a reduction of the required volume to achieve the same conversion. Parallel reactors with even-split feeds are required to obtain volumes beyond 82% conversions.

As we feature a adiabatic reactor, the temperature rise is an important parameter to control. The adiabatic temperature rise is obviously affected by the initial reactor temperature and conversion, but is marginally affected by the fraction of hydrogen in the recycle/purge and the molar ratio. This is expected as larger amounts of methane in the recycle feed loop result in a larger heat capacity of the stream and therefore reduce the stream temperature.

2.6: Separation System Design

The MATLAB design for both distillation columns makes a constant molal overflow assumption so the molar flow rates of the liquid and vapor are constant throughout the columns. Another assumption made for the distillation system was the assumption of saturated liquid feed.

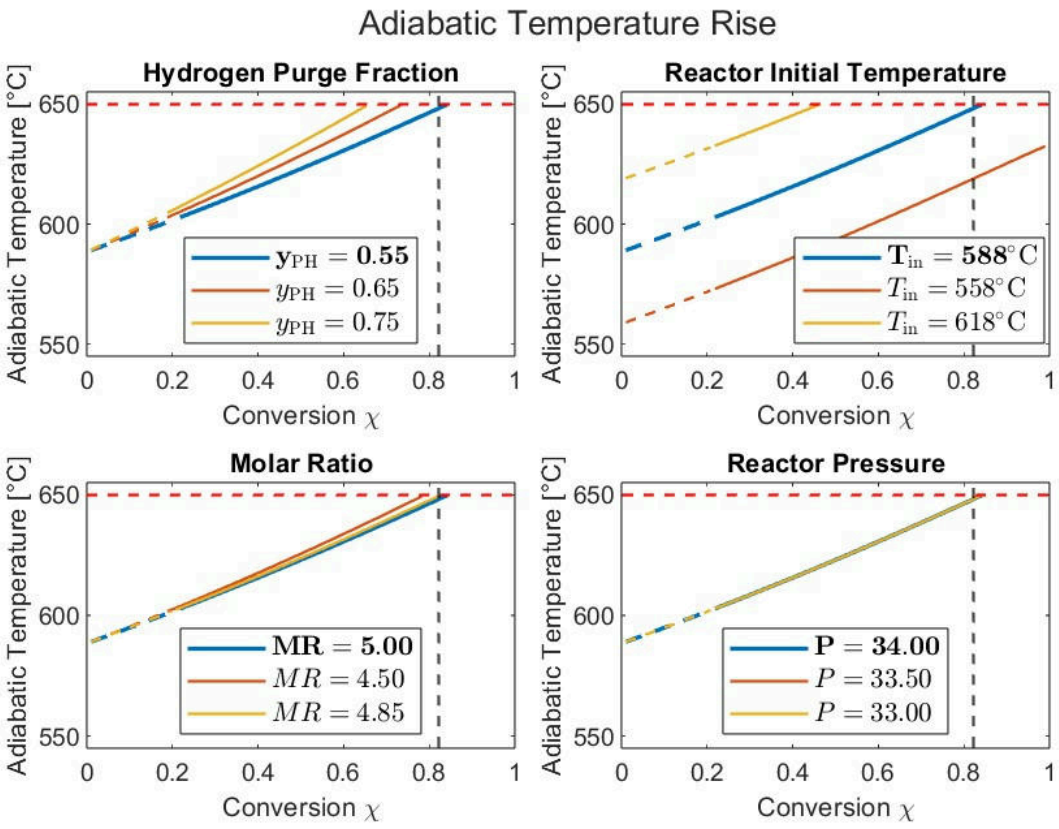


Figure 2.5 : Adiabatic temperature rise predictions for our MATLAB conceptual design as a function the four design variables. The bolded lines represent the optimized values used in the HYSYS simulation and the dashed line represents the optimal conversion.

No azeotropes were found between the components of this design. The separation of the ternary group benzene, diphenyl, and toluene was performed using 2 distillation columns in series, first separating benzene as the light key from the 2 heavier components, and then separating toluene from diphenyl. The first column has 17 theoretical stages, a vapor to feed molar flow rate (V/F) ratio of 1.6, and a specified 0.99 recovery of benzene in the distillate. The column is operated at 15 bar assuming a 1 bar pressure drop across the reactor. The second column has 13 theoretical stages, a V/F ratio of 1.3, and a specified 0.9999 recovery of toluene in the distillate, operated at 10 bar. Both columns are operating at greater than atmospheric pressure as it reduces the duties for condensers and keeps outlet feeds at a reasonable temperature for later compressor input.

To maximize benzene production a wet packed-bed scrubber was used to separate the benzene and toluene coming out of the top of the flash column. A stabilizer was added to the bottom of the flash to separate the methane and hydrogen from the rest of the components.

2.7: Heat Exchanger Network Design

The reactor outlet primarily consists of methane and hydrogen and exits at 654°C. In order to ensure safe operation of the process units following the reactor, it is necessary to develop a heat exchanger network to efficiently cool down the high heat outlet to the flash temperature of 35°C. To do this, the reactor effluent is allowed to heat exchange with the reactor feed after the furnace, the gas recycle, and additional cooling provided through condensate steam and cooling water. In this scheme, the furnace is a trim heating utility

and trim cooling is provided by the condensate steam and water utilities. This system separately heats each inlet stream to prevent spontaneous mixing in the furnace and heat exchangers prior to the reactor. Heat integration was determined for the streams between the reactor feeds and the effluent of the plug flow reactor.

The heat exchanger network was designed based on a stream table of available process streams, which if implemented reduce operating costs across the plant. The integration performed reduces the furnace duty of 18 MW to 7.5 MW (a 59% reduction) and a 19.2 MW cooling duty to 8.9 MW (a 54% reduction in cooling costs). Together this reduces the total heating and cooling utility energy consumption from 37.2 MW to 16.4 MW (a 56% reduction in total duty used).

The heat exchangers used for the reactor feed and effluent were priced as Grade 316 stainless steel to effectively guard against hydrogen embrittlement while sustaining high temperature use. As 2 of these heat exchangers do surpass the given material limitation of 120°C, this steel type has been proven to withstand extreme heating temperatures[7]. No heat exchangers were costed for the reactor as it operates adiabatically.

In our MATLAB optimization, we modeled the HEN by imagining the network as a single, large heat exchanger with a total area that was then costed. This approximation is thought to provide a smaller than required area for the HEN network.

Section 3: Economic Analysis

3.1: Economic Metrics & Analysis

The final process economic analysis (Table 3.4) was calculated using the HYSYS simulation feed inlets, product outlets, heater duty and reactor volume at the ideal conditions determined through MATLAB optimization.

Equipment included in installation costs for this plant are the furnace, reactor, separation system, and heat exchanger network. The mixers and expander are not included in these economic calculations because they are beyond the scope of this report. The land cost for this plant is also not included under the assumption that overall plant size and exact location are unknown. The total ISBL cost for this process is $9.5 MM and the OSBL cost is $3.8 MM. The direct cost is $17.7 MM and the indirect cost is $4.0 MM. These values bring the fixed capital investment to a total of $21.7 MM when including a 25% contingency factor to ensure all unexpected costs during construction can be covered. The working capital value is $8.1 MM and the start-up cost is $1.7 MM, bringing the total capital investment to $31.5 MM.

Table 3.2 : Final economic calculations determined using HYSYS parameters.

Economic Metric	Value
NPV ($MM)	104.9
IRR (%)	70
TCI ($MM)	31.5
OPEX ($MM/yr)	60.9

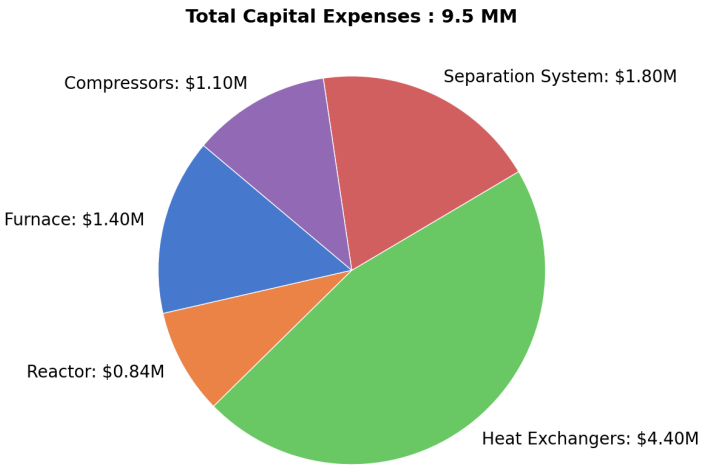


Figure 3.6 : Pie chart of the capital costs for the plant design proposal.

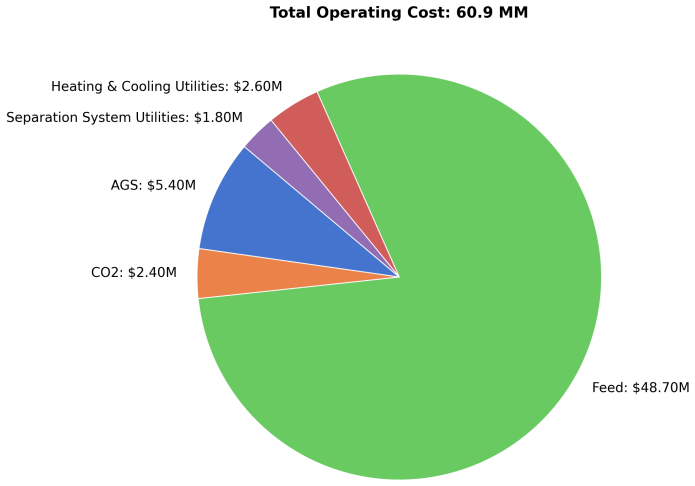


Figure 3.7 : Pie chart of the operating costs for the plant design proposal.

The operating costs for this plant include utilities used to power the process units, administrative & general services (AGS), carbon production charge and raw material costs (See Appendix B.4 for cost correlations). The cost of the inlet feed is $48.7 MM. The total cost of operational utilities is $4.4 MM. The administrative and general service costs total to $5.4 MM and include administration costs, product selling prices, and plant research & development, assuming these values include salary, overhead, maintenance costs, and financing interest. The carbon charge fee is $40/MT and costs $2.4 MM. We assume that the carbon produced from powering the plant is equivalent to the production from burning methane (Appendix C.8.2). This brings the overall plant operating cost to $60.9 MM.

3.2: Furnace and Reactor Costs

The furnace and reactor costs were determined under the assumption that both process units consisted of Grade 321 stainless steel as it is highly resistant to high temperature operations and hydrogen embrittlement in comparison to carbon or other stainless steels due to its austenitic structure[8]. The installed cost of the furnace is $1.4 MM and is based on the heat load as the sum of the energy necessary to heat the feed to reaction temperature without influencing the adiabatic operation of the reactor. To determine the

installation cost of the reactor, we calculated the heat load as the sum of required feed-heating energy as well as to maintain adiabatic operation within the reactor, a cost of $0.84 MM. To determine the operating cost of the reactor and furnace, the total heat load is also used, making the assumption that the heat comes from natural gas combustion. The heat load multiplied by the price of fuel calculates the operating cost for both process units.

3.3: Compressor Costs

The compressor installed cost is determined using the cost correlation in Appendix C.6. The compressor is assumed to be a motor drive centrifugal compressor and has an installed cost of $1.1 MM. The operating cost is based on the braking horsepower of the compressor and assumes that the compressor is electricity powered. This sets the operating cost as a function of the braking horsepower and the price of electricity in kWh, valued at $0.06/kWh, to be $1.7K.

3.4: Separation System Costs

The plant separation system consists of a flash drum, 2 distillation columns, a scrubber, and a stabilizer. For any process unit with high levels of hydrogen, vessels were built from Grade 316 or 321 stainless steel to combat potential hydrogen embrittlement. The installed cost of the flash drum is calculated using the HYSYS simulation as it is not part of the MATLAB model due to its assumptions. The first column contains 17 theoretical stages with a 99% fractional recovery of benzene in the distillate and an installed cost of $1.1M. The second distilla.tion column has 13 theoretical stages with a 99.99% fractional recovery of toluene in the distillate and an installed cost of $570K

The scrubber and stabilizer were added to the HYSYS simulation to maximize the benzene production and keep consistency between the HYSYS design and the MATLAB design. Preliminary costing calculations determined that the scrubber has an installed cost of approximately $2.6 MM and the stabilizer has an installed cost of $0.97 MM using minimum work capital correlations. As these are preliminary calculations, we recommend FEL-2 conduct a more detailed analysis to refine the installed cost estimations.

3.5: Sensitivity Analysis

To produce viable economic analysis based on our FEL-1 plant design, we must rely on multiple assumptions regarding various economic factors including toluene, benzene, and fuel prices holding constant with time, as well as the tax and enterprise rates staying at their current rates of 27% and 15% respectively. To evaluate the sensitivity of the net present value to these economic fluctuations, we calculated the range of changes possible based on expected variations in different factors.

Although the NPV was impacted by the variations for every factor, none were significant enough to produce negative values. While these values are key to retaining high profitability, they have minimal influence on the overall profitability of the process.

The factor with the most influence over NPV is the price of benzene. NPV when the price of benzene is altered has the greatest fluctuations, in a range of $24MM to $172MM. This range of sensitivity is an indicator that this value should be continually monitored during plant construction and operation, as the profitability of this plant could be severely limited.

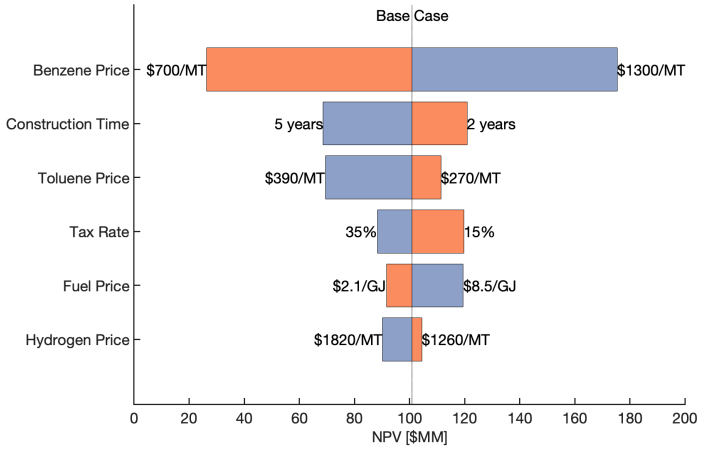


Figure 3.8 : Variations in NPV as a result of fluctuations in key economic variables.

Furthermore, we must also consider the economic impact of variations in fuel prices, which could result in a profitability range of $90MM to $122MM, a staggering change in financial returns.

These factors are similar in that their prices are dictated by global supply and demand for petrochemicals. While the market for benzene is predicted to grow by 6.1% from 2025 to 2034[1], its value could skyrocket further with proposed tariffs targeting the largest exporters of benzene to the United States[9], including South Korea[10], India[11], and the Netherlands[12]. If these economic restrictions are enacted, it can become far more lucrative to produce benzene domestically. For fuel prices, while our plant design does sell fuel at a rate of $4.25/GJ, is unlikely to sustain plant operations in its entirety, so natural gas will need to be purchased for this system. These fuel prices could see varying economic changes in the near future, as Canada, a top energy importer to the United States, recently enacted a major tariff on electricity, increasing natural gas prices by 21% since the beginning of 2025[13]. From these findings, it appears that political decisions will be integral in evaluating the continued viability of this plant design.

3.6: Second-Order Effects: Monte Carlo Simulation

To further quantify the risks involved in this plant's design with respect to the economic viability, we applied a Monte Carlo simulation to measure the financial risk associated with our chosen conceptual design, which can be found in Figure 3.9. In our simulation, we applied a triangular distribution to model the underlying distribution of three factors: the prices of benzene, toluene, and the construction time of the project.

Triangular distributions were employed for all variables as our goal was to measure deviations around the most frequent or common value (mode-based distribution). In addition, the population distribution for most variables are often unknown and the triangular distribution provides a well-known simple and effective solution to modeling said parameters. We do not account for recent tariff prices in our simulation as those effects are short term and likely wont effect the plant lifetime of 15 years.

We conclude that our design is risk-adverse as there is no case were we at at financial risk of becoming unprofitable. In general, we have a 72% chance of obtaining a NPV less than our base case of $100 MM and a 28% chance of obtaining a higher NPV.

(a) **(b)**

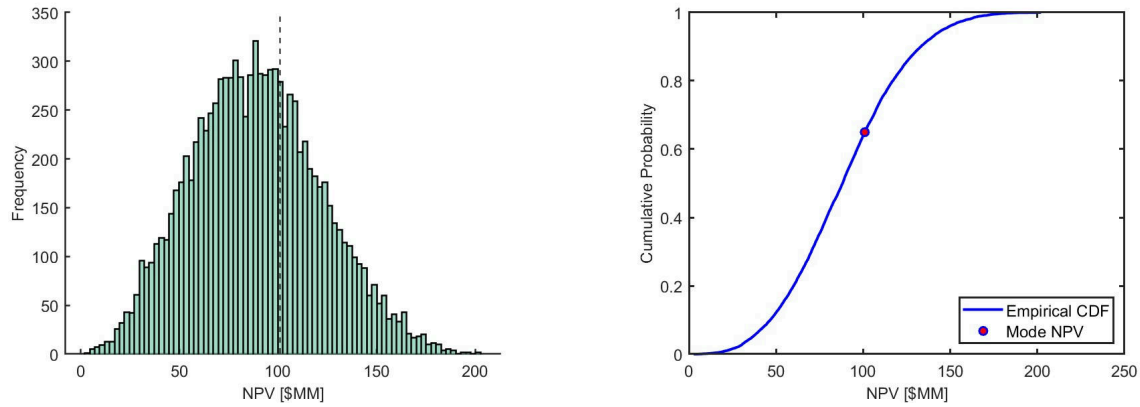


Figure 3.9 : **(a)** Distribution of our NPV after applying Monte Carlo simulation featuring changes construction time and prices of benzene and toluene. **(b)** The calculated probability distribution which measures the probability of achieving a certain NPV given the applied perturbations to our design.

Section 4: Safety and Environmental Impact

4.1: Safety Hazards

There are important safety considerations necessary for plant function, involving the operation conditions, chemical components, and reactions within this process. The major safety hazards faced in the use of the chemical components and their reactions are the flammability of the components, as they pose an explosion hazard and health risks upon exposure to the chemicals. The chemicals involved in the process pose health risks as they have relatively low occupational exposure limits, ranging from an average of 20 ppm over 8 hours for toluene to an average of 0.2 ppm over 8 hours for diphenyl (see Appendix E).

As hydrogen is utilized in significant amounts throughout the plant, precautions should be taken during operation. All process units where hydrogen is present are constructed from Grade 316 or 321 Stainless Steel to reduce the risk of hydrogen embrittlement. This phenomenon causes material susceptibility to cracking, fatigue and even explosions when involving reactive environments[14].

Hazards around the smaller compounds primarily center around the operating parameters of the process, as a leak from the high-pressure and high-temperature equipment, such as the furnace and reactor can cause severe burns for operators. Operators must always wear the proper protective equipment when they are within the vicinity of the process (see Appendix E for PPE).

Operating parameters and equipment also present hazards to operators due to the high pressure, high temperature, and high flow rate operating conditions. For example, pressure can increase if there is a blockage in pipes that can lead to buildup, and the pressure will decrease after the blockage. To ensure the low possibility of these events occurring, independent control systems for pressures, temperatures, and flow rates should be installed throughout the process to decrease the probability of these events happening.

4.2: Environmental Impact

To evaluate the extent of this chemical plant's environment impact, it is important for us to consider the carbon dioxide being produced and then released. In total, 800K GJ are used for an overall production of 60 kt CO_2 annually, where 1.094 kg CO_2 is produced for every 1 kg of C_6H_6. We must also consider that

while our product streams do not directly emit carbon, those used as fuel may be burned to release carbon dioxide after being sold, such as the diphenyl outlet.

Table 4.3 : Environmental hazards that product streams may contribute to the environment.

Component	Environmental Hazard
Toluene (C_7H_8)	Toluene has moderate chronic acute toxicity for aquatic life, but its extensive effects on birds or land animals have not been determined.[15]
Benzene (C_6H_6)	Benzene has a high acute toxicity for aquatic life. It can cause death and damage in plants and other agricultural crops.[16]
Methane (CH_4)	Methane is a greenhouse gas that can cause extreme damage to the earth ozone layer. It has 80 times the potency of carbon dioxide over a 20-year period.[17]
Hydrogen Gas (H_2)	Hydrogen gas has no major negative impact on wildlife in concentrations up to 100 ppm.
Diphenyl ($C_{12}H_{10}$)[18]	Diphenyl has high acute toxicity but low chronic toxicity for aquatic life. Diphenyl is generally non-toxic to terrestrial organisms.[19]

The product streams may pose additional environmental hazards beyond carbon dioxide release, detailed in Table 4.5. In summary, most of our products are short-term environmental hazards that can cause acute harm, excepting methane as a intense greenhouse gas.

Section 5: Process Alternatives and FEL-2 Considerations

The main process alternative considered was the cold shot method, where the reaction takes place in multiple reactors in series with interstage cooling. The feed is split into multiple streams to cool the reactor effluent before it is sent to the next reactor in the series. This process resulted in an NPV of $104.9 MM, which is identical to our optimized case using a single reactor.

Ultimately, our team decided that although this process alternative generates the same NPV as our chosen base case, it is a far more complicated design. A more complex system requires more monitoring to ensure all parts are functioning as intended while giving the same NPV. There is a potential to provide a more efficient Heat Exchanger Network (HEN), but this would require better modelling of the HEN network in MATLAB.

Section 6: Conclusion

Our FEL-1 Analysis demonstrated that the proposed hydrodealkylation process has the potential to be profitable at a production rate of 100 kta benzene. The NPV of the plant is $104.9M and the IRR is 70%. The TCI of the plant is $31.5M. Based on our sensitivity analysis, it is likely these calculations will fluctuate with market pricing and economic cycles in the petrochemical industry.

The 565 m^3 plug-flow reactor for this plant operates at 598.9°C, 34 bar, and a hydrogen-to-toluene inlet ratio of 5.4 to produce 100 kta of benzene. The reactor conversion is 82%, and unreacted toluene and hydrogen is separated and recycled back to the reactor feed.

Ultimately, we recommend that APL moves forward to a FEL-2 design of this plant due to favorable profitability metrics.

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(25) *Diphenyl*. https://www.inchem.org/documents/icsc/icsc/eics0106.htm.

(26) Compatible with Carbon Steel.

(27) Compatible with Stainless Steel.

Appendix A: Reactor Design and Mole Balances

A.1: Overall Plant Balances and Degree of Freedom Analysis

$$\begin{aligned} &\text{Reaction 1} && \mathrm{C_7H_8 + H_2 \longrightarrow C_6H_6 + CH_4} \\ &\text{Reaction 2} && \mathrm{2C_6H_6 \longrightarrow C_{12}H_{10} + H_2} \end{aligned} \tag{A.1}$$

We also have the follow shorthand symbols used throughout this appendix to describe all balances:

$$T \equiv \mathrm{C_7H_8}, B \equiv \mathrm{C_6H_6}, H \equiv \mathrm{H_2}, M \equiv \mathrm{CH_4}, D \equiv \mathrm{C_{12}H_{10}} \tag{A.2}$$

A material balance on each of the components involved in these reactions produces the following system of equations, where ξ_i is defined as the extent of the ith reaction. Using the steady-state process balance *in - out + gen - cons = 0*, the following mole balances are produce

$$\begin{aligned} T &: F_T = \xi_1 \\ B &: P_B = \xi_1 - 2\xi_2 \\ H &: P_H - F_H = \xi_2 - \xi_1 \\ M &: P_M - F_M = \xi_1 \\ D &: P_D = \xi_2 \end{aligned} \tag{A.3}$$

where F_i is the molar flow rate of component i into the plant and P_i is the molar flow rate of component i out of the plant. For 2 reactions, we require 2 reference compounds, toluene and diphenyl. Hence the remaining balances of benzene, hydrogen, and methane can be rearranged to determine the global input-output balances (IO Level):

$$\begin{aligned} F_T &= P_B + 2P_D \\ P_H - F_H &= P_D - F_T \\ P_M - F_M &= F_T \end{aligned} \tag{A.4}$$

Here F_G is the total molar flow rate of feed of gas (hydrogen + methane) and P_G is the total molar flow rate of purge gas leaving the plant. In addition, S is defined as the selectivity of benzene with respect to toluene.

$$\begin{aligned} F_H &= F_G y_{\mathrm{FH}} \\ F_M &= F_G(1 - y_{\mathrm{FH}}) \\ P_H &= P_G y_{\mathrm{PH}} \\ P_M &= P_G(1 - y_{\mathrm{PH}}) \\ S &= \frac{P_B}{F_T} \end{aligned} \tag{A.5}$$

Rewriting the mole balances in Equation (A.4) with Equation (A.5) above and reducing to just design variables (S, y_{PH}, and P_B) gives the following IO Level balances

$$\begin{aligned}
F_T &= \frac{P_B}{S} \\
F_H &= \frac{P_B y_{\mathrm{FH}}(1 + S + y_{\mathrm{PH}} - S y_{\mathrm{PH}})}{2S(y_{\mathrm{FH}} - y_{\mathrm{PH}})} \\
F_M &= \frac{P_B(-1 + y_{\mathrm{FH}})(-1 + S(-1 + y_{\mathrm{PH}}) - y_{\mathrm{PH}})}{2S(y_{\mathrm{FH}} - y_{\mathrm{PH}})} \\
P_H &= \frac{P_B(1 + S + y_{\mathrm{FH}} - S y_{\mathrm{FH}}) y_{\mathrm{PH}}}{2S(y_{\mathrm{FH}} - y_{\mathrm{PH}})} \\
P_M &= P_B\left(\frac{(-1 + S(-1 + y_{\mathrm{FH}}) - y_{\mathrm{FH}})(-1 + y_{\mathrm{PH}})}{2S(y_{\mathrm{FH}} - y_{\mathrm{PH}})}\right) \\
P_D &= \frac{P_B(1 - S)}{2S}
\end{aligned} \tag{A.6}$$

A.2: Recycle and Reactor Mole Balances

Let E_i be the effluent of specie i out of the reactor and R_i to be the recycle flow rate of specie i.

The reactor per-pass conversion X can be defined using the recycle stream of toluene (our limiting reactant) using the relationship below:

$$\mathrm{X} = \frac{\xi_1}{F_T + R_T} = \frac{F_T}{F_T + R_T} \tag{A.7}$$

Another design variable of interest is the molar ratio of toluene to hydrogen into the reactor:

$$\mathrm{MR} = \frac{F_H + R_H}{F_T + R_T} \tag{A.8}$$

Finally, we can do a molar balance around the reactor with all outgoing flow rate written in E_i notation. Just like the IO Level balances, only the methane, hydrogen, and benzene are required. Note that only the methane and hydrogen balances are non-trivial.

$$\begin{aligned}
T &: E_T = R_T \\
B &: E_B = P_B = F_T - 2P_D \\
H &: E_H = R_H + P_D + F_H - F_T \\
M &: E_M = F_M + R_M + F_T
\end{aligned} \tag{A.9}$$

Additional, we have the balance around the purge stream splitter:

$$\begin{aligned}
H &: E_H = R_H + P_H \\
M &: E_M = R_M + P_M
\end{aligned} \tag{A.10}$$

Put all together and written in terms of purely design variables (S, X, y_{PH}, MR, P_B) we have the following expressions:

$$
\begin{aligned}
R_T &= P_B\left(\frac{1-\mathrm{X}}{S\mathrm{X}}\right) \\
R_H &= \frac{P_B}{2S}\left(\frac{2\ \mathrm{MR}}{\mathrm{X}} + \frac{y_{\mathrm{FH}}(-1+S(-1+y_{\mathrm{PH}})-y_{\mathrm{PH}})}{y\mathrm{FH}-y_{\mathrm{PH}}}\right) \\
R_M &= P_B\frac{(-1+y_{\mathrm{PH}})(2\ \mathrm{MR}\ (y_{\mathrm{PH}}-y_{\mathrm{FH}})+y_{\mathrm{FH}}\mathrm{X}(1+S+y_{\mathrm{PH}}-Sy_{\mathrm{PH}}))}{2S\mathrm{X}y_{\mathrm{PH}}(y_{\mathrm{PH}}-y_{\mathrm{FH}})} \\
E_H &= P_B\left(\frac{2\ \mathrm{MR}-\mathrm{X}-S\mathrm{X}}{2S\mathrm{X}}\right) \\
E_M &= P_B(1-y_{\mathrm{PH}})\left(\frac{2\ \mathrm{MR}-\mathrm{X}-S\mathrm{X}}{2S\mathrm{X}y_{\mathrm{PH}}}\right)
\end{aligned}
\tag{A.11}
$$

A.3: Derivation of Reactor Design Equations

A.3.1: Reaction System Rate Laws

The reaction rates for chemical reactions 1 & 2 are:

$$r_1 = k_{1,\mathrm{f}}C_T C_H^{0.5} \tag{A.12}$$

$$r_2 = k_{2,\mathrm{f}}P^2\left(y_B^2 - \frac{(y_D y_H)}{K_{\mathrm{eq}}}\right) \tag{A.13}$$

with pre-factors $k_{1,\mathrm{f}}$ and $k_{2,\mathrm{f}}$ determined by the expressions

$$k_{1,\mathrm{f}} = k_{1,0}\exp\left(-\frac{E_{\mathrm{a1}}}{RT}\right) \tag{A.14}$$

$$k_{2,\mathrm{f}} = k_{2,0}\exp\left(-\frac{E_{\mathrm{a2}}}{RT}\right) \tag{A.15}$$

To obtain concentrations or partial pressures, we assume ideal gas law behavior and solve for each using the following equations:

$$[i] = \frac{P}{RT}\left(\frac{F_i}{F_{\mathrm{tot}}}\right) \qquad P_i = RT\left(\frac{F_i}{F_{\mathrm{tot}}}\right) \tag{A.16}$$

Here, T is in K, the activation energies are in kcal/gmol, and concentrations are in gmol/L. The rate and rate constant equations are viable over a temperature range of $550°C < T < 650°C$, a pressure range of $30\ \mathrm{bar} < P < 35\ \mathrm{bar}$ and a toluene to hydrogen molar ratio range of 4.5–5.5.

A.3.2: Reactor Material and Energy Design Equations

In order to find the conversion, volume, and projected adiabatic temperature rise in the PFR at desired design variable conditions, we developed an series that combined the PFR material and energy design equations with our given reaction rates. The component material balance equations take the generic form:

$$\frac{dF_i}{dV} = -r_i \tag{A.17}$$

where P is the total pressure, R is the universal gas constant, T is the incoming reactor temperature, and F_{tot} is the total molar flow rate at a specific length/volume along the reactor. This produces the following ODEs for the change in molar flow rate for each component.

$$\begin{aligned} B &: \frac{dF_B}{dV} = r_1 - 2r_2 \\ T &: \frac{dF_T}{dV} = -r_1 \\ H &: \frac{dF_H}{dV} == -r_1 + r_2 \\ M &: \frac{dF_M}{dV} = r_1 \\ D &: \frac{dF_D}{dV} = r_2 \end{aligned} \tag{A.18}$$

The energy balance design equation takes the form,

$$\frac{dT}{dV} = -\frac{\sum_i^R \Delta H_{\text{rxn},i} r_i}{\sum_i^C F_i \left\{C_{\text{p}}(T)\right\}_i}) \tag{A.19}$$

where C is the number of components (5) and R is the number of reactions (2). Here, the heat capacities for each specie are modelled by the Allen-Lee-Dipper approximation as a function of temperature. Individual parameters for each species are reported in and use the following formula:

$$\left\{C_{\text{p}}(T)\right\}_i = A_i + B_i \left(\frac{C_i/T}{\sinh(C_i/T)}\right)^2 + D_i \left(\frac{E_i/T}{\cosh(E_i/T)}\right)^2 \tag{A.20}$$

Table A.4 : Allen-Lee-Dippr parameters for the major components in the reactor. Note that parameters for diphenyl were difficult to source in addition to the fact that only a small fraction of the total molar flow rate in the reactor was diphenyl. In other words, its heat capacity contribution was found to be minimal.

Species	***A***	***B***	***C***	***D***	***E***
Benzene	13.0616	53.9027	1691.83	42.5244	819.552
Toluene	17.5653	66.1975	1712.33	50.2196	795.271
Methane	8.00318	19.2633	2148.79	10.4423	1017.07
Hydrogen	6.66789	2.33458	2584.98	0.749019	559.656

A.3.3: Reactor Design

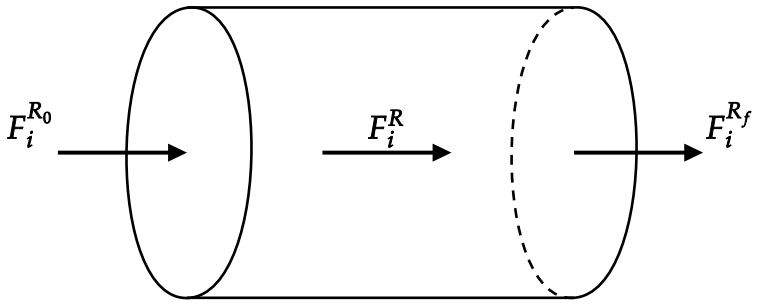


Figure A.10 : Illustration of the reactor with feed flow rate $F_i^{R_0}$ and outlet flow rate F_i^{R}.

The relationship between selectivity and conversion are independent of reactor scale, which allows us to calculate the volume of the reactor and its conversion and selectivity based on a reactor feed rate basis. In this case, the toluene is taken at a basis of 1 mol/s with all basis flow rates indicated with a prime notation: $F_i^{R'}$. It is assumed that only toluene, methane, and hydrogen are recycled and are the only components at the feed of the reactor.

$$\begin{aligned} F_T^{R_0'} &= 1 \\ F_H^{R_0'} &= \mathrm{MR} \\ F_M^{R_0'} &= \frac{2\ \mathrm{MR}\ S(1-y_{\mathrm{PH}}) - P_B(1+S+y_{\mathrm{PH}}-Sy_{\mathrm{PH}})}{2Sy_{\mathrm{PH}}} \end{aligned} \tag{A.21}$$

These initial conditions can be used to solve the system of ODEs for the flow rates of each component and temperature along the reactor. Integration over a range of volumes produces the flow rates for each component across the reactor, which can then be used to calculate selectivity and conversion as follows

$$X = \frac{F_T^{R_0'} - F_T^{R_f'}}{F_T^{R_0'}} \qquad S = \frac{F_B^{R_f'}}{F_T^{R_0'}} \tag{A.22}$$

The basis volume V_{basis} of the reactor is obtained by solving the ODE design equations and treating the volume V as a function of conversion X. The actual or true volume is obtained by considering the multiplicative increase in total molar flow rate over that of the total basis flow rate. Note that C is the number of components.

$$V_{\text{true}} = \left(\frac{\sum_i^C F_i^{R_f}}{\sum_i^C F_i^{R_f'}} \right) V_{\text{basis}} \tag{A.23}$$

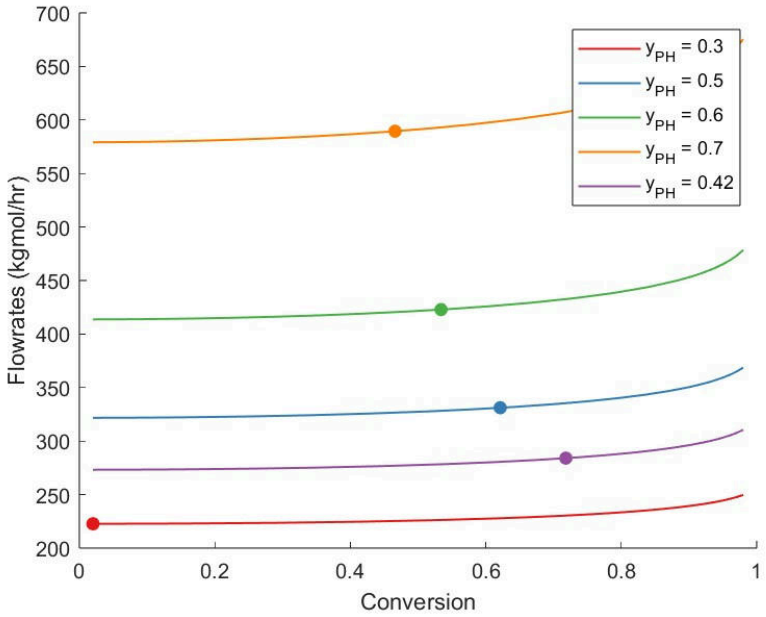
A.4: Product Component Destinations

Table A.5 : Component Classifications and Destinations

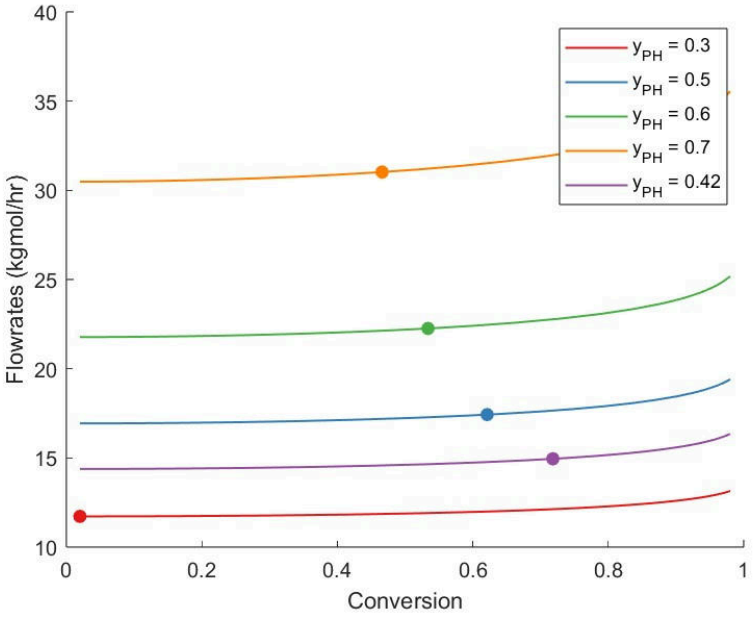
Component	Classification	Destination
Toluene	Primary Reactant	Recycle Stream back to Mixer
Hydrogen	Primary Reactant	Recycle Stream back to Mixer
Benzene	Primary Product	Sold as Benzene
Methane	Co-Product	Purge & Recycle
Diphenyl	By-Product	Sold with Methane as Fuel

A.5: Consistency Checks

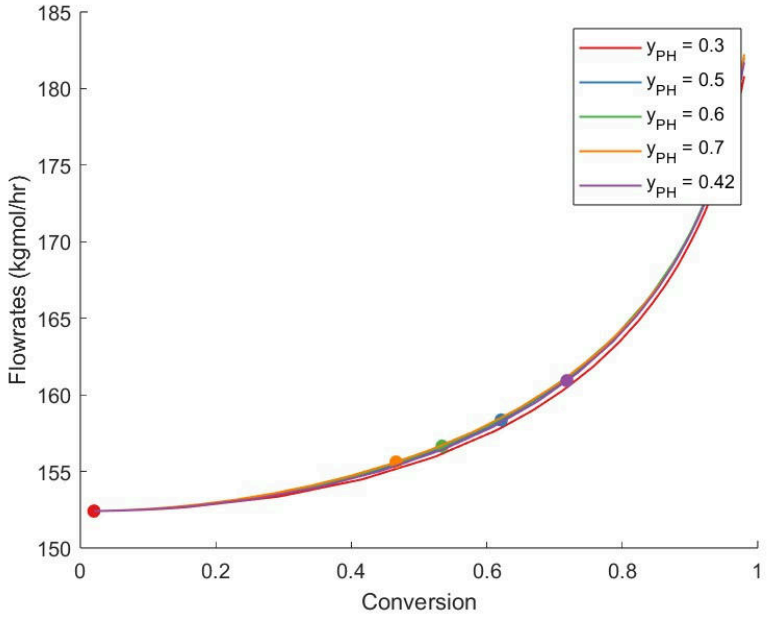
A.5.1: Feed to Process



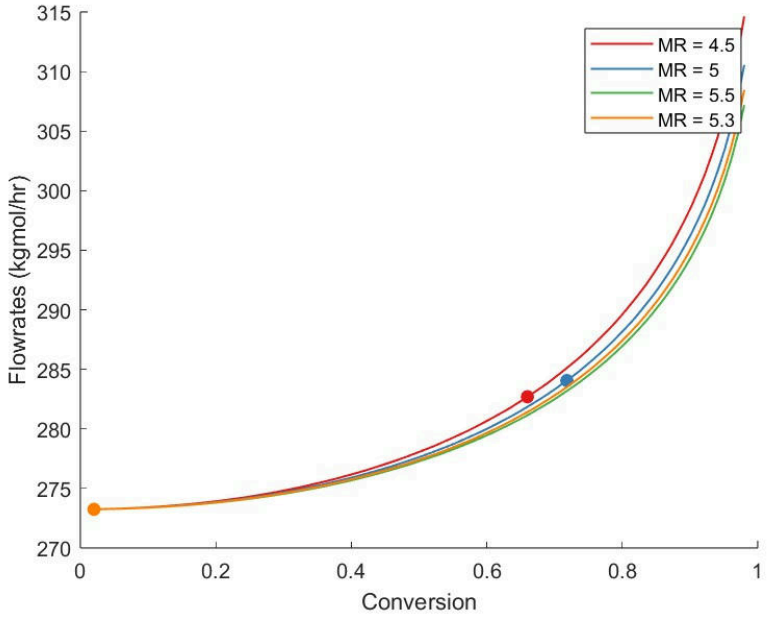
(5.1.1.a) Fresh feed of toluene versus reactor conversion for bounds of y_{PH} at 598.9°C and an MR of 5.4.



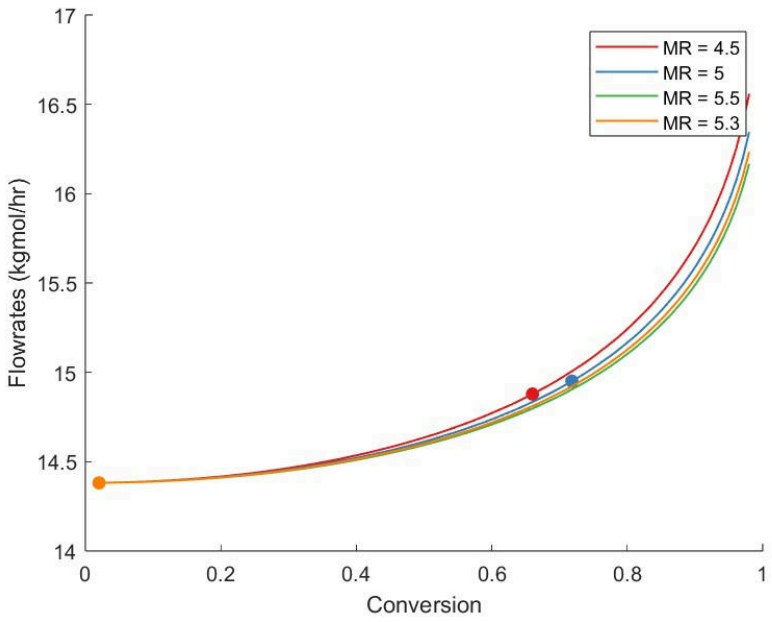
(5.1.1.b) Fresh feed of hydrogen versus reactor conversion for bounds of y_{PH} at 598.9°C and an MR of 5.4.



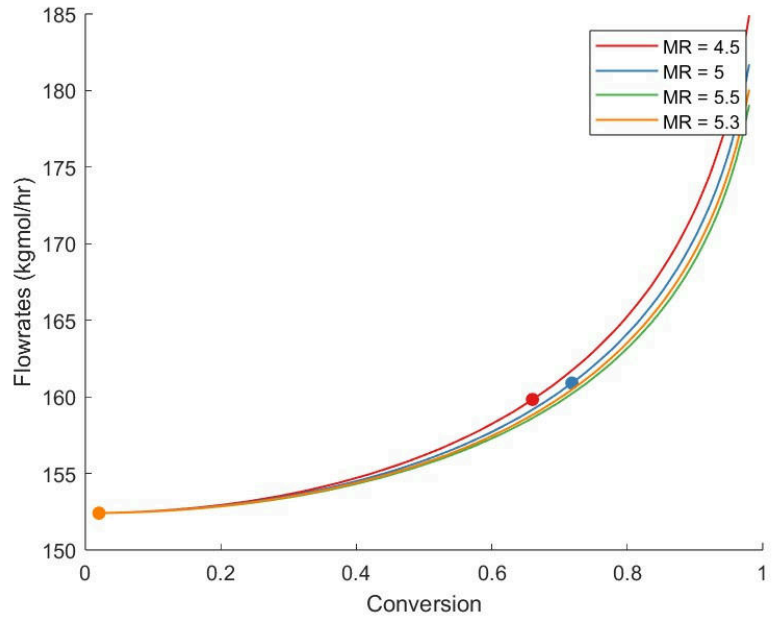
(5.1.1.c) fresh feed of methane versus reactor conversion for bounds of y_{PH} at 598.9°C and an MR of 5.4.



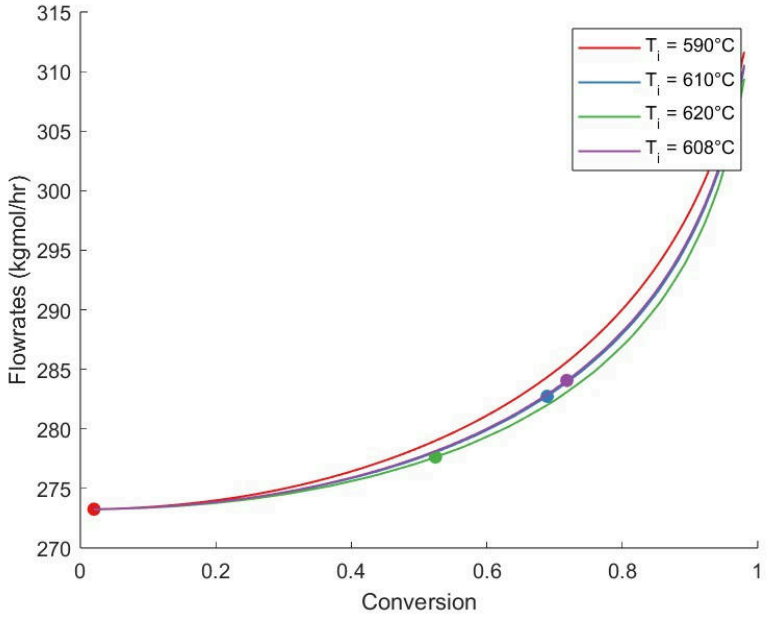
(5.1.2.a) Fresh feed of toluene versus reactor conversion for bounds of MR at 598.9°C and a y_{PH} of 0.42.



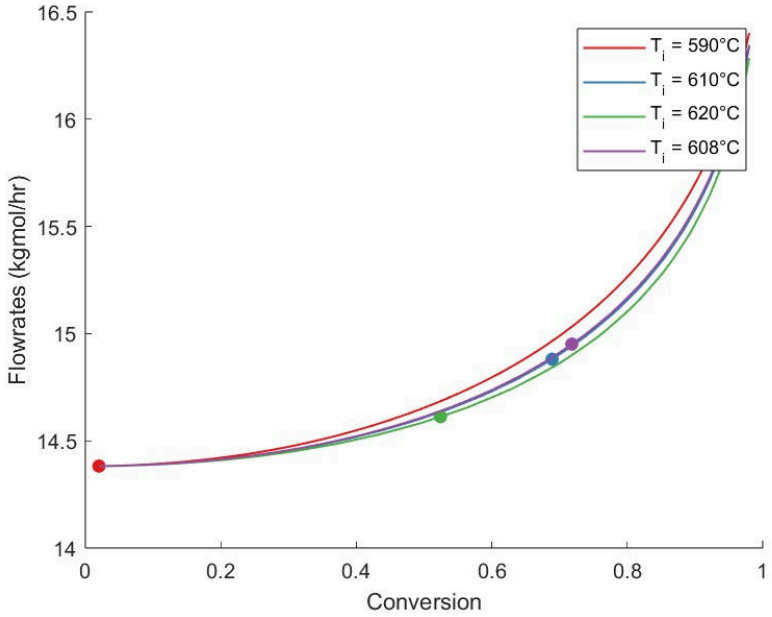
(5.1.2.b) Fresh feed of hydrogen versus reactor conversion for bounds of MR at 598.9°C and a y_{PH} of 0.42.



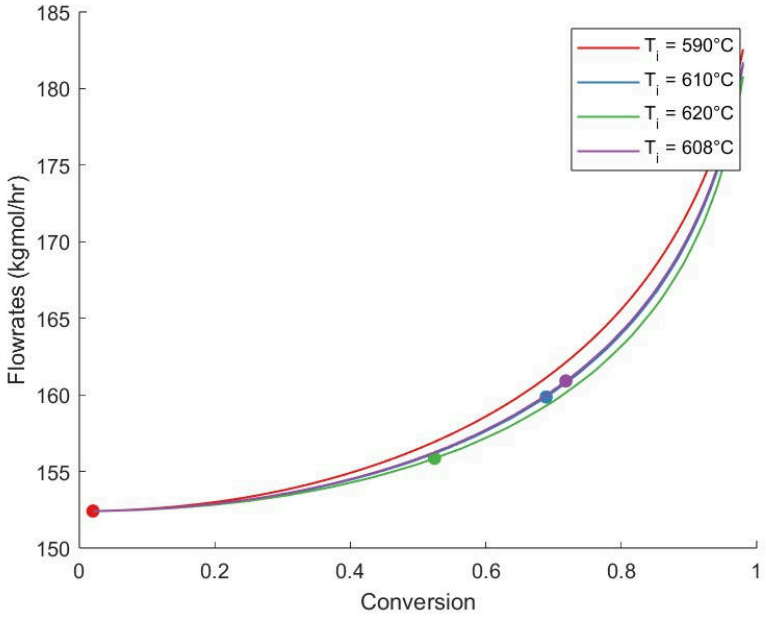
(5.1.2.c) Fresh feed of methane versus reactor conversion for bounds of MR at 598.9°C and a y_{PH} of 0.42.



(5.1.3.a) Fresh feed of toluene versus reactor conversion for bounds of T_i at an MR of 5.4 and a y_{PH} of 0.42.

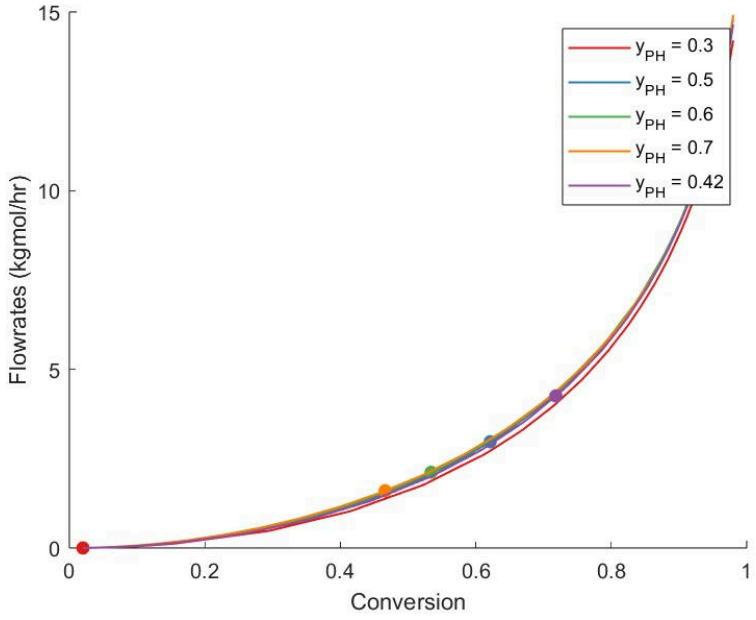


(5.1.3.b) fresh feed of hydrogen versus reactor conversion for bounds of T_i at an MR of 5.4 and a y_{PH} of 0.42.

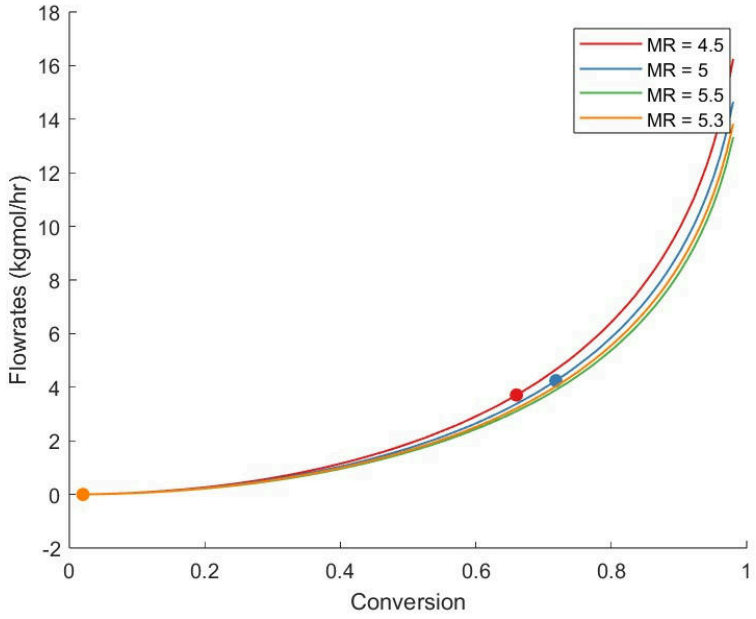


(5.1.3.c) Fresh feed of methane versus reactor conversion for bounds of T_i at an MR of 5.4 and a y_{PH} of 0.42.

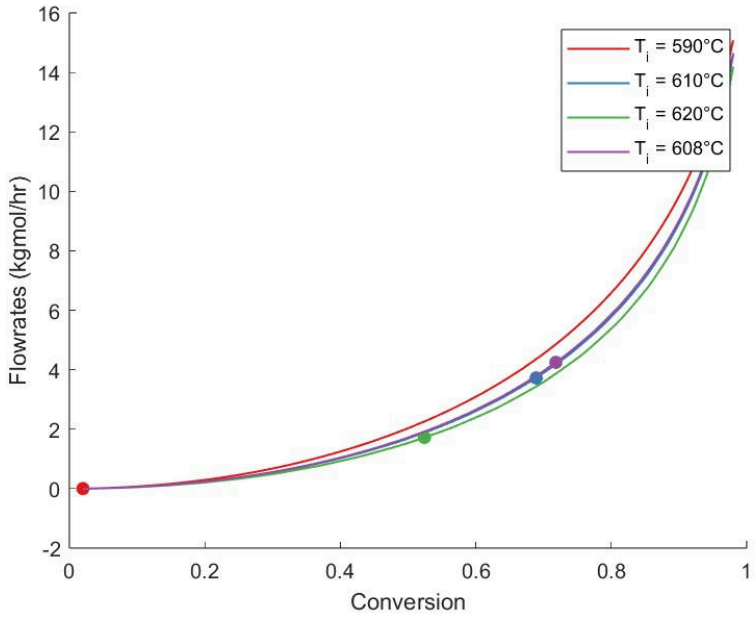
A.5.2: Diphenyl Production Rate



(5.2.1.a) Production rate of diphenyl versus reactor conversion for bounds of y_{PH} at 598.9°C and an MR of 5.4.

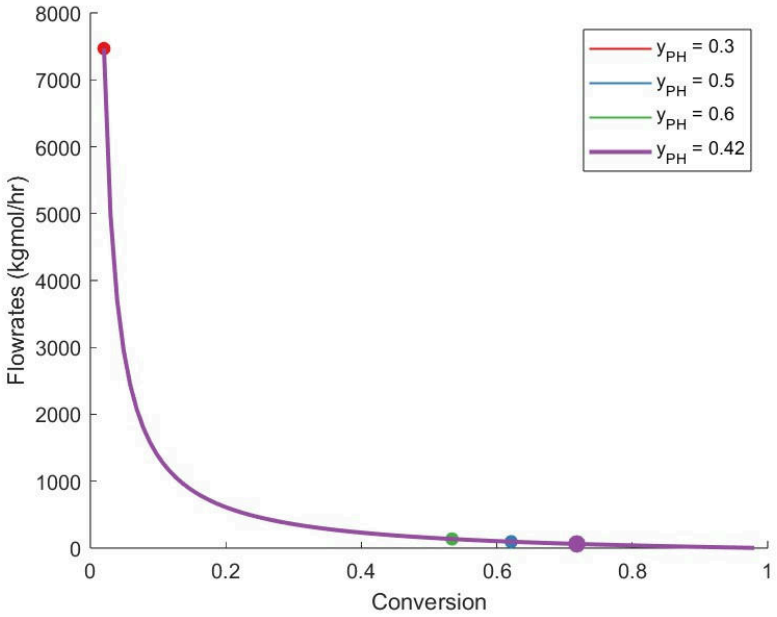


(5.2.1.b) Production rate of diphenyl versus reactor conversion for bounds of MR at 598.9°C and a y_{PH} of 0.42.

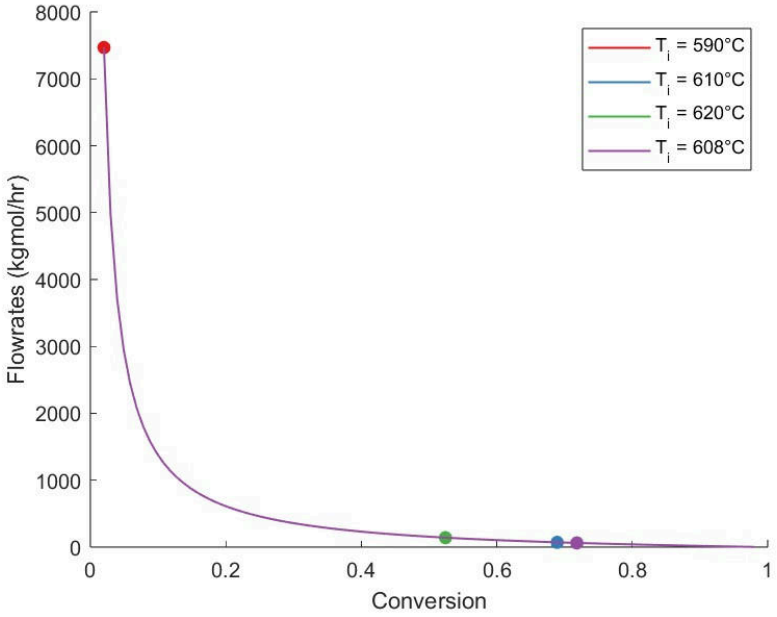


(5.2.1.c) Production rate of diphenyl versus reactor conversion for bounds of T_i at an MR of 5.4 and a y_{PH} of 0.42.

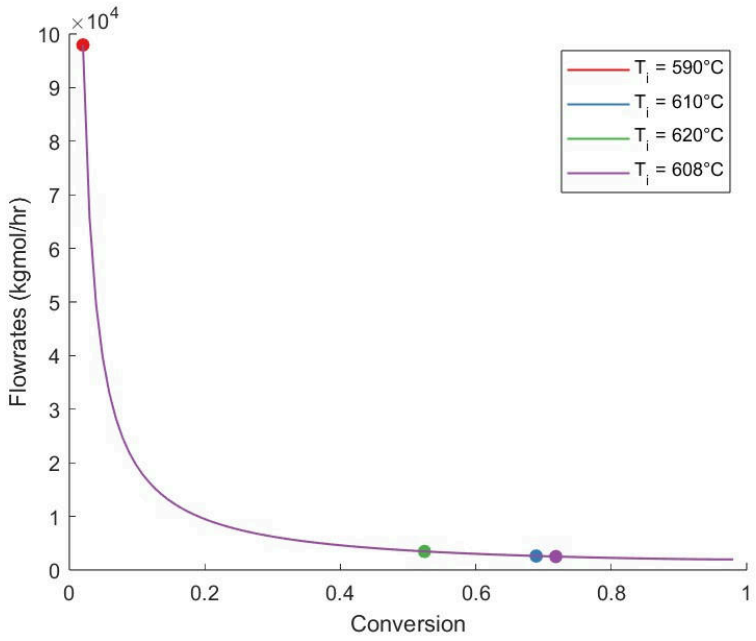
A.5.3: Recycle Flow



(5.3.1.a) Recycle flow rates of toluene versus reactor conversion for bounds of y_{PH} at 598.9°C and an MR of 5.4.

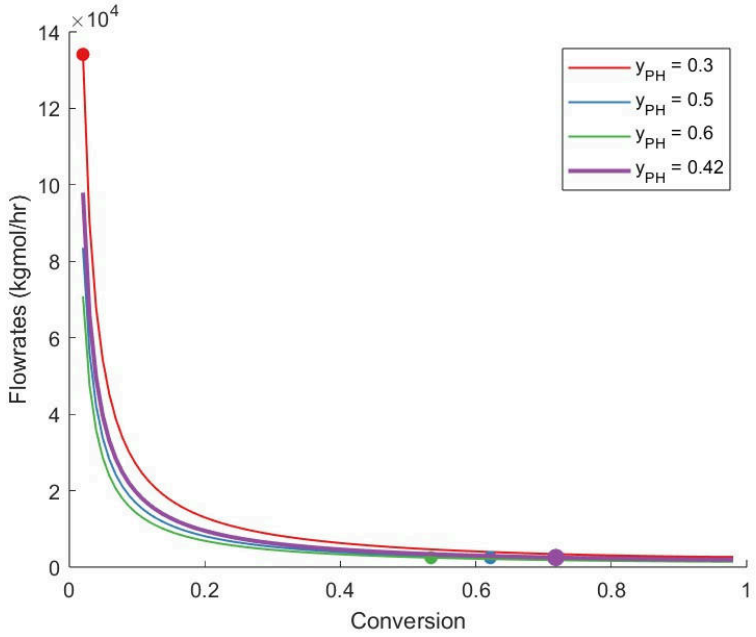


(5.3.1.b) Recycle flow rates of toluene versus reactor conversion for bounds of MR at 598.9°C and a y_{PH} of 0.42.

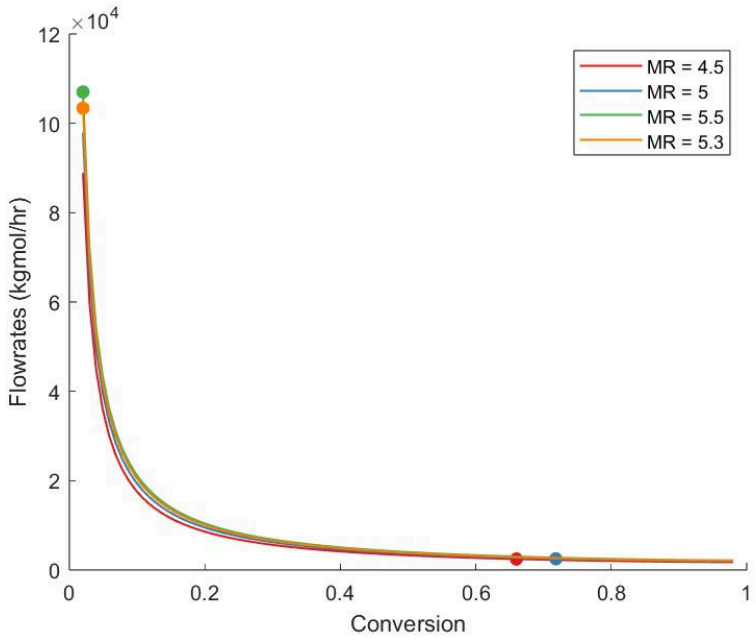


(5.3.1.c) Recycle flow rates of toluene versus reactor conversion for bounds of T_i at an MR of 5.3 and a y_{PH} of 0.42.

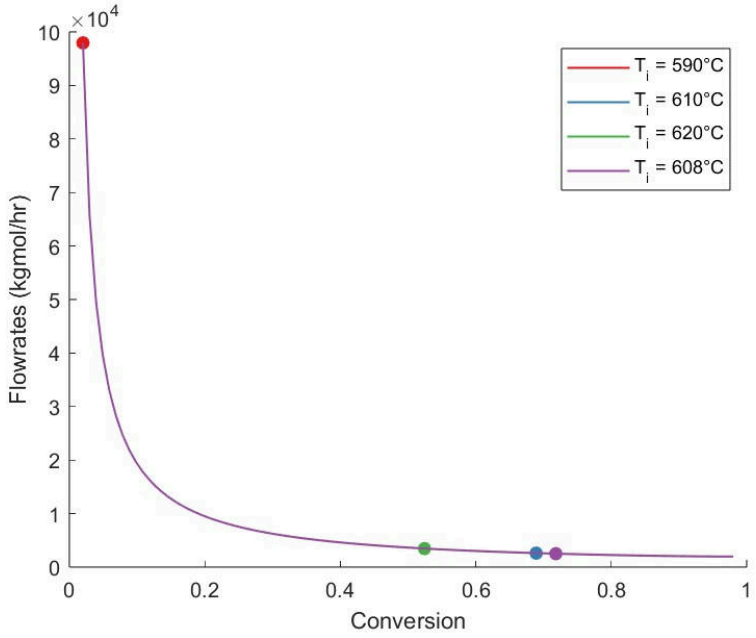
A.5.4: Reactor Feed



(5.4.1.a) Total flow rate into the reactor versus reactor conversion for bounds of y_{PH} at 598.9°C and an MR of 5.4.

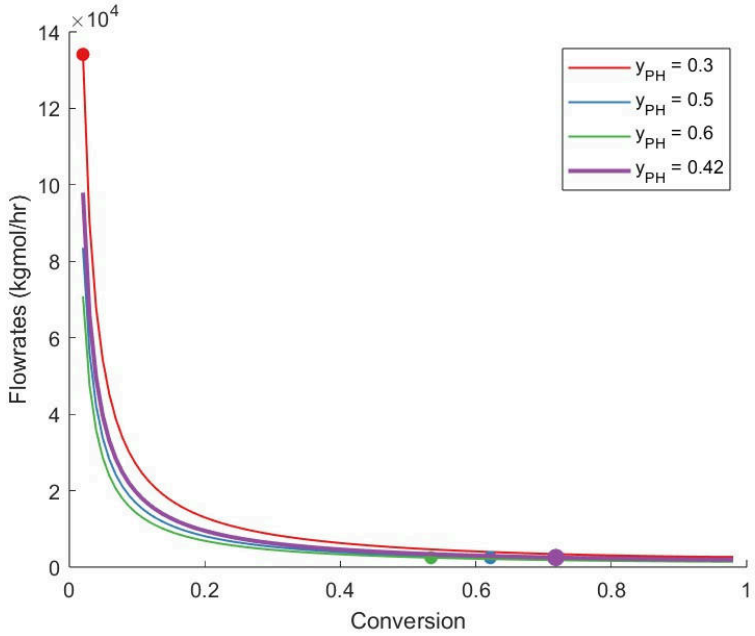


(5.4.1.b) Total flow rate into the reactor versus reactor conversion for bounds of MR at 598.9°C and a y_{PH} of 0.42.

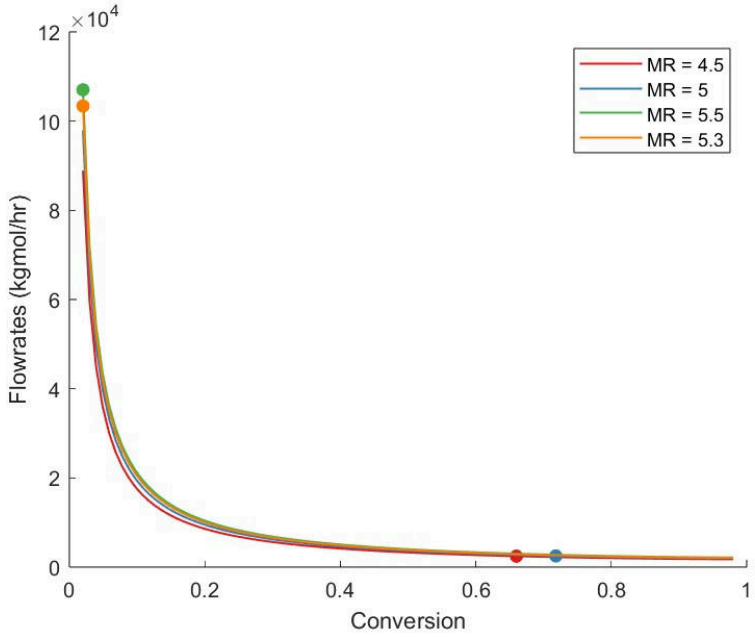


(5.4.1.c) Total flow rate into the reactor versus reactor conversion for bounds of T_i at an MR of 5.4 and a y_{PH} of 0.42.

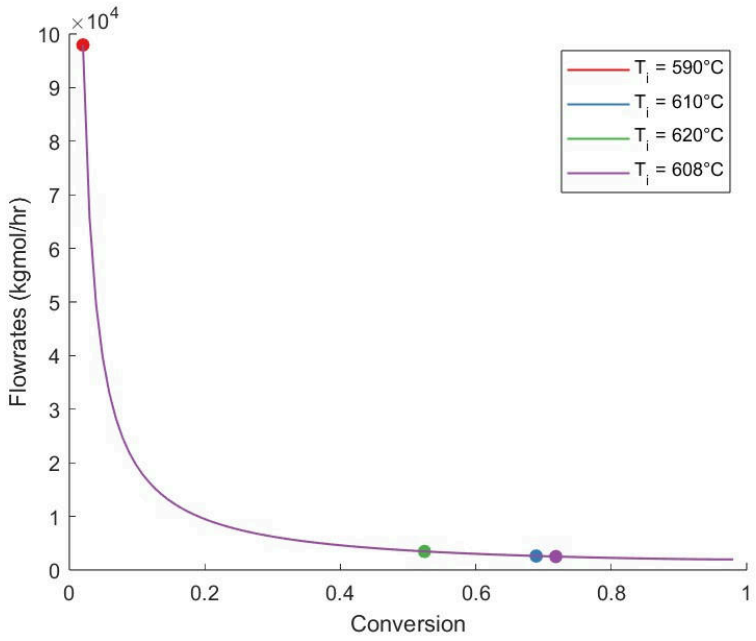
A.5.5: Separation Subsystem Feed



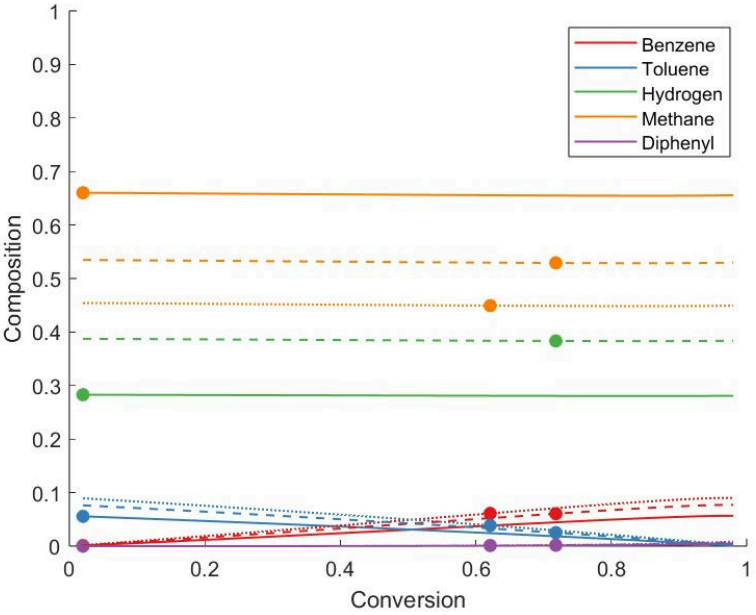
(5.5.1.a) Total flow rate into the separator versus reactor conversion for bounds of y_{PH} at 598.9°C and an MR of 5.4.



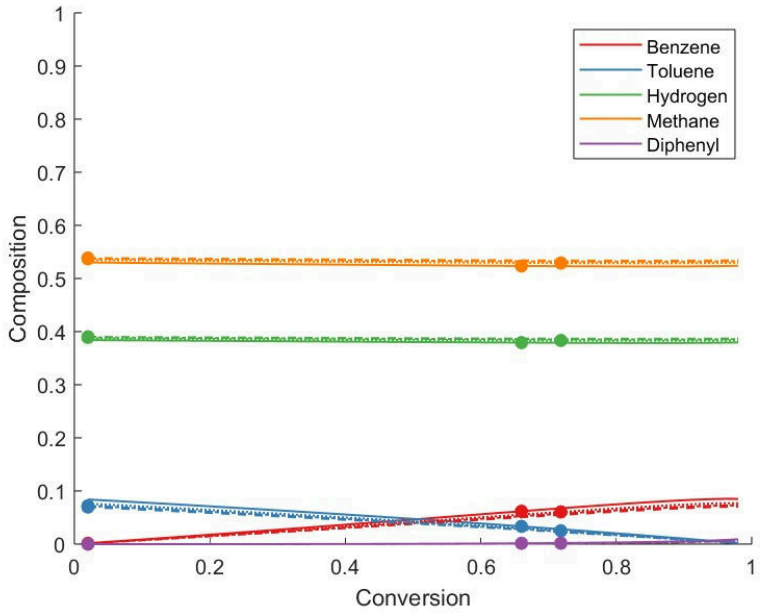
(5.5.1.b) Total flow rate into the separator versus reactor conversion for bounds of MR at 598.9°C and a y_{PH} of 0.42.



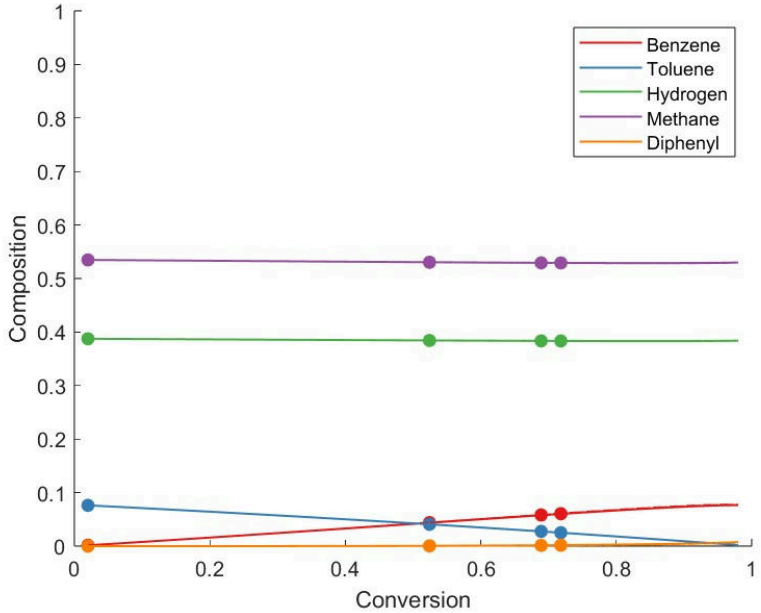
(5.5.1.c) Total flow rate into the separator versus reactor conversion for bounds of T_i at an MR of 5.4 and a y_{PH} of 0.42.



(5.5.2.a) Molar compositions of the compounds entering the separator versus reactor conversion for bounds of y_{PH} represented by the colored dashed lines at 598.9°C and an MR of 5.4.



(5.5.2.b) Molar compositions of the compounds entering the separator versus reactor conversion for bounds of MR represented by the colored dashed lines at 598.9°C and a y_{PH} of 0.42.



(5.5.2.c) Molar compositions of the compounds entering the separator versus reactor conversion for bounds of T_i represented by the colored dashed lines at an MR of 5.4 and a y_{PH} of 0.42.

Appendix B: Separation Design References

B.1: Key Values for a Flash Drum

In order to find the split ratio, vapor, and liquid compositions after our flash drum, the Rachford-Rice equations were used. This process, detailed in section 2.5 of Doherty & Malone's Conceptual Design of Distillation Processes[20], allows us to effectively separate the methane & hydrogen recycle stream from key component benzene and other hydrocarbons toluene and diphenyl. For the purposes of solving the mole balances of the HDA plant, we assumed a "perfect" flash where heavier components (benzene, toluene, and diphenyl) concentrate in the liquid while the light gases remain in the vapor (methane and hydrogen). This is a reasonable assumption for our system operating at 26 bar and 35°.

B.2: Key Values for a Distillation Column

In order to solve for the key variables of a distillation column we used a combination of Fenske-Underwood-Gilliand (abr. FUG) and the full set of Underwood's equations. While FUG is known to produce a optimal number of stages, we have found FUG to have poor convergence in the general case without a saturated feed. Thus, optimization and evaluation of this system was determined via Underwood's equations. Equations and additional details can be found on pages 127-136 of Doherty & Malone's Conceptual Design of Distillation Processes[20]. This includes number of stages, feed stage, reflux ratio, & reboil ratio, and can be extended to find column dimensions.

Appendix C: Economic Analysis

C.1: Assumptions and Constants

- There is a 3 year construction period for the plant.
- Assets will depreciate linearly over 10 years.
- The plant will have a life cycle of 15 years.
- All annual income and expenses are accounted at the end of the year.
- The total tax rate on profits is 27%.
- Administration and General Services (AGS) costs approximately 5% of annual revenue.
- The Chemical Engineering Plant Cost Index in 2025 for cost correlations is 800.
- The enterprise rate is 15% annually.
- The charge for carbon emissions is accounted as a cost and is not taxed.
- Following the life cycle of the plant, the salvage value of equipment is 5%.

C.2: Component Costs

The prices of toluene, methane, benzene, and hydrogen as commodity chemicals were calculated from current market prices. The prices of diphenyl, methane, hydrogen as fuels were calculated from the current price of energy and the heats of combustion of each component when burned.

Table C.6 : Process Components and their Costs

Components	Cost per Tonne	Energy Cost
Benzene (C_6H_6)	\$1000/MT	N/A
Toluene (C_7H_8)	\$300/MT	N/A
Hydrogen Gas Feed: $H_2 + CH_4$	\$1400/MT	N/A
Diphenyl ($C_{12}H_{10}$) (Fuel)	\$172/MT	\$4.25/GJ
Hydrogen Purge (H_2) (Fuel)	\$603/MT	\$4.25/GJ
Methane Purge (CH_4) (Fuel)	\$236/MT	\$4.24/GJ

C.3: Equations for Economic Analysis

C.3.1: Inside and Outside Battery Limits

The ISBL is the total installation cost for all required equipment for the chemical components of the plant, and IC is the installation cost for each separate unit of the plant.

$$\text{ISBL} = \text{I}C_{\text{heater}} + \text{I}C_{\text{reactor}} + \text{I}C_{\text{separation system}} + \text{I}C_{\text{compressor}} \tag{C.1}$$

The OSBL is the cost of installation for infrastructure outside of the chemical units of the plant, including loading facilities, administrative offices, storage tanks, power generation and waste treatment. The relation between the ISBL and the OSBL is defined as:

$$\text{OSBL} = 0.4\text{ISBL} \tag{C.2}$$

C.3.2: Direct and Indirect Costs

Direct costs refer to the sum of all expenses that can be tied to a product or specified aspect of the project, while indirect costs cover all expenses not tied to one. They are calculated using the formulas:

$$\text{Direct Costs} = \text{ISBL} + \text{OSBL} \tag{C.3}$$

$$\text{Indirect Costs} = 0.3(\text{ISBL} + \text{OSBL}) \tag{C.4}$$

C.3.3: Fixed Capital Investment

The fixed capital investment covers additional money as a possibility, which is approximately 25% for this project. The FCI can then be determined by the relation:

$$\text{FCI} = (1 + \text{Contingency}) \setminus (\text{Direct Costs} + \text{Indirect Costs}) \tag{C.5}$$

Inputting the relation involving ISBL produces

$$\text{FCI} = 2.28\text{ISBL} \tag{C.6}$$

C.3.4: Total Capital Investment

The total capital investment is calculated as

$$\text{TCI} = \text{FCI} + \text{WC} + \text{SU} \tag{C.7}$$

where working capital WC is costed as 2 months of feed, and the start-up cost SU is 10% of the FCI.

C.3.5: Revenue and Operating Cost

The revenue of the plant is defined as the profit earned from selling the streams produced. Prices used are taken from Table B.1. The revenue equation is then

$$R = \sum_{i=1}^{\text{components}} P_i \cdot c_i \tag{C.8}$$

where P_i is the product flow rate for each component and c_i is the component price. The plant operating cost is the total of the feedstock prices, the equipment operating costs within the ISBL, Administrative & General Services (AGS), and the CO_2 fee, summed as

$$\begin{aligned}\text{COM} = \text{OP}_{\text{heater}} + \text{OP}_{\text{reactor}} + \text{OP}_{\text{separation system}} + \text{OP}_{\text{compressor}} + \text{AGS} + C_{\text{C0}_2} + \\ \sum_{i=1}^{\text{components}} P_i * c_i\end{aligned} \tag{C.9}$$

C.3.6: Asset Depreciation

Using the assumption that company assets will linearly depreciate over a 10-year span, the annual depreciation can be calculated using the relation

$$d = \frac{\text{FCI - SU + Salvage}}{10} \tag{C.10}$$

where the salvage value is stated to be 5% of the fixed capital investment.

C.3.7: Cash Flows

Annual post-tax cash flows are determined by the equation

$$\mathrm{ATCF} = (R - \mathrm{COM} - d)(1 - t) + d \tag{C.11}$$

where t is equal to the sum of federal, state, and local tax rates and assumed to be 27%. The discounted cash flow is based on a discount rate of 15% and for each year i it is calculated as

$$\mathrm{DCF}_i = \frac{\mathrm{ATCF}}{(1+r)^i} \tag{C.12}$$

C.3.8: Net Present Value and Internal Rate of Return

The net present value, or NPV for the plant is determined from the sum of each discounted cash flow over the plant lifetime, as

$$\mathrm{NPV} = \sum_{i=0}^{\text{plant lifetime}} \mathrm{DCF}_i \tag{C.13}$$

The internal rate of return, or IRR is the discount rate r where NPV is 0 in the equation for DCF_i.

C.4: Capital and Operating Expenses Calculations

All equipment cost correlations are from Appendix E of Douglas' *Conceptual Design of Chemical Processes*.

C.4.1: Furnace Installed Cost

To determine the cost of the feed furnace, we calculated the heat load, as this furnace must increase the temperature of the incoming stream of toluene, methane and hydrogen to the temperature of the first heat exchanger in series T_R while maintaining an adiabatic reaction. The total heat duty is

$$\begin{aligned} Q = F_{C_7H_8}\left(\Delta H_{\mathrm{vap},C_7H_8} + C_{p,C_7H_8}\left(T_R - T_{\mathrm{in},C_7H_8}\right)\right) + \\ F_{\mathrm{CH_4},H_2}\left(C_{p,\mathrm{CH_4},H_2}\right) - \left(T_R - T_{\mathrm{in},\mathrm{CH_4},H_2}\right) \end{aligned} \tag{C.14}$$

where the heat of vaporization of toluene ΔH_{vap} = 33180 J/mol, the C_p of toluene and the combined feed of methane and hydrogen gas are 157.09 J/mol*K and 3.518 J/mol*K respectively. The heat duty is then converted to MM Btu / hour to calculate the installed cost of the heater $\mathrm{IC}_{\mathrm{Heater}}$ using the equation

$$\mathrm{IC}_{\mathrm{Heater}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right)(5.52{*}10^3)Q^{0.85(1.27+F_c)} \tag{C.15}$$

where F_c is a correction factor calculated from the design type, F_d the radiant tube material, F_m, and the design pressure, F_p.

F_d = Design Type: A-Frame Furnace

F_m = Radiant Tube Material: 321 Stainless Steel

F_p = Design Pressure

$F_c = F_d + F_m + F_p$

C.4.2: Reactor Installed Cost

The cost of the reactor, modeled as a pressure vessel was determined using the vessel dimensions with the assumption that it is a cylinder with a height 5 times greater than its diameter. The cost correlation for the installed cost of the reactor, $\mathrm{IC}_{\mathrm{reactor}}$ is

$$\mathrm{IC}_{\mathrm{reactor}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right) 101.9 D^{1.066} H^{0.82(2.18+F_c)} \tag{C.16}$$

Where F_c, the correction factor is

F_m = Shell Material: Stainless Steel

F_p = Design Pressure F_c = F_m *F_p

C.5: Distillation Tower Installed Cost

The overall installed cost for the distillation column includes the installed cost of the tower, modeled as a pressure vessel, the trays within the tower, and the condenser and reboiler. Although the pressure vessel cost is calculated using the reactor installed cost correlation, the height and diameter are defined by the number of stages and vapor flow rate through the tower rather than a fixed ratio.

Using the condenser and reboiler heat duties and the estimated heat transfer coefficient from Table 6.2 in Doherty & Malone[20], the total and areas of heat exchangers needed for each portion of the distillation column were calculated. The installed cost of all the heat exchangers was then used to determine the installed cost for the condenser and reboiler. The installed cost of the distillation tower trays is calculated from

$$\mathrm{IC}_{\mathrm{trays}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right) 4.7 D^{1.55} H F_c \tag{C.17}$$

Where F_c is a correction factor calculated from F_s, determined by the tray spacing F_m, determined by the tray material, and F_t, determined by the tray type.

C.5.1: Furnace and Reactor Operating Cost

The operating costs of the furnace and reactor are determined from the heat duty of the heater Q. Using the $4.25/GJ value for fuel, the total operating cost for the furnace and reactor in tandem is

$$\mathrm{OP}_{\mathrm{reactor}} + \mathrm{OP}_{\mathrm{heater}} = Q^* \text{ Fuel Price} \tag{C.18}$$

C.6: Compressor Installed Cost

The compressor installed cost is dependant on the braking horsepower of the compressor. The type of compressor used in this process is a centrifugal motor compressor, with a correction factor F_c of 1.0. The total installed cost for a compressor is

$$\mathrm{IC}_{\mathrm{compressor}} = \left(\frac{\mathrm{CEPCI}}{113.7}\right) (517.5)(\mathrm{bhp})^{0.82} F_c \tag{C.19}$$

C.6.1: Compressor Operating Cost

The compressor operating cost is calculated from its braking horsepower given from HYSYS. Under the assumption that compressors are powered via electricity, and the price of electricity is $0.08/kWh, the compressor operating cost is

$$\mathrm{OP}_{\text{compressor}} = \text{bhp} * \text{price of electricity} \tag{C.20}$$

C.7: Heat Exchanger Operating Cost

The operating cost for a heat exchanger where 2 process streams make contact with each other is zero. The operating cost of the heat exchanger is taken into account if a process stream is contacted by an external cold or hot stream. If the process stream makes contact with a hot stream, the operating cost of the heat exchanger is calculated by the heat duty production via natural gas combustion producing the equation

$$\mathrm{OP}_{\text{heat exchanger}} = \dot{Q} * (\text{fuel price}) \tag{C.21}$$

To determine the cooler costs, the price of the coolant is calculated as a function of the temperature of the coolant

$$\text{Cost } \left(\frac{\$}{\mathrm{GJ}}\right) = \varepsilon(0.047 * T^2 - 2.97T + 105.88) \tag{C.22}$$

where T is in °C and ε is the price of electricity in $/kWh. At temperatures between and 5 and 35°C, chilled water with a cost per GJ of $4.9 is used.

C.8: Distillation Operating Cost

The operating cost for a distillation column is the sum of the operating costs of the reboiler, acting as a heater, and the condenser, acting as a cooler. Cost correlations are as described in Appendix C.7.

C.8.1: Minimum Separation Power

The minimum separation power can be derived from the exit streams of each component entering the separation system, as defined in Table A1. Assuming temperature and pressure are constant, the minimum work, W_{min} to overcome ΔG of mixing is

$$\dot{\mathrm{W}}_{\text{min}} = \Sigma_{j=1}^{\text{exit streams}} \left(F_j \Sigma_{i=1}^{\text{components}} x_{\text{i,j}} \ln\left(\frac{x_{\text{i,j}}}{z_i}\right)\right) \tag{C.23}$$

C.8.2: Carbon Charge of Process

The carbon charge for this process is stated as $\$\frac{40}{\mathrm{MT}}$ and can be converted to a charge in terms of energy used

$$\frac{\$40}{\mathrm{MT}} * \frac{1\mathrm{MT}}{10^3 \text{ kg}} * \frac{44.01 \text{ kg}}{1 \text{ kgmol}} * \frac{1 \text{ kgmol } \mathrm{CO_2}}{1 \text{ kgmol } \mathrm{CH_4}} * \frac{1 \text{ kgmol } \mathrm{CH_4}}{1.125 \text{ GJ}} = \$\frac{1.56}{\mathrm{GJ}} \tag{C.24}$$

The carbon fee for this process can then be calculated as the sum of the energy used by the plant, as well as the cost of burning the diphenyl outlet as fuel.

C.9: Sizing Flash Drums

Flash drums were designed as pressure vessels with a diameter-to-height ratio of 1:4. The residence time of the vessel was set at 15 minutes, so the volume of the flash drum was calculated as

$$V = \dot{Q} * \tau \tag{C.25}$$

where V is the flash drum volume in m^3, $\dot{Q}$ is the volumetric flow rate entering the flash in $\frac{m^3}{s}$, and τ is the residence time in seconds.

C.10: Sizing Compressors

The braking horsepower (bhp) of the compressor was calculated from values of streams from HYSYS. Using equation 6.5-1 from Doherty & Malone's *Conceptual Design of Distillation Systems*[20],

$$\text{bhp} = \left(\frac{1}{0.8}\right)\frac{3.03 * 10^{-}5}{\gamma} P_{\text{in}} Q_{\text{in}} \left[\left(\frac{T_{\text{out}}}{T_{\text{in}}}\right) - 1\right] \tag{C.26}$$

where P_{in} and P_{out} are pressures in $\frac{\text{lbf}}{\text{ft}^2}$, Q_{in} is in $\frac{\text{ft}^3}{\text{min}}$, and the ratio of absolute temperatures and ratio of specific heat γ are

$$\frac{T_{\text{out}}}{T_{\text{in}}} = \left(\frac{P_{\text{out}}}{P_{\text{in}}}\right)^{\gamma} \qquad \gamma = \frac{R}{C_p} \tag{C.27}$$

where C_p is the heat capacity of the stream at constant pressure.

C.11: Sizing Heat Exchangers

From the necessary heat duties and the estimated heat transfer coefficients in Table 6.2 of Doherty & Malone[20], we can calculate the areas of the heat exchangers.

The heat transfer coefficients for coolers was determined via the required coolant temperature. We assumed the cold side of heat exchanger would be liquid water or high pressure steam.

The heat transfer coefficients for heaters was determined via the required hot stream temperature. Condensing steam was used for hot streams in these heat exchangers.

To remain within the sizing range of the correlations in Appendix D for the installed and operating costs of heaters and coolers within this system, the maximum heat exchanger size was set to be 2,300 m^3. If the heat exchanger was found to be larger than this, it was divided into a series of heat exchangers. Heat exchangers did not exceed a temperature difference of 120°C to ensure safe operation and remain within material constraints, unless the material provided (Grade 316/321 SS) was proven to adequately withstand extreme thermal environments.

C.12: Estimating Scrubber & Stabilizer Costs

To evaluate potential installation costs for the scrubber and stabilizer in our MATLAB simulation, the minimum work $\dot{W}_{\text{min}}$ was multiplied by the estimated ratio of the real work to the minimum work, γ and the capital cost correlation factor, $C_{\$/\text{W}}$ allowing us to account for the necessary costs of both process units.

$$\text{IC}_{\text{Sc/St}} = \dot{W}_{\text{min}} * \gamma * C_{\$/\text{W}} \tag{C.28}$$

C.13: Cash Flow Analysis

C.13.1: Discounted & Non-discounted Cash Flow Graphs

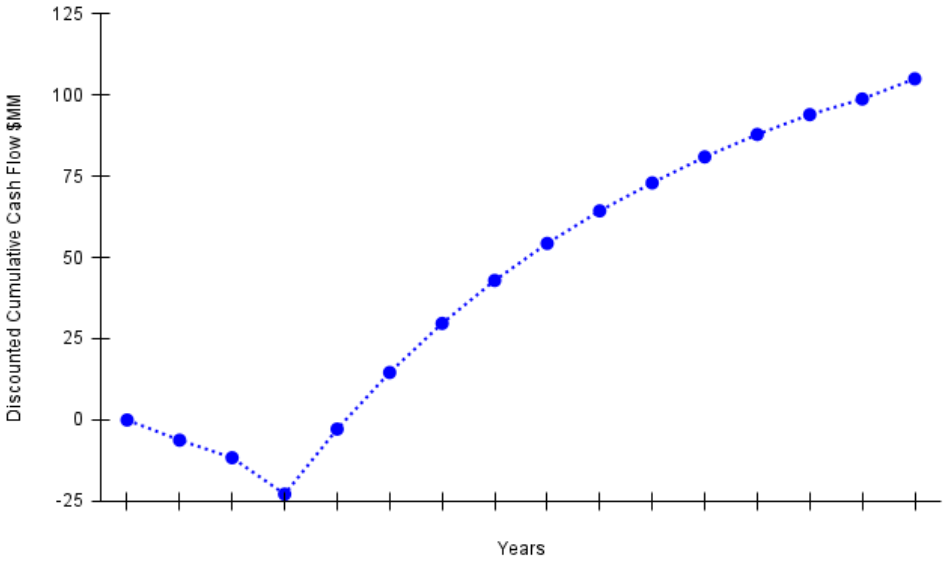


Figure B.1: The cumulative discounted cash flow for the plant over its construction and lifetime.

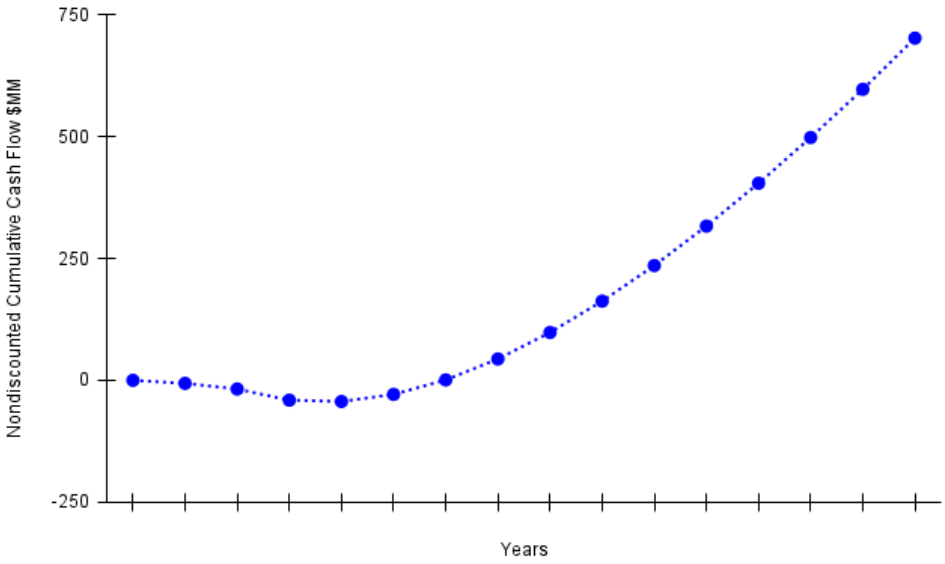


Figure B.2: The cumulative non-discounted cash flow for the plant over its construction and lifetime.

C.13.2: Discounted Cash Flow Spreadsheet

APL, Inc.
123 Lagoon Road
Santa Barbara, CA

ECONOMIC ANALYSIS

Plant Location: Freeport, TX
Case Description: Hydrodealkylization Plant for Benzene Production

Project Name: FEL-1 Toluene Hydrodealkylization Plant

Rev	Date	BY	APVD	Rev	Date	BY	APVD
V1.12	4/30/25	D9					

REVENUES AND PRODUCTION COSTS

	$MM/yr
Main product revenue	108.0
Byproduct revenue	0.0
Raw materials cost	48.7
Utilities cost	4.4
Consumables cost	0.0
CO_2 sustainability charge	2.4
VCOP	55.5
Salary and overheads	0.0
Maintenance	0.0
Interest	0.0
AGS	5.4
FCOP	5.4

CAPITAL COSTS

	$MM
ISBL Capital Cost	9.5
OSBL Capital Cost	3.8
Indirect Cost	4.0
Contingency	2.4
Total Fixed Capital Cost	21.7
Working Capital	8.1
Start-up Costs	1.7
Land	0.0
Total Capital Investment	31.5

CONSTRUCTION SCHEDULE

Year	% FC	% WC	% SU	% FCOP	% VCOP
0					
1	33%				
2	33%				
3	33%	100%	100%		
4				100%	100%
5				100%	100%
6+				100%	100%

ECONOMIC ASSUMPTIONS

On Stream	8400 hr/yr	Discount Rate	15%	Depreciation method	Straight-Line
	350 day/yr	Tax Rate	27%	Depreciation period	10 yrs
Project Life	15 yr	Salvage Value	5%		

CASH FLOW ANALYSIS

All figures in $MM unless indicated

Project Year	Cap. Ex.	Revenue	COM	Gr. Profit	Deprcn.	Taxable Inc	Taxes Paid	Cash Flow	PV of CF	NPV
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	7.1	0.0	0.0	0.0	0.0	0.0	0.0	-7.1	-6.2	-6.2
2	7.1	0.0	0.0	0.0	0.0	0.0	0.0	-7.1	-5.4	-11.6
3	17.0	0.0	0.0	0.0	0.0	0.0	0.0	-17.0	-11.2	-22.8
4	0.0	108.0	60.9	47.1	2.2	44.9	12.1	35.0	20.0	-2.8
5	0.0	108.0	60.9	47.1	2.2	44.9	12.1	35.0	17.4	14.6
6	0.0	108.0	60.9	47.1	2.2	44.9	12.1	35.0	15.1	29.7
7	0.0	108.0	60.9	47.1	2.2	44.9	12.1	35.0	13.2	42.9
8	0.0	108.0	60.9	47.1	2.2	44.9	12.1	35.0	11.4	54.3
9	0.0	108.0	60.9	47.1	2.2	44.9	12.1	35.0	9.9	64.3
10	0.0	108.0	60.9	47.1	2.2	44.9	12.1	35.0	8.6	72.9
11	0.0	108.0	60.9	47.1	10.1	37.0	10.0	37.1	8.0	80.9
12	0.0	108.0	60.9	47.1	10.1	37.0	10.0	37.1	6.9	87.8
13	0.0	108.0	60.9	47.1	10.1	37.0	10.0	37.1	6.0	93.9
14	0.0	108.0	60.9	47.1	0.0	47.1	12.7	34.4	4.9	98.7
15	-4.8	112.8	60.9	51.9	0.0	51.9	14.0	50.5	6.2	104.9

ECONOMIC ANALYSIS

NPV	15 years	104.9	$MM	IRR	70%
NPV at yr	12	87.8	$MM		

NOTES

1. All cash flows are assumed to occur at the end of the project year.
2.
3.

Appendix D: Conceptual MATLAB Design vs HYSYS Analysis

D.1: MATLAB Process Flow Diagram

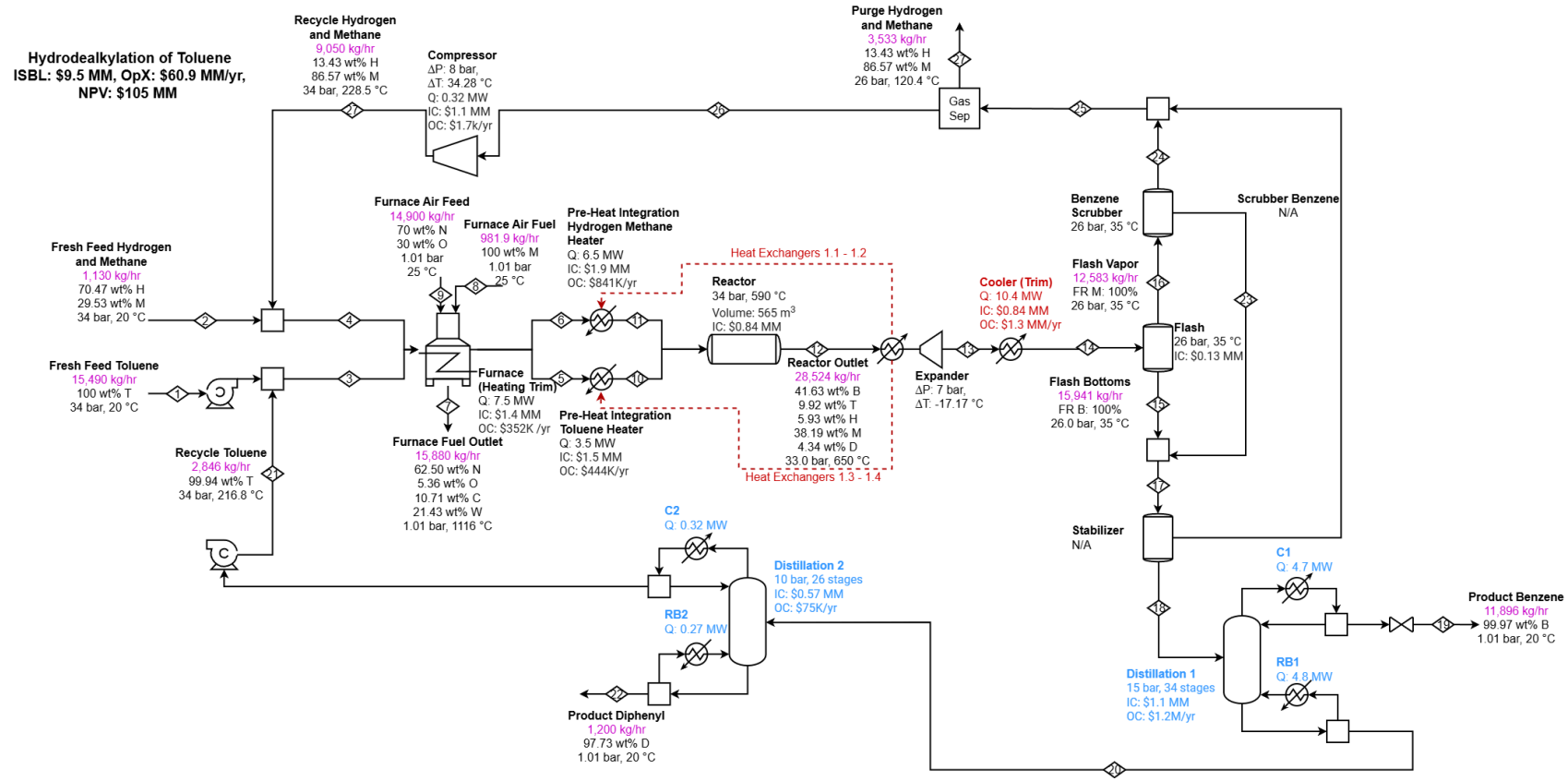


Figure D.14 : Schematic for MATLAB Design of HDA Process

D.2: MATLAB Stream and Equipment Tables

Table D.7 : Component streams as modeled on MATLAB under the assumption that the process operates on plant years (8400/yr)

Stream	Name	Temperature (°C)	Pressure (bar)	Mass Composition	Mass Flow Rate (kg/hr)	Mass Flow Rate (kta)
1	Toluene Feed	20.00	34.0	100% T	15,490	130
2	Gas Feed	20.00	34.0	70.5% H, 30.5% M	1,130	9.5
3	Toluene Mixer Outlet	58.77	34.0	100% T	18,400	155
4	Gas Mixer Outlet	134.6	34.0	80.24% M, 19.76% H	10,180	86
5	Toluene Furnace Outlet	310	34.0	100% T	18,400	155
6	Gas Furnace Outlet	310	34.0	80.24% M, 19.76% H	10,180	86
7	Furnace Flue Outlet	1116	1.01	62.98%, N 6.17% O, 16.96% C, 13.89% W	15,880	133
8	Furnace Fuel Gas	25.00	1.01	100% M	981.9	8.2
9	Furnace Air Feed	25.00	1.01	67.1% N, 32.9% O	14,900	125
10	Toluene HX Outlet	598.9	34.0	100% T	18,400	155
11	Gas HX Outlet	589.9	34.0	80.24% M, 19.76% H	10,180	86
12	Reactor Products	653.5	33.0	38.19% M, 5.93% H, 9.92% T, 41.63% B, 4.34% D	28,524	
13	Reactor Cooler System Outlet	356.2	26.0	38.19% M, 5.93% H, 9.92% T, 41.63% B, 4.34% D	28,524	239
14	Flash Drum Inlet	35.00	26.0	38.19% M, 5.93% H, 9.92% T, 41.63% B, 4.34% D	28,524	239
15	Flash Bottoms	35.00	26.0	17.76% T, 74.49% B, 7.75% D	15,900	134
16	Flash Vapor	35.00	26.0	86.57% M, 13.43% H	12,590	106

Stream	Name	Temperature (°C)	Pressure (bar)	Mass Composition	Mass Flow Rate (kg/hr)	Mass Flow Rate (kta)
17	Stabilizer Feed	n/a	n/a	n/a	n/a	n/a
18	Distillation Column A Feed	211.7	15.0	17.76% T, 74.49% B, 7.75% D	15,900	134
19	Benzene Outlet	20	1.01	0.024% T, 99.97% B, 0.0019% D	11,896	100
20	Distillation Column A Bottoms	247.7	15	69.63% T, 0.29% B, 30.08% D	4,114	36
21	Toluene Pump Inlet	216.8	10.0	99.94% T, 0.06% D	2,846	24
22	Diphenyl Outlet	20	1.01	2.27% T, 97.73% D	1,200	10
23	Scrubbed Volatiles	n/a	n/a	n/a	n/a	n/a
24	Methane & Hydrogen Recycle	35.00	26.0	86.57% M, 13.43% H	12,590	106
25	Purge Inlet	120.4	26.0	86.57% M, 13.43% H	12,590	106
26	Purge Recycle	120.4	26.0	86.57% M, 13.43% H	12,590	106
27	Purge Outlet	120.4	26.0	86.57% M, 13.43% H	3,533	30
28	Compressed Gas Recycle	228.5	34.0	86.57% M, 13.43% H	9,050	76

D.3: MATLAB vs HYSYS Comparison and Analysis

MATLAB and HYSYS mass and energy flows for this system differ due to thermodynamic assumptions made in the MATLAB simulation. MATLAB treats all gasses as ideal and assumes that the heats of reaction are constant, while HYSYS uses the UNIQUAC model to determine complex relationships for these variables. Due to these discrepancies, the HYSYS simulation results should produce more accurate calculations.

Variable	HYSYS	MATLAB
Reactor Inlet Temperature (°C)	598.9	598.9
Reactor Outlet Temperature (°C)	654.3	648
Pressure (bar)	34	34
Hydrogen-to-Toluene Molar Ratio	5.4	5
Reactor Volume (m^3)	565	565
Conversion	82.2	84.4
Selectivity of Benzene	93	90
Stream 1 (kg/hr)	15,110	15,400
Stream 2 (kg/hr)	907.6	1,130
Streams 3,5,10 (kg/hr)	18,070	18,400
Stream 4,6,11 (kg/hr)	15,610	10,180
Streams 12,13,14 (kg/hr)	33,700	28,524
Stream 15 (kg/hr)	15,020	15,900
Stream 16 (kg/hr)	18,680	12,590
Stream 18 (kg/hr)	15,800	15,900
Stream 19 (kg/hr)	11,960	11,896
Stream 20 (kg/hr)	3,842	4,114
Stream 21 (kg/hr)	3,041	2,846
Stream 22 (kg/hr)	802.3	1,200
Stream 23 (kg/hr)	969.9	N/A
Stream 24 (kg/hr)	17,720	12,590
Stream 25 (kg/hr)	17,920	12,590
Stream 26 (kg/hr)	14,740	12,590
Stream 27 (kg/hr)	3,172	3,533
Stream 28 (kg/hr)	14,740	9,050

Table D.9 : Energy streams as modeled by HYSYS

Stream	Unit Name	Energy Flow (MW)
E1	Furnace	7.5
E2	HX 1.1-1.2	6.5
E3	HX 1.3-1.4	3.5
E4	Trim Cooler	10.4
E5	Compressor	0.32
E6	Distillation Column A	9.5
E7	Distillation Column B	0.59

Table D.10 : Equipment units as modeled by HYSYS. The mixers and expanders are not included as they are beyond the scope of this report.

Name	Unit Operation	Material	Design Parameters	Capital Cost ($MM)
FR-1	Furnace	Grade 321 Stainless Steel	Heats reactants for Heat Exchanger Set 1 to inlet temperature of 310°C	1.4
PFR-100	Plug-Flow Reactor	Grade 321 Stainless Steel	Operates adiabatically at 598.9°C and 33 bar, sized at 565 m^3	0.84
E-103	Heat Exchanger	Grade 316 Stainless Steel	Heats the reactor inlet and cools the reactor outlet via countercurrent exchange. Minimizes utility energy needed for the furnace. The heat transfer coefficient is 239 $\frac{\mathrm{W}}{m^2 \cdot K}$ and surface area is 54.34 m^2	0.87

Name	Unit Operation	Material	Design Parameters	Capital Cost ($MM)
E-104	Heat Exchanger	Grade 316 Stainless Steel	Heats the reactor inlet and cools the reactor outlet via countercurrent exchange. Minimizes utility energy needed for the furnace. The heat transfer coefficient is 147 $\frac{W}{m^2 \cdot K}$ and surface area is 718.3 m^2	1.2
HX 1	Heat Exchanger	Grade 316 Stainless Steel	Heats the reactor inlet and cools the reactor outlet via countercurrent exchange. Minimizes utility energy needed for the furnace. The heat transfer coefficient is 239 $\frac{W}{m^2 \cdot K}$ and surface area is 60.2 m^2	0.62
HX 2	Heat Exchanger	Grade 316 Stainless Steel	Heats the reactor inlet and cools the reactor outlet via countercurrent exchange. Minimizes utility energy needed for the furnace. The heat transfer coefficient is 147 $\frac{W}{m^2 \cdot K}$ and surface area is 167 m^2	0.69
E-110	Heat Exchanger	Grade 316 Stainless Steel	Cools the reactor outlet via countercurrent exchange with condensate steam. The heat transfer coefficient is 480 $\frac{W}{m^2 \cdot K}$ and surface area is 289.6 m^2	0.29
E-105	Heat Exchanger	Grade 316 Stainless Steel	Cools the reactor outlet via countercurrent exchange with the purge feed. The heat transfer coefficient is 133 $\frac{W}{m^2 \cdot K}$ and surface area is 862 m^2	0.28
E-100	Heat Exchanger	Grade 316 Stainless Steel	Cools the reactor outlet via countercurrent exchange with chilled water. The heat transfer coefficient is 749 $\frac{W}{m^2 \cdot K}$ and surface area is 236 m^2	0.27

Name	Unit Operation	Material	Design Parameters	Capital Cost ($MM)
Flash	Flash Drum	Grade 321 Stainless Steel	Flash separates H/M from the remaining hydrocarbons at 35°C. Has a volume of 9.9 m^3	0.13
T-100	Distillation Column A	Grade 321 Stainless Steel	Separates benzene from diphenyl and toluene using 17 stages of sieve trays with spacing of 24 in. The reflux ratio is 3.5 and reboil ratio is 16.0. Operates at 15 bar. Has an area of 18.2 m^2 and a height of 12.2 m.	1.1
T-101	Distillation Column B	Grade 321 Stainless Steel	Separates toluene from diphenyl using 13 stages of sieve trays with spacing of 24 in. The reflux ratio is 0.234 and reboil ratio is 3.55. Operates at 10 bar. Has an area of 14.7 m^2 and a height of 9.8 m.	0.57
K-100	Compressor	N/A	Centrifugal motor compressor. Compresses recycled gas products to 34 bar before entering FR-100. Braking horsepower of 4.6 hp.	1.1

Table D.11 : MATLAB vs. HYSYS Comparison: Number of Stages, Reflux & Reboiler Ratios, Feed Stage

Parameter	T-100 HYSYS	T-100 MATLAB	T-101 HYSYS	T-101 MATLAB
Number of Stages	17	18	13	13
Reflux Ratio	3.5	1.06	0.234	0.30
Reboiler Ratio	16	7.97	3.55	5.03
Feed Stage	9	9	3	3

D.4: Heating Composite Curve

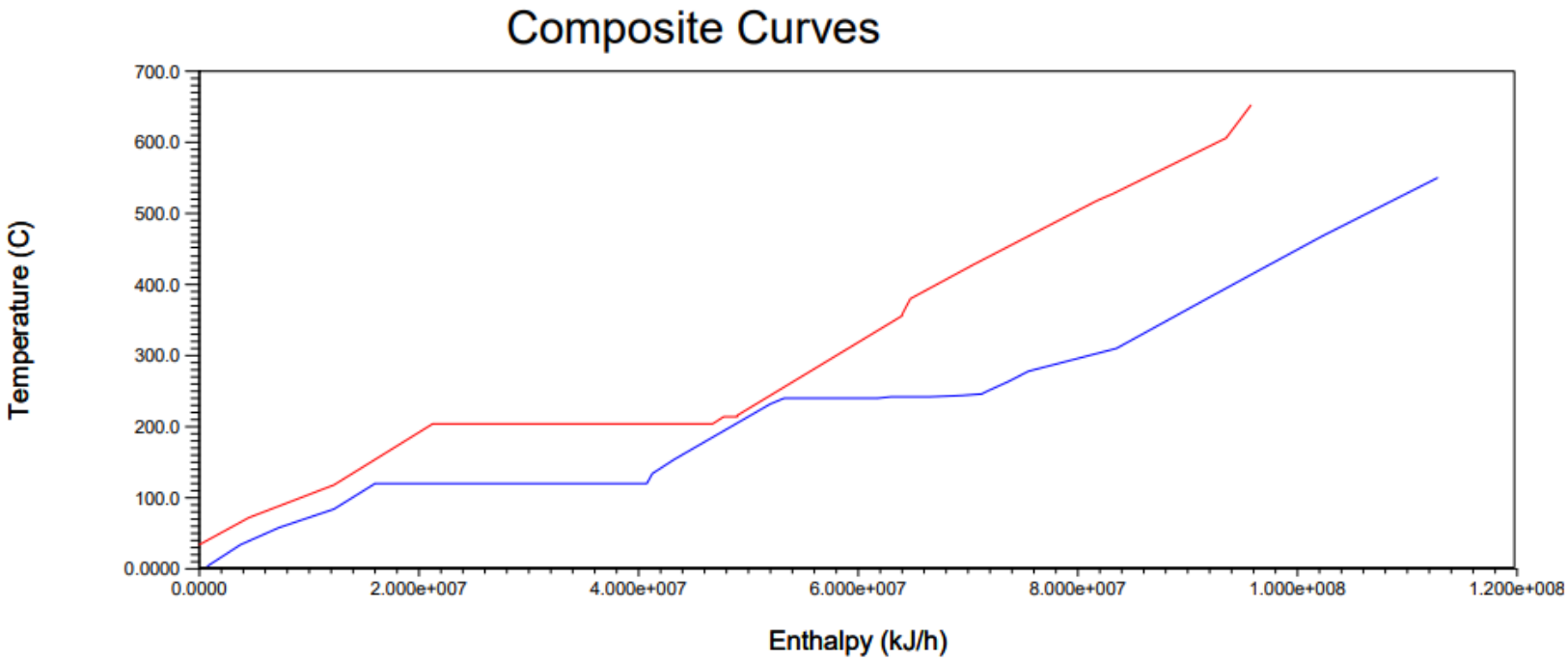


Figure D.15 : Temperature (C) vs. Enthalpy (kJ/h) Graph for Heat Exchanger Network

Appendix E: HAZOP Analysis

Table E.1: Summary of safety concerns and recommendations the address these concerns for specific units within our process

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
Furnace	Fuel Flow	More	Valve fully open Incorrectly calibrated flow meter and control system for incoming flow	The furnace may overheat and over pressurize, causing an explosion	Temperature control system with a fail-close fuel feed valve, and an emergency venting and cooling system
	Fuel Flow	Less	Valve closed Blockage in pipes feeding fuel to furnace Leakage in the inlet flow	The furnace may not heat the reactants enough to drive the reaction May impact worker's health	Temperature control system with alarm and control system for reactant valve to control the fuel flow
	Air Flow	More	Issue with air intake location Incorrect air-to-fuel ratio settings	Excess air may increase availability of oxygen for combustion leading to more heat generated Increase the availability of oxygen for explosions	Air-to-fuel ratio monitoring Air feed concentration sensor that controls fuel feed valve
	Air Flow	Less	Blocked air intake	May lead to incomplete combustion, carbon monoxide formation, and risk of explosion due to	Air-to-fuel ratio monitoring Air feed concentration sensor that controls fuel feed valve

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
				unburned fuel accumulation	
Furnace	Reactant Flow	More	Flow control valve failure Incorrectly calibrated flow meter	Reduced residence time in furnace leading to incomplete heating of reactants May stop reaction from occurring	Flow monitoring system with an automated control valve and high flow alarms
	Reactant Flow	Less	Valve close Leakage in inlet stream Pipeline blockage	Reactants remain in the furnace longer, leading to potential overheating and coking of reactants With little to no reactants the furnace may overheat and, which can lead to an explosion	Control system for reactant feed valve, emergency shutdown for overheating, pressure monitoring to detect blockages
	Temperature	More	Reduced reactant feed Increase air-to-fuel ratio Increased flow rate of natural gas Blockage of inlet or exit stream of reactant	Increase the likelihood of explosion Coking formation may occur on the walls of the furnace	Temperature control system with alarms Emergency cooling system Fail-close valve on feed or gas
	Temperature	Less	Increased reactant feed	Reactants do not reach the necessary	Temperature control system with alarm

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
			Decreased air-to-fuel ratio Decreased flow rate of natural gas	temperature for reaction to occur, leading to no product formation	Redirect reactant feed stream to a storage Schedule maintenance for cleaning any coke formation
Furnace	Pressure	More	Increased reactant feed Increased air-to-fuel ratio Increased flow rate of natural gas Blockage of inlet or exit stream of reactant	Increase the likelihood of an explosion or structural damage	Emergency pressure relief system with fail-open valve on piped connected to furnace Pressure sensors with automatic trip mechanisms
	Pressure	Less	Decreased reactant feed Decreased air-to-fuel ratio Decreased flow rate of natural gas	Reduces the temperature and can lead to inadequate conditions for reaction to occur	Pressure monitoring with pressure control valves
Mixer	Pressure	More	Blockage in reactant exit stream Excess recycle flow Faulty pressure regulator	Potential overpressure leading to pipe rupture	Pressure relief valves High pressure alarms Real time monitoring Fail-open valve out of mixer Fail-close valve going into mixer

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
Mixer	Pressure	Less	Blockage in reactant inlet stream Leak in pipeline Insufficient fresh feed	Insufficient flow to furnace which can lead to an explosion	Low pressure alarms Backup pressure control system
	Flow Rate	More	Completely open valve Excess recycle stream	Overloading the furnace and possibly leading to no reaction occurring	Flow meters with alarms Fail-close valve
	Flow Rate	Less	Closed valve	Temperature may rise too high in the furnace and cause and explosion	Flow meters with alarms
	Speed	More	Mixer motor over speeding Faulty speed controller Incorrect setpoint	Excessive mixing can lead to damages to the mechanical parts of the mixer	Speed control system with alarms
	Speed	Less	Mixer motor under speeding Faulty speed controller Incorrect setpoint	Poor mixing Non-homogenous mixture fed to the furnace	Speed control system with alarms Back up motor Routine maintenance
PFR	Temperature	More	Excess flow of heated reactants Highly exothermic reaction	Reaction can become a runway reaction and cause and explosion Can damage the equipment is the	Temperature monitoring Fail-close valve for the reactant feed

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
				reaction surpassed 650 °C	
	Temperature	Less	Low amount of reactants Insufficient feed temperature from furnace	No reaction if temperature is below 550 °C	Temperature monitoring with alarm to notify personnel if temperature decreases past a certain setpoint
	Pressure	More	Increased reactant flow rate into the reactor Blockage in outlet flow of the reactor	May cause the vessel to over pressurize and rupture An increase in partial pressure of the species can lead to an increase in reaction rate outside of the modeled rate	Pressure relief on the reactor High pressure alarms
	Pressure	Less	Decreased reactant flow rate into the reactor Blockage in inlet flow of the reactor	Decrease in partial pressure if species may result in a lower reaction rate than modeled	Pressure monitoring system System control for feed coming into the reactor
Flash	Temperature	More	Overheating of inlet stream Faulty temperature controller	Poor vapor-liquid separation Can cause pressure build up inside the column	Temperature control system with alarms

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Temperature	Less	Inlet flow is insufficiently heated	Incomplete vaporization leading to off-spec products	Temperature control system with alarms
	Pressure	More	Blockage at outlet High feed flow rate	Pressure build up may cause the vessel to explode	Pressure relief valves
	Pressure	Less	Blockage at inlet Leak in the column Decreased flow into the column	Poor separation die to lower vapor pressure	Automatic pressure control valve Continuous pressure monitoring system
	Flow Rate	More	Valve fully open Incorrect flow meter calibration	May lead to flooding and poor separation	Fail-close valve at feed line Temperature control system with alarms
	Flow Rate	Less	Valved closed Blockage in feed line	May lead to temperature rise in column	Temperature control system with alarms Continuous feed monitoring
Distillation Column	Temperature	More	Incoming feed is excessively hot Incorrect temperature controller	May lead to decomposition of components Increased pressure due to excess vapor	Temperature control system with alarms
	Temperature	Less	Fouling in reboiler	Incorrect separation of components Bottoms may be off spec	Temperature control system with alarms Routine reboiler cleaning

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Pressure	More	Blockage in exit streams High vapor flow rate	Pressure build up may lead to column rupture	High pressure relief valve High pressure alarm system
	Pressure	Less	Leak in column	May cause poor separation due to disruption in internal flow patterns	Pressure monitoring system with alarms
	Flow Rate	More	Feed valve fully open	Overloading of distillation trays Poor separation Weeing may occur if too much feed enters the column	Flow control system between the flash and distillation column Fail-close valve on distillation feed
	Flow Rate	Less	Blockage in feed valve	Insufficient feed into column may lead to unstable operating conditions	Continuous feed flow monitoring
Compressor	Temperature	More	Inadequate cooling system Recycle stream at an excessively high temperature	Damage to compressor internal mechanism	Temperature control system with alarms and emergency shutdown
	Temperature	Less	Recycle stream at an excessively low temperature	Can lead to condensation of gases at extremely low temperatures and cause corrosion to compressor	Temperature control system with alarms

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Pressure	More	Malfunctioning control valve Operating above setpoint	Over pressurized vessel can lead to a rupture or explosion	Pressure relief valve High pressure alarms Automatic shut down if pressure exceeds a certain value
	Pressure	Less	Compressor wear Incorrect setpoint	Insufficient compression Reduced recycle flow	Pressure monitoring Routine maintenance to ensure compressor has none to minimal wear
Reboiler	Temperature	More	Steam valve fully open Blockage at reboiler outlet	Overheating of liquid May lead to pressure build up in the column	Temperature control system with alarms Fail-close steam supply valve
	Temperature	Less	Low steam pressure used Heat exchanger fouling	Incomplete vapor generation Poor separation	Temperature control system with alarms Regular cleaning of reboiler tubes
	Pressure	More	Blocked vent line	Overpressurization of reboiler increases the risk of equipment rupture	Pressure relief valve Pressure monitoring system with alarms
	Pressure	Less	Leak in reboiler shell	Inefficient heat exchanger	Pressure monitoring system with alarms
Condenser	Temperature	More	Cooling water valve closed	Incomplete condensation of vapor	Temperature control system Fail-open valve for cooling water

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
	Temperature	Less	Cooling water valve fully open	Product may cool past condensation state	Temperature control system with alarms
	Pressure	More	Vapor overload Condenser tube blockage	Overpressurized overhead system	Pressure relief system with alarms
	Pressure	Less	Low vapor load Leak in condenser	May affect condensation rate	Pressure monitoring system with alarms
Condensate Steam Heat Exchanger	Temperature	More	Steam flow valve fully open Fouling on cold side of tube and shell exchanger Faulty temperature control system	Overpressurization may occur and lead to equipment rupture	Temperature control system with alarms
	Temperature	Less	Steam pressure drop Valve closed	Inadequate heat exchange Poor energy recovery	Temperature control system with alarms
	Pressure	More	Blocked pipes Valve malfunction	Equipment rupture Hazard from possible steam leak	Pressure relief valve Regular inspection
	Pressure	Less	Insufficient steam supply Lower pressure used	Poor heat transfer	Pressure monitoring system
	Steam Flow Rate	More	Steam valve fully open	Overheating of process stream	Fail-close valve Flow rate monitor
	Steam Flow Rate	Less	Valve closed Supply line leak	Insufficient heating	Steam flow monitors with alarms

Item	Parameter	Deviations	Possible Cause	Consequences	Safeguards and Recommendations
Heat Exchangers	Temperature	More	Excess of hot stream	Inadequate heat transfer	Temperature system with alarms
	Temperature	Less	Excess of cold stream	Incomplete heating	Temperature monitoring with alarms
	Pressure	More	Blocked pipes Valve malfunction	Equipment rupture Hazard from possible leak	Pressure relief valve Regular inspection
	Pressure	Less	Insufficient supply	Poor heat transfer	Pressure monitoring system
	Flow Rate	More	Valve fully open	Inadequate heat transfer	Flow monitoring with alarms Fail-close valve
	Flow Rate	Less	Valve fully closed Pipe blockage	Overheating of process stream	Flow sensors with alarms

Table E.2: Summary of safety concerns for chemicals of interest and how to safely store and handle each chemical. Safety data regarding reactant chemicals,product, co-product, and by-product is available here.

Species	Flash Point [°C]	Auto-Ignition Temperature [°C]	Fire Rating	Explosive Limits [vol%]	Safety Concerns	Storage and Handling
Toluene $[C_7H_8]$[21]	4	480	3	1.1-7.1	Limit of exposure of 20 ppm over an eight-hour shift. Flammable in the presence of open flames, sparks, smoke, electrostatic charges, and contact with strong oxidants.	Store in a well-closed, fireproof area away from strong oxidants. Store only in original container and away from areas without drain or sewer access. PPE for handling toluene includes protective gloves and safety goggles.
Hydrogen $[H_2]$[22]	< −150	560	4	4-75	Flammables in the presence of open flames, sparks, and smoke. High concentrations of expanding gas can cause a deficiency of oxygen, leading to asphyxiation.	Store separately from oxidizing materials in a cool, dry, and fireproof area. Ensure there is ventilation along the floor and ceiling to prevent risk of suffocation. PPE

Species	Flash Point [°C]	Auto-Ignition Temperature [°C]	Fire Rating	Explosive Limits [vol%]	Safety Concerns	Storage and Handling
						for handling hydrogen includes cold-insulating gloves and face shields.
Methane $[CH_4]$[23]	−183	537	4	5-15	Flammable in the presence of open flames, sparks, and smoke. High concentrations of expanding gas can cause frostbite or cause a deficiency of oxygen, leading to asphyxiation.	Store in a cool and fireproof area. Ensure there is ventilation along the floor and ceiling to prevent the risk of suffocation. PPE for handling methane includes cold-insulating gloves and safety goggles.
Benzene $[C_6H_6]$[24]	−11	498	3	1.2-8	Limit of exposure of 0.5 ppm over an eight-hour shift and a limit of 2.5 ppm over a short range. Fla mmable in the presence of open flames, sparks, and smoke.	Store in a fireproof area away from feedstuffs, oxidants, and halogens. Store in an area without drain or sewer access. PPE for benzene includes

Species	Flash Point [°C]	Auto-Ignition Temperature [°C]	Fire Rating	Explosive Limits [vol%]	Safety Concerns	Storage and Handling
					Benzene is confirmed human carcinogen. Long-term or repeated exposure may affect the central nervous system and immune system	protective clothing, and a face shield with breathing protection.
Diphenyl $[C_{12}H_{10}]^{25}$	113	540	1	0.6-5.8	Limit of exposure of 0.2 ppm over an eight-hour shift. Combustible in the presence of open flames. Long-term or repeated exposure may affect the liver and nervous system and result in impaired functions.	Store in a fireproof area away from feedstuffs and oxidants. Store in an area without drain or sewer access. PPE for handling diphenyl includes protective gloves and safety goggles with breathing protection.

Table E.14 : Material properties of chemicals under normal conditions of 25°C and 1 atm.

Species	Molecular Weight [g/mol]	Boiling Point [°C]	Melting Point [°C]	Density [kg/m^3]
Toluene [$C_6H_5CH_3$]	92.1	111	−95	870
Hydrogen [H_2]	2.0	−253	−259	0.0813
Methane [CH_4]	16.0	−161	−183	0.648
Benzene [C_6H_6]	78.1	80	6	880
Diphenyl [$C_{12}H_{10}$]	154.2	256	70	1040

Table E.15 : Species compatibility with possible plant material

Species	Carbon Steel[26]	Chrome/Moly	Stainless[27]
Toluene [$C_6H_5CH_3$]	Excellent	Excellent	Excellent
Hydrogen [H_2]	Excellent	Excellent	Excellent
Methane [CH_4]	Severe Effect	Excellent	Excellent
Benzene [C_6H_6]	Excellent	Excellent	Good
Diphenyl [$C_{12}H_{10}$]	Good	Excellent	Good

Appendix F: MATLAB CODE

F.1: Material Balances

Code that calculated the IO Level and RR Level mole balances. Mass balance checks were also implemented to verify that the resulting balances were correct.

```
% All this does is return a new material stream object
% A stream object simply encapulates data assoicated with each flowsheet
% stream, such as its composition and temperature
%
% Note that we only store molar compositions and total flowrate. This is
% much easiler to work with when doing the mole balances, but we can
% calculate mass versions simply by calling calculate_to_mass function on
% our stream object
function stream = mole_stream(constants)
    stream.totalmolarflowrate = 0;
    stream.molefraction = zeros(1,length(constants.componentdata.molarmass));
    stream.temperature = 0;
    stream.pressure = 0;
end

function [mole_streams, residuals] = mole_balances(specifications, plant_parameters,
constants)
% CRATESYSTEM at the end of this function, we should have the complete
% stream information related to this plant. This is where most things are
% updated relating to both energy and material streams

    % List our specifications
    yFH = specifications.yFH;
    yPH = specifications.yPH;
    s = specifications.s;
    x = specifications.x;
    PB = specifications.PB;
    mr = specifications.mr;
    reactor_Ti = specifications.reactor_initial_temperature;
    reactor_Tf = plant_parameters.reactor_Tf;
    reactor_P = specifications.reactor_pressure;

    % Define all streams
    % Outer Envelope
    mole_streams.FeedToluene = mole_stream(constants);
    mole_streams.FeedGas = mole_stream(constants);
    mole_streams.ProductBenzene = mole_stream(constants);
    mole_streams.ProductDiphenyl = mole_stream(constants);
    mole_streams.PurgeGas = mole_stream(constants);
    % Reactor and Recycle
    mole_streams.TotalReactorFeed = mole_stream(constants);
    mole_streams.TotalReactorEffluent = mole_stream(constants);
    mole_streams.RecycleGas = mole_stream(constants);
    mole_streams.RecycleToluene = mole_stream(constants);
    % Seperation System
    mole_streams.FlashLiquid = mole_stream(constants);
```

```
mole_streams.FlashVapor = mole_stream(constants);
mole_streams.Column1D = mole_stream(constants);
mole_streams.Column1B = mole_stream(constants);

% Specify known stream properties, such as composition
mole_streams.ProductBenzene.totalmolarflowrate = PB;

mole_streams.ProductBenzene.molefraction = [1 0 0 0 0];
mole_streams.FeedToluene.molefraction = [0 1 0 0 0];
mole_streams.FeedGas.molefraction = [0 0 yFH 1-yFH 0];
mole_streams.RecycleToluene.molefraction = [0 1 0 0 0];
mole_streams.ProductDiphenyl.molefraction = [0 0 0 0 1];

gas_composition = [0 0 yPH 1-yPH 0];
mole_streams.PurgeGas.molefraction = gas_composition;
mole_streams.RecycleGas.molefraction = gas_composition;

% Outer evelope
% Calculate Toluene Feed
FT = PB ./ s;
% Calculate Hydrogen Feed
FH = PB.*yFH.*(1 + s + yPH - s.*yPH)/(2.*s.*(yFH - yPH));
% Calculate Methane Feed
FM = (PB.*(1 - yFH).*(1 + s + yPH - s.*yPH))./(2.*s.*(yFH - yPH));
% Calculate Diphenyl
PD = -PB.*(-1 + s)./(2.*s);
% Calculate Hydrogen in Purge
PH = (PB.*(1 + s + yFH - s.*yFH).*yPH)./(2.*s.*(yFH - yPH));
% Calculate Methane in Purge
PM = (PB.*(1 + s + yFH - s.*yFH).*(1 - yPH))./(2.*s.*(yFH - yPH));

% Reactor system

% Calculate the flowrate of effulent hydrogen at the reactor exit
EH = -((PB.*(-2.*mr + x + s.*x))/(2.*s.*x));
% Calculate the flowrate of effulent methane at the reactor exit
EM = (PB.*(-2.*mr + x + s.*x).*(-1 + yPH))/(2.*s.*x.*yPH);
% By definition of single-pass conversion of toluene
RT = (PB - PB.*x)/(s.*x);
% Calculate the total feed to reactor of toluene
RF_T = PB ./ (s.*x);
% Calculate the total feed to reactor of hydrogen
RF_H = PB .* mr / (s.*x);
% Calculate the total feed to reactor of methaane
RF_M = -((PB.*(2.*mr.*(-1 + yPH) + x.*(1 + s + yPH - s.*yPH)))./(2.*s.*x.*yPH));
% MUST be in the same order as the component data
TE_stream =  [PB RT   EH   EM   PD];
TFR_stream = [0  RF_T RF_H RF_M 0];
RG_stream =  [0  0    EH   EM   0];
PG_stream =  [0  0    PH   PM   0];
% Calculates the total flowrate
TE_total = sum(TE_stream);
```

```
    TFR_total = sum(TFR_stream);
    RG_total = sum(RG_stream);
    PG_total = sum(PG_stream);
    % Finds the molar fraction of each components
    yTE = TE_stream ./ TE_total;
    yTFR = TFR_stream ./ TFR_total;

    % IO Level
    mole_streams.FeedToluene.totalmolarflowrate = FT;
    mole_streams.FeedGas.totalmolarflowrate = FM + FH;
    mole_streams.ProductDiphenyl.totalmolarflowrate = PD;
    mole_streams.PurgeGas.totalmolarflowrate = PG_total;

    % Reactor + Recycle
    mole_streams.TotalReactorFeed.totalmolarflowrate = TFR_total;
    mole_streams.TotalReactorFeed.molefraction = yTFR;
    mole_streams.TotalReactorEffluent.totalmolarflowrate = TE_total;
    mole_streams.TotalReactorEffluent.molefraction = yTE;
    mole_streams.RecycleGas.totalmolarflowrate = RG_total;
    mole_streams.RecycleToluene.totalmolarflowrate = RT;

    % Flash

    % P_flash = specifications.flash_pressure;
    % T_flash = specifications.flash_temperature;
   % [vaporFlash, liquidFlash, yV, xL] = calculate_flash(yTE, TE_total, T_flash, P_flash,
logical([1,1,1,1,1]), constants);

    % Perfect flash...
    liquidFlash = sum(TE_total.*yTE([1,2,5]));
    vaporFlash = sum(TE_total.*yTE([3,4]));
    comps_vap = zeros(1,5);
    comps_liq = zeros(1,5);
    comps_vap([1,2,5]) = yTE([1,2,5]);
    comps_liq([3,4]) = yTE([3,4]);
    xL = TE_total*comps_vap ./ liquidFlash;
    yV = TE_total*comps_liq ./ vaporFlash;

    mole_streams.FlashLiquid.totalmolarflowrate = liquidFlash;
    mole_streams.FlashVapor.totalmolarflowrate = vaporFlash;
    mole_streams.FlashLiquid.molefraction = xL;
    mole_streams.FlashVapor.molefraction = yV;

    % Temperatures
    mole_streams.TotalReactorFeed.temperature = reactor_Ti(1);
    mole_streams.TotalReactorEffluent.temperature = reactor_Tf(end);
    mole_streams.ProductDiphenyl.temperature = 293;
    mole_streams.ProductBenzene.temperature = 293;
    mole_streams.RecycleToluene.temperature = specifications.flash_temperature;
    mole_streams.RecycleGas.temperature = specifications.flash_temperature;

    % Pressures
```

```
    mole_streams.TotalReactorFeed.pressure = reactor_P;
    mole_streams.TotalReactorEffluent.pressure = specifications.flash_pressure;
    mole_streams.RecycleGas.pressure = specifications.flash_pressure;
    mole_streams.RecycleToluene.pressure = specifications.flash_pressure;
    mole_streams.ProductDiphenyl.pressure = constants.units.pressure.atm; % CHANGE LATER
    mole_streams.ProductBenzene.pressure = constants.units.pressure.atm; % CHANGE LATER
    mole_streams.FeedToluene.pressure = reactor_P;
    mole_streams.FeedGas.pressure = reactor_P;

    % Mass Checks
        [residuals.envelope_res, ~] = mass_check(constants, [mole_streams.PurgeGas,
mole_streams.ProductBenzene, mole_streams.ProductDiphenyl], [mole_streams.FeedToluene,
mole_streams.FeedGas]);
    [residuals.reactor_res, ~] = mass_check(constants, [mole_streams.TotalReactorFeed],
[mole_streams.TotalReactorEffluent]);
                                        [residuals.seperation_res, ~] =
mass_check(constants, [mole_streams.TotalReactorEffluent], [mole_streams.RecycleToluene,
mole_streams.ProductBenzene, mole_streams.ProductDiphenyl, mole_streams.RecycleGas]);
end
```

F.2: Reactor Design and Kinetics

Solves the reactor design equation using the closed mole balance formulas at the reactor feed. This is returned as a function of theoretical volume. The reactor's conversion and selectivity is then computed.

The selectivity is iterated to ensure the mole balance is valid. The selectivity used to calculated at the reactor feed must match the calculated reactor selectivity.

General PFR reactor ODE. As we have a adiabatic reactor, we must account for the reactor energy balance for accurate prediction of the reactor volume as a function of conversion. Heat capacity estimates are handled using the Allen-Lee-Dipper approximation as a function of temperature. Individual parameters are included inline with the heat capacity data. For the temperature range were are considering, the heat of reactions were found to be weakly a function of temperature and hence are considered constant.

```
function reactors = reactor_configurations(theoretical_reactors, constants)
    reactors = [];
    for i = 1:length(theoretical_reactors)
        theory_volume = theoretical_reactors(i).volume;
        N_splits = 1;
        if theory_volume > 565
            N_splits = ceil(theory_volume / 565);
            while true
                V_split = theory_volume / N_splits;
                % Turns out that for reactors its almost always best just
                % to maximize their length rather than use the rule of
                % thumb ratio due to how the costing works...
                H_split = 20;
                D_split = 2*sqrt(V_split / (pi * H_split));
                if D_split <= 6
                    V = V_split;
                    D = D_split;
                    H = H_split;
                    break
```

```
                else
                    N_splits = N_splits + 1;
                end
            end
        else
            V = theory_volume;
            H = 20; % Use the largest length...
            A = V / H;
            D = sqrt(A / pi);
        end
        % Copy over the data from the theory one...
        % Remember that reactors split in this way should have the same s
        % and x cause they must be in parallel
        for j = 1:N_splits
            reactor = theoretical_reactors(i);
            reactor.name = sprintf("Stage %d Reactor", i);
            reactor.volume = V;
            reactor.diameter = D;
            reactor.length = H;
            reactors = [reactors reactor];
        end
    end

    for i = 1:length(reactors)
            fprintf("Reactor %d: V=%0.0fm^3, D=%0.0fm, L=%0.0fm, x=%0.2f, s=%0.2f\n",
i, reactors(i).volume, reactors(i).diameter, reactors(i).length, reactors(i).conversion,
reactors(i).selectivity)
    end
end

% Recall that conversion is specified as part of the
% optimization where selectivity is some function of conversion. But the
% initial conditions to the reactor is in fact a function of both! Hence,
% we can remindy this by iterating the reactor design function until the
% selectivity convergences at the input to the reactor and that the output
% has the same selectivity.
function [reactors, S] = reactor_design(specs, constants, s0)
    objective_spec = specs;

    objective = @(s) objective_selectivity(s, objective_spec, constants);
    options = optimset('Display', 'off');
    S = fsolve(objective, s0, options); % Actual selectivity

    objective_spec.s = S;
    [reactors, ~] = specs.reactor_design_fun(objective_spec, constants);
end

% This mini function acts as a f(s) = 0 setup for fsolve where s is
% selectivity. It also functions as a way to specify the method to
% providing a reactor solver.
function residual = objective_selectivity(s_in, specs, constants)
    % Specify the reactor solver function here...
```

```
    specs.s = s_in;
    [~, s_out] = specs.reactor_design_fun(specs, constants);
    residual = s_in - s_out;
    if constants.debug
        fprintf("Selectivity (in): %0.2f\n", s_in);
        fprintf("Selectivity (out): %0.2f\n", s_out);
        fprintf("Selectivity residual: %0.2e\n", residual);
    end
end

function [reactors, S] = reactor_design_cold_shoot(specs, constants)
    options = optimset('Display', 'off');

    % Partition converisons
    x1 = specs.x * specs.conversion_split;

    % Total reactor feed
    F0 = reactor_feed_ic(specs.x, specs.s, specs.mr, specs.PB, specs.yPH)';
    Fs = F0 .* specs.reactor_cold_shot_ratio;
    Ts = specs.reactor_initial_temperature + specs.delta_temperature_cold_shot;

    % First reactor with the split feed. Allow the volume to max out at the
    % upper temperature limit.
      [v1, T1, F1, x1, s1] = solve_single_reactor(Fs, Ts, specs.reactor_pressure, x1,
false, constants);

    if ~isempty(v1)

        F3 = F1 + F0 .* (1 - specs.reactor_cold_shot_ratio);

        mr2 = F3(3) / F3(2);
        mr_tol = 0.75;

        if mr2 <= specs.mr+mr_tol && mr2 >= specs.mr-mr_tol

            % The meaning of the cold-shot actually is that there is a
            % heater
            T2 = specs.reactor_initial_temperature;

            % Solve the non-linear energy balance, resulting temperature must exist
            % at some temperature between Ti and exit of reactor, T1
            enthalpy_in = sum(F1 .* integrated_dipper(T1, T2, true(1,5), constants)');
               energy_balance = @(t3) enthalpy_in - sum(F3 .* integrated_dipper(t3, T2,
true(1,5), constants)');

            if constants.debug
                T = linspace(273, 900, 1e5);
                ebal = arrayfun(energy_balance, T);
                plot(T, ebal);
                xline([T2 T1])
                yline(0)
            end
```

```
            % Seems like the integrated functions arent perfect, so strictly
            % speaking the temperature is bounded by T1 and T2, but we should
            % relax this a bit to account for the numerical flaws
            T_tol = 2;
            T3 = fzero(energy_balance, [T2-T_tol T1+T_tol]);

            % Solve second reactor with the new conversion
            x2 = specs.x - x1;

            % solve for the correct selectivity

              [v2, T4, F4, ~, s2] = solve_single_reactor(F3, T3, specs.reactor_pressure,
x2, false, constants);

            if ~isempty(v2)

                % Collect reactor details
                reactor1.volume = v1;
                reactor1.effluent_temperature = T1;
                reactor1.inlet_temperature = Ts;
                reactor1.conversion = x1;
                reactor1.selectivity = s1;

                reactor2.volume = v2;
                reactor2.effluent_temperature = T4;
                reactor2.inlet_temperature = T3;
                reactor2.conversion = x2;
                reactor2.selectivity = s2;

                reactors = [reactor1 reactor2];
                S = F4(1) / (F0(2) - F4(2));
                return
            end
        end
    end

    reactors = [];
    S = 0;
end

function [reactors, S] = reactor_design_single(specs, constants)
    F0 = reactor_feed_ic(specs.x, specs.s, specs.mr, specs.PB, specs.yPH)';
      [V, Tf, ~, ~, S] = solve_single_reactor(F0, specs.reactor_initial_temperature,
specs.reactor_pressure, specs.x, false, constants);
    if ~isempty(V)
        reactor.volume = V;
        reactor.effluent_temperature = Tf;
        reactor.conversion = specs.x;
        reactor.selectivity = S;
        reactors = reactor;
    else
```

```
        reactors = [];
    end
end

function F0 = reactor_feed_ic(x, s, mr, PB, yPH)
    T = PB ./ (s .* x);
    H = mr .* PB ./ (s .* x);
    M = -((PB.*(2.*mr.*(-1 + yPH) + x.*(1 + s + yPH - s.*yPH)))./(2 .*s.*x.*yPH));

    F0(:,1) = zeros(size(T));
    F0(:,2) = T;
    F0(:,3) = H;
    F0(:,4) = M;
    F0(:,5) = zeros(size(T));
end

% 1 = Benzene molar flow (mol / s)
% 2 = Toluene molar flow (mol / s)
% 3 = Hydrogen molar flow (mol / s)
% 4 = Methane molar flow (mol / s)
% 5 = Diphenyl molar flow (mol / s)

function dydv = reactor_ode(~, y, P, constants)
    % Initialize dFdV as a column vector
    dydv = zeros(6, 1);
    T = y(1);
    F = y(2:6);
    F_total = sum(F);
    y = F ./ F_total;  % Normalizing f to get mole fraction

    r1 = rxn1_rate(y, T, P, constants);
    r2 = rxn2_rate(y, T, P, constants);

    % Benzene
    dydv(2) = r1 - 2*r2;

    % Toluene
    dydv(3) = -r1;

    % Hydrogen
    dydv(4) = -r1 + r2;

    % Methane
    dydv(5) = r1;

    % Diphenyl
    dydv(6) = r2;

    % Temperature
    cps = dipper_heat(T, true(5,1), constants)';
    CpT = sum(F.*cps);
    %[deltaH1, deltaH2] = hess_law(T, constants);
```

```
    deltaH1 = -42.0e3; % J / mol
    deltaH2 = 14.8e3; % J / mol
    %  mol/(m^3*s) J/mol = J/(m^3*s)
    H =  r1.* deltaH1 + r2.*deltaH2; % J / (m^3 * s)
    %         J/(m^3*s) / (J / (K*s)) = K/m^3
    dydv(1) = -H ./ CpT; % K / m^3
end

function [X, S, V, T, F] = reactor_scaleup(F0, Ti, P, constants)
    [X, S, V, T, F] = solve_reactor(F0, Ti, P, constants);

    X = X(2:end);
    S = S(2:end);
    T = T(2:end);
    F = F(2:end,:);
    V = V(2:end);
end

function rate = rxn1_rate(y, T, P, constants)
% Arguments are in SI, convert appropriately
% Returns in mol/(m^3 * s)
    c = y .* P .* constants.units.volume.litre ./ (constants.R .* T); % mol/L
    k1f = rxn1_rate_coeff(T);
    % Calulate rate in mol/(L * hr)
    rateNonSI = k1f .* c(2) .* c(3).^0.5;
    % Convert to mol/(m^3 * s)
    rate = 1./(constants.units.volume.litre .* constants.units.time.hour) .* rateNonSI;
end

function k1f = rxn1_rate_coeff(T)
    k10 = 1.188e14;
    Ea1 = 52.0; % kcal / gmol
    R_s = 1.987204259e-3;
    k1f = k10.*exp(-Ea1 ./ (R_s.*T));
end

function [K, log10_K] = rxn2_equalibrium(T)
   log10_K = (-734.92 ./ T) - 8.1046 + 3.1294 .* log10(T) - 7.0804e-4 .* T + 8.523e-8 .* T.^2;
    K = 10.^log10_K;
end

function rate = rxn2_rate(y, T, P, constants)
% Arguments are in SI, convert appropriately
% Returns in mol/(m^3 * s)
    k2f = rxn2_rate_coeff(T);
    [K, ~] = rxn2_equalibrium(T);
      rate  =  1./(constants.units.volume.litre  .*  constants.units.time.hour)  .*  k2f  .*
(P.^2) .* (y(1).^2 - (y(5) .* y(3))./K);
end

function k2f = rxn2_rate_coeff(T)
    k20 = 1.1647e-5;
```

```
    Ea2 = 30.19;
    R_s = 1.987204259e-3;
    k2f = k20.*exp(-Ea2 ./ (R_s.*T));
end

function [X, S, V, T, F] = solve_reactor(F0, Ti, P, constants)
    Vspan = linspace(0, 5e3, 1e5);
    [V, Y] = ode45(@(v,f) reactor_ode(v, f, P, constants), Vspan, [Ti; F0]); % Evaluate
the reactor system
    F = real(Y(:,2:6));

    % moles of LR (T) reacted = F0_basis(2) - F_basis(:,2)
    % moles of LR (T) fed to reactor = F0_basis(2)
    % moles of product (B) leaving = F_basis(:,1)
    X = real((F0(2) - F(:,2)) ./ (F0(2)));
    S = real((F(:,1) - F0(1)) ./ (F0(2) - F(:,2)));

    % Columns are specie, so sum along columns
    T = real(Y(:,1));

    % Prune conversion in excess, NEVER let the optimization go past this
    % limit otherwise the interpolation will never work.
    indices = X < 0.9999;

    S = S(indices);
    X = X(indices);
    V = V(indices);
    T = T(indices);
    F = F(indices,:);
end

function [V, Tf, F, X, S] = solve_single_reactor(F0, Ti, P, x, allow_interstage_volume,
constants)

   [X_solve, S_solve, V_solve, T_solve, F_solve] = reactor_scaleup(F0, Ti, P, constants);

    % Check if all quanities actually make sense...
    if ~isempty(X_solve) && all(V_solve > 0) && min(T_solve) <= constants.reactor_T_max

        Tf = interp1(X_solve, T_solve, x, 'linear');
        S = interp1(X_solve, S_solve, x, 'linear');

        if constants.reactor_T_max > Tf
            V = interp1(X_solve, V_solve, x, 'linear');
            X = x;

            N_comp = size(F_solve, 2);
            F = zeros(N_comp, 1);
            for col = 1:N_comp
                F(col) = interp1(X_solve, F_solve(:, col), x, 'linear');
            end
            return
```

```
        elseif allow_interstage_volume
            V = interp1(T_solve, V_solve, constants.reactor_T_max, 'linear');
            X = interp1(T_solve, X_solve, constants.reactor_T_max, 'linear');

            N_comp = size(F_solve, 2);
            F = zeros(N_comp, 1);
            for col = 1:N_comp
            F(col) = interp1(T_solve, F_solve(:, col), constants.reactor_T_max, 'linear');
            end
            return
        end
    end

    V = [];
    Tf = constants.reactor_T_max;
    X = x;
    S = 0;
    F = zeros(1,5);
end
```

F.3: Optimization using NPV

In order to obtain the best NPV, we using a objective function that returns the NPV given select optimization parameters, which includes the inlet reactor feed temperature T_i, molar ratio MR, and purge fraction of hydrogen y_{PH}.

$$\text{NPV} = f(x, y_{\text{PH}}, T_i) \tag{F.1}$$

This is done using the `fmincon` function. This is a minimization function, but we can seek for a maximization by considering the negative of the objective and then taking the negative of its minimization value. The objective function in this case is called `objective_npv`. Other minor variables were briefly simulated, include the distillation column pressures, but were generally excluded from the typical optimization setup.

MATLAB's `fmincon` function also allows upper and lower bounds, which were used to limit that the

$$\begin{aligned} g(x, y_{\text{PH}}, T_i) &= -f(x, y_{\text{PH}}, T_i) \\ [-x, -y_{\text{PH}}, -T_i] &= \overset{\text{fmincon}}{\max}(-f(x, y_{\text{PH}}, T_i)) \end{aligned} \tag{F.2}$$

```
% Remove variables that may mess us up...
clear;
clc;
close('all');

% run_two_reactor_cold_shot();
run_one_reactor()

function run_one_reactor()

    persistent s_opt;
    options = optimset('MaxFunEvals',Inf,'MaxIter',5000,...
        'Display','none', ...
        'PlotFcn', {@optimplotfval});
    constants = get_constants();
```

```
    specs = initial_spec_single_reactor(constants);

    % initial condition to maximizer
    specs.reactor_design_fun = @reactor_design_single;
            P_BT   =   (6*constants.units.pressure.bar   -   specs.column_pressures(1))/
((35-1)*constants.units.pressure.bar);
            P_TD   =   (6*constants.units.pressure.bar   -   specs.column_pressures(1))/
((35-1)*constants.units.pressure.bar);
    T_reactor = (650+273.15-specs.reactor_initial_temperature)/(650-550);
    x0 = [specs.x, T_reactor, specs.yPH, P_BT, P_TD];

    %     x     Ti    yPH  P_column_BT P_column_DT
    lb = [0.01, 0.001 0.01 0.01 0.01];
    ub = [0.999, 0.999 0.99 0.99, 0.99];
    %lb = [0.01, 550+273.15 0.01];
    %ub = [0.999, 650+273.15 0.99];

    % for maximumization, objective is the negative of f: g = -f and maximize g
    [sol, ~] = fmincon(@(x_sol) -objective_npv(x_sol, specs, constants), x0, [], [], [],
[], lb, ub, [], options);

    % Keep the selectivity
    sol(end+1) = s_opt;

    save('sol_single_reactor.mat', "sol");

    function NPV = objective_npv(x_sol, spec, constants)
      fprintf("Simulating parameters: x=%0.2f Ti=%0.2f yPH=%0.2f P_BT=%0.2f P_DT=%0.2f\n",
x_sol');
        %fprintf("Simulating parameters: x=%0.2f Ti=%0.0fK yPH=%0.2f\n", x_sol');

        obj_spec = objective_spec_one_reactor([x_sol spec.s], spec, constants);
        [plant_econ, ~, ~, s_opt] = solve_HDA_plant(obj_spec, constants);
        NPV = plant_econ.npv;
    end
end

function run_two_reactor_cold_shot()

    persistent s_opt;
    constants = get_constants();
    specs = initial_spec_two_reactor_cold_shot(constants);

    % initial condition to maximizer
    options = optimset('MaxFunEvals',Inf,'MaxIter',5000,...
        'Display','none', ...
        'PlotFcn', {@optimplotfval});
    specs.reactor_design_fun = @reactor_design_cold_shoot;
                P_BT    =    (specs.column_pressures(1)-1*constants.units.pressure.bar)/
((35-1)*constants.units.pressure.bar);
                P_TD    =    (specs.column_pressures(2)-1*constants.units.pressure.bar)/
((35-1)*constants.units.pressure.bar);
```

```
    T_reactor = (specs.reactor_initial_temperature - (550+273.15))/(650-550);
    delta_T = (specs.delta_temperature_cold_shot)/(120);
     x0 = [specs.x, T_reactor, specs.yPH, P_BT, P_TD, delta_T, specs.conversion_split,
specs.reactor_cold_shot_ratio];

    lb = 0.001.*ones(1,length(x0));
    ub = 0.999.*ones(1,length(x0));

    % for maximumization, objective is the negative of f: g = -f and maximize g
    [sol, ~] = fmincon(@(x_sol) -objective_npv(x_sol, specs, constants), x0, [], [], [],
[], lb, ub, [], options);

    % Keep the selectivity
    sol(end+1) = s_opt;

    save('sol_cold_shot.mat', "sol");

    function NPV = objective_npv(x_sol, spec, constants)
        fprintf("Simulating parameters\n");
        obj_spec = objective_spec_cold_shot([x_sol spec.s], spec, constants);
        [plant_econ, ~, ~, s_opt] = solve_HDA_plant(obj_spec, constants);
        NPV = plant_econ.npv;
    end
end

function run_isotermal_reactor()

    persistent s_opt;
    constants = get_constants();
    specs = initial_spec_two_reactor_cold_shot(constants);

    % load("sol.mat", "sol");
    % specs = objective_spec(sol, initial_spec(constants));

    % initial condition to maximizer
    specs.reactor_design_fun = @reactor_design_cold_shoot;
    x0 = [specs.x, specs.reactor_Ti, specs.yPH];

    %     x     Ti          yPH
    lb = [0.01, 550+273.15, 0.01];
    ub = [0.999,640+273.15, 0.99];

    % for maximumization, objective is the negative of f: g = -f and maximize g
    [sol, ~] = fmincon(@(x_sol) -objective_npv(x_sol, specs, constants), x0, [], [], [],
[], lb, ub);

    % Keep the selectivity
    sol(end+1) = s_opt;

    save('sol_isothermal.mat', "sol");

    function NPV = objective_npv(x_sol, spec, constants)
```

```
        %fprintf("Simulating parameters: x=%0.2f Ti=%0.0fK yPH=%0.2f\n", x_sol');
        obj_spec = objective_spec_one_reactor([x_sol spec.s], spec);
        [plant_econ, ~, ~, s_opt] = solve_HDA_plant(obj_spec, constants);
        NPV = plant_econ.npv;
    end
end
```

F.4: Separation System and Distillation Solver

```
function Tbub = calculate_bubble_point(x, Ptot, Tbp_g, constants, mask)
    opts = optimoptions('fsolve', 'Display', 'off');
   Tbub = fsolve(@(t) sum(x .* vapor_pressure(t, constants, mask) ./ Ptot) - 1, Tbp_g, opts);
end

function Tdew = calculate_dew_point(y, Ptot, Tbp_g, constants, mask)
    opts = optimoptions('fsolve', 'Display', 'off');
   Tdew = fsolve(@(t) sum(y .* Ptot ./ vapor_pressure(t, constants, mask)) - 1, Tbp_g, opts);
end

function Hvap = latent_heat(T, constants, mask)
    factors = constants.componentdata.latent_heats;
    Tr = T ./ constants.componentdata.critical_temperature(mask);
    % J/mol
    Hvap = 1e-3 .* factors.C1(mask) .* (1 - Tr).^(factors.C2(mask) + factors.C3(mask).*Tr
+ factors.C4(mask).*Tr.^2);
end

function [V, L, y, x, phi] = calculate_flash(z, F, T_flash, P_flash, mask, constants)
    Psat = vapor_pressure(T_flash, constants, mask);
    K = Psat ./ P_flash; % Raoult's Law

    objective_phi = @(phi) rachford_rice(phi, z, K);
    options = optimset('Display','off');
    phi = fsolve(objective_phi, 0.5, options);

    V = F * phi;
    L = (1 - phi)*F;
    y = K .* z ./ (1 + phi.*(K - 1));
    x = z ./ (1 + phi.*(K - 1));

    % Remove the limiting components which will barely affect the balance
    % but usually have numerical problems with the seperation processes.
    % x(x < 1e-6) = 0;
    % y(y < 1e-6) = 0;
end

function f = rachford_rice(phi, z, K)
    f = sum(z .* (K - 1) ./ (1 + phi*(K - 1)));
end

function [diameter_top, diameter_bottom] = column_diameter(B, D, xB, xD, T_B, T_D, r, s,
spacing, Ptot, constants, mask)
```

```
    % Distillation Column Diameter Sizing

    MW = constants.componentdata.molarmass(mask); % g/mol
    % Weighted rho and sigma of the components. Usually this is identical
    % to the pure component values of the LK and HK.

    % Top of column
    L = r * D .* xD; % mol / s
    V = (r+1) * D .* xD; % mol / s
    L_total_mass = sum(L .* MW); % g/s
    V_total_mass = sum(V .* MW); % g/s
    mass_ratio_top = L_total_mass / V_total_mass;
    % At the top its a total condensor, so xL = xV
     diameter_top = get_diameter(mass_ratio_top, sum(V), D, T_D, xD, xD, spacing, Ptot,
constants, mask);

    % Bottom of column
    L_prime = s * B .* xB; % mol / s
    V_prime = (s - 1) .* B .* xB; % mol / s
    L_total_mass_prime = sum(L_prime .* MW);
    V_total_mass_prime = sum(V_prime .* MW);
    mass_ratio_bottom = L_total_mass_prime / V_total_mass_prime;
    % At the bottom its a partial reboiler, so xL != xV
    % To do this, use Raoults Law equalibrium
    masked = mask(logical([1 1 0 0 1]));
    xV_B = xB .* (vapor_pressure(T_B, constants, masked) ./ Ptot);
     diameter_bottom = get_diameter(mass_ratio_bottom, sum(V_prime), B, T_B, xB, xV_B,
spacing, Ptot, constants, mask);
end

function diameter = get_diameter(mass_ratio, V, L, T, xL, xV, spacing, Ptot, constants,
mask)
    MW = constants.componentdata.molarmass(mask);
    eta = 0.90; % efficiency approximation
    pfrac = 0.75; % Percent flooding
    R = 8.314; % m^3 Pa / (mol K)

    % The liquid density via Rackett is not the most accurate...will
    % replace

    density_mask = logical([1 1 0 0 1]);
    rho_liq = sum(MW .* xL .* molar_density(T, constants, density_mask(mask)));
    rho_vap = sum(xV .* (MW .* Ptot) ./(R*T)); % g / m^3
    %surface_tension = sum(xL .* brock_bird(T, constants, mask)); % mN/m
    surface_tension = 20 * 0.8;

    F_LV = mass_ratio*((rho_vap/rho_liq)^0.5); % Flow parameter
    csb = Csb(F_LV, spacing); % Tray height correction factor, ft/s
    % ft/s -> m/s
     u_flood = constants.units.length.ft * csb * ((surface_tension/20)^0.2) * ((rho_liq-
rho_vap)/rho_vap)^0.5; % Flooding Flow Rate
    % m -> ft
```

```
    diameter = (((4*V*R*T)/(pi*eta*Ptot*u_flood*pfrac))^0.5) / constants.units.length.ft;
end

function csb = Csb(F_LV, spacing)
    switch spacing
        case 12
            C1 = 1.0674;
            C2 = 0.5578;
            C3 = 0.17919;
        case 18
            C1 = 1.0262;
            C2 = 0.63513;
            C3 = 0.2097;
        case 24
            C1 = 0.94506;
            C2 = 0.70234;
            C3 = 0.22618;
    end
    csb = 10^(-C1 - C2*log10(F_LV) - C3*(log10(F_LV))^2);
end

function [D, B, xD, xB] = distillation_balance(F, z, LK_idx, HK_idx, FR_LK_D, FR_HK_B,
FR_non_HK_B, FR_non_LK_D, alpha)
% DISTILLATION_BALANCE
% Inputs:
%   F              - Feed flowrate (scalar)
%   z              - Feed composition (1×n array, must sum to 1)
%   LK_idx         - Index of light key
%   HK_idx         - Index of heavy key
%   FR_LK_D        - Fractional recovery of light key in distillate
%   FR_HK_B        - Fractional recovery of heavy key in bottoms
%   FR_non_HK_B    - Fractional recovery of low vol non-key components in bottoms
%   FR_non_LK_D    - Fractional recovery of high vol non-key components in distillate
%
% Outputs:
%   D  - Distillate flowrate
%   B  - Bottoms flowrate
%   xD - Distillate composition vector
%   xB - Bottoms composition vector

    n = length(z);
    assert(abs(sum(z) - 1) < 1e-6, 'Feed composition must sum to 1');

    flow_D = zeros(1, n);
    flow_B = zeros(1, n);

    for i = 1:n
        zi = z(i);
        if i == LK_idx
            frac_D = FR_LK_D;
        elseif i == HK_idx
            frac_D = 1 - FR_HK_B;
```

```
        elseif alpha(i) < alpha(HK_idx)
            frac_D = 1 - FR_non_HK_B;
        elseif alpha(i) > alpha(LK_idx)
            frac_D = FR_non_LK_D;
        else
            frac_D = 0.5; % sandwitch components, between LK and HK
        end

        flow_D(i) = frac_D * zi * F;
        flow_B(i) = (1 - frac_D) * zi * F;
    end

    D = sum(flow_D);
    B = sum(flow_B);

    xD = flow_D / D;
    xB = flow_B / B;
end

function column = solve_distillation_column(F, D, B, z, xD, xB, r, s, column_idx,
column_LK_idx, column_HK_idx, alphas, specs, mask, constants, method)
    % Uses either solver to find the number of stages

    switch method
        case "FUG-CS"
         [N, NF] = solve_stages_fug(B, D, z, xD, xB, r, s, r/2, constants.stage_efficiency,
alphas, column_LK_idx, column_HK_idx, "CS");
        case "FUG-KR"
         [N, NF] = solve_stages_fug(B, D, z, xD, xB, r, s, r/2, constants.stage_efficiency,
alphas, column_LK_idx, column_HK_idx, "KR");
        case "UW"
            est_N = zeros(2,1);
            est_NF = zeros(2,1);
                [est_N(1), est_NF(1)] = solve_stages_fug(B, D, z, xD, xB, r, s, r/2,
constants.stage_efficiency, alphas, column_LK_idx, column_HK_idx, "CS");
                [est_N(2), est_NF(2)] = solve_stages_fug(B, D, z, xD, xB, r, s, r/2,
constants.stage_efficiency, alphas, column_LK_idx, column_HK_idx, "KR");

            NT_g = est_N - est_NF;
            NB_g = est_NF;

            isSolved = false;
            for i = 1:2
                    [N, NF, exitflag] = solve_stages_underwood(z, xD, xB, r, s,
constants.stage_efficiency, alphas, NT_g(i), NB_g(i));

                if exitflag == 1
                    isSolved = true;
                    break
                end
            end
```

```
            if ~isSolved
                fprintf("Could not solve column stages using Underwood Method! Returning
FUG guess...\n")
                N = mean(est_N);
                NF = mean(est_NF);
            end
        otherwise
            error("Please choose a correct solver method!")
    end

    % Get the total pressure for this column given in the specs
    Ptot = specs.column_pressures(column_idx);

    % First guess for the reboiler and condensor temperatures
    Tbp = constants.componentdata.nominal_boiling_temperature(mask);

    % Since everything is saturated, we use Antonie's equation at a
    % specified pressure to find the corresponding temperatures of
    % distallate and bottoms assuming pure components, which corresponds to
    % the LK and HK. This is essentially a dew and bubble point
    % calculation.

    % Total condensor -> vapor and liquid compositions are equal
    % Use due point temperature of mixture
    under_mask = logical([1 1 0 0 1]);
    T_D = calculate_dew_point(xD, Ptot, Tbp(column_LK_idx), constants, under_mask(mask));
    % Use bubble point temperature of mixture
          T_B   =   calculate_bubble_point(xB,   Ptot,   Tbp(column_HK_idx),   constants,
under_mask(mask));

    % Solve for the column size
    spacing = specs.column_spacing(column_idx);
    height = (spacing/12)*(N + 3); %ft
   [diameter_top, diameter_bottom] = column_diameter(B, D, xB, xD, T_B, T_D, r, s, spacing,
Ptot, constants, mask);

    % Calculate the condenser and reboiler requirements
    distillate_vapor_flow = D*(1 + r);
    bottoms_vapor_flow = B*s;

          [area_D,   duty_D,   flow_working_D,   log_mean_delta_T_D,   cost_per_mol_D]   =
calculate_latent_heat_exchange_fluid(distillate_vapor_flow, xD, Ptot, T_D, T_D, "steam",
"cooling", 770, mask, constants);
          [area_B,   duty_B,   flow_working_B,   log_mean_delta_T_B,   cost_per_mol_B]   =
calculate_latent_heat_exchange_fluid(bottoms_vapor_flow,  xB,  Ptot,  T_B,  T_B,  "steam",
"heating", 820, mask, constants);

    condenser.area = area_D;
    condenser.duty = duty_D;
    condenser.flow_working = flow_working_D;
    condenser.log_mean_deltaT = log_mean_delta_T_D;
    condenser.cost_per_mol = cost_per_mol_D;
```

```
    reboiler.area = area_B;
    reboiler.duty = duty_B;
    reboiler.flow_working = flow_working_B;
    reboiler.log_mean_deltaT = log_mean_delta_T_B;
    reboiler.cost_per_mol = cost_per_mol_B;

    % Collect column information
    column.stages = round(N/2);
    column.feed_stage = round(NF/2);
    column.real_stages = round(N);
    column.real_feed_stage = round(NF);
    column.pressure = Ptot / constants.units.pressure.kPa;
    column.reflux_ratio = r;
    column.reboiler_ratio = s;
    column.feed_flow = F;
    column.bottoms_flow = B;
    column.distillate_flow = D;
    column.vapor_frac = D*(1 + r)/F;
    column.distillate_vapor_flow = D*(1 + r);
    column.bottoms_vapor_flow = B*s;
    column.distillate_comp = xD;
    column.bottoms_comp = xB;
    column.condenser = condenser;
    column.reboiler = reboiler;
    column.distillate_temperature = T_D;
    column.bottoms_temperature = T_B;
    column.height = height * constants.units.length.ft; % m
    column.diameter_top = diameter_top * constants.units.length.ft; % m
    column.diameter_bottom = diameter_bottom * constants.units.length.ft; % m
    column.spacing = spacing * constants.units.length.in; % m
end

function [columns, flashs] = solve_seperation_system(mole_streams, specs, constants)
    [columns, flashs] = direct_method(mole_streams, specs, constants);
end

function [columns, flashs] = direct_method(mole_streams, specs, constants)

    Flash_molarflow = mole_streams.TotalReactorEffluent.totalmolarflowrate;
    Flash_molarfrac = mole_streams.TotalReactorEffluent.molefraction;
     Flash_liquidflow = sum(Flash_molarflow .* Flash_molarfrac(logical([1,1,0,0,1])) ./
(molar_density(specs.flash_temperature, constants, logical([1,1,1]))));

    % Only use the liquid flowrate for the residence time
    residence_time = 15 * constants.units.time.min;
    Flash_Volume = Flash_liquidflow * residence_time;

    flash_split.volume = Flash_Volume;
    flash_split.vapor_comp = mole_streams.FlashVapor.molefraction;
    flash_split.liquid_comp = mole_streams.FlashLiquid.molefraction;
    flash_split.pressure = specs.flash_pressure;
```

```
    flash_split.pressure = specs.flash_temperature;

    flashs = flash_split;

    % B/T split, expect almost no methane and hydrogen from here onwards
    % due to stablizer. Expect dipenyl to drop to bottoms always

    column_BTD_idx = 1;
    BTD_mask = logical([1 1 0 0 1]);

    F = mole_streams.FlashLiquid.totalmolarflowrate;
    z = mole_streams.FlashLiquid.molefraction;

    [F1, z1] = mask_and_normalize(F, z, BTD_mask);
    alphaBTD = get_relative_volitilities(specs.column_pressures(column_BTD_idx));
    alphaBTD = alphaBTD(BTD_mask);
              [D1,    B1,    xD1,    xB1]    =    distillation_balance(F1,    z1,    1,    2,
specs.column_FR_D(column_BTD_idx), specs.column_FR_B(column_BTD_idx), 1, 1, alphaBTD);

    %f = 1 + 0.01*z(2);
       %rmin1  =  alphaBTD(2)*(z(1)  +  z(2))/(f*z(1)*(alphaBTD(1)-alphaBTD(2)))  +  z(3)/
(f*z(1)*(alphaBTD(1)-1));
    rmin1 = calculate_rmin(z1, 1, xD1, alphaBTD, 1, 2);
    r1 = 2 * rmin1;
    s1 = (D1 / B1)*(r1 + 1);
     column_BTD = solve_distillation_column(F1, D1, B1, z1, xD1, xB1, r1, s1, 1, 1, 2,
alphaBTD, specs, BTD_mask, constants, "FUG-CS");
    column_BTD.name = "B/TD Column";
    column_BTD.reboiler.name = "B/TD Column";
    column_BTD.condenser.name = "B/TD Column";

    % T/D Split
    column_TD_idx = 2;
    TD_mask = logical([0 1 0 0 1]);
    [F2, z2] = mask_and_normalize(F, z, TD_mask);
    alphaTD = get_relative_volitilities(specs.column_pressures(column_TD_idx));
    alphaTD = alphaTD(TD_mask);
              [D2,    B2,    xD2,    xB2]    =    distillation_balance(F2,    z2,    1,    2,
specs.column_FR_D(column_TD_idx), specs.column_FR_B(column_BTD_idx), 1, 1, alphaTD);
    %f = 1 + 0.01*z(2);
       %rmin1  =  alphaBTD(2)*(z(1)  +  z(2))/(f*z(1)*(alphaBTD(1)-alphaBTD(2)))  +  z(3)/
(f*z(1)*(alphaBTD(1)-1));
    rmin2 = calculate_rmin(z2, 1, xD2, alphaTD, 1, 2);
    r2 = 2 * rmin2;
    s2 = (D1 / B1)*(r2 + 1);
   column_TD = solve_distillation_column(F2, D2, B2, z2, xD2, xB2, r2, s2, column_TD_idx,
1, 2, alphaTD, specs, TD_mask, constants, "FUG-KR");
    column_TD.name = "T/D Column";
    column_TD.reboiler.name = "T/D Column";
    column_TD.condenser.name = "T/D Column";

    columns = [column_BTD column_TD];
```

```
    for i = 1:length(columns)
            fprintf("%s:  N=%0.0f,  Nf=%0.0f,  Dia  Top=%0.1fm,  Dia  B=%0.1fm,  H=%0.1fm\n",
columns(i).name,   columns(i).stages,   columns(i).feed_stage,   columns(i).diameter_top,
columns(i).diameter_bottom, columns(i).height)
    end
end

function [N, NF, exitflag] = solve_stages_underwood(z, xD, xB, r, s, stage_efficiency,
alpha, NT_g, NB_g)
    c = length(alpha);
    % find the underwood roots
    [phi, phi_bar] = find_underwood_roots(xD, xB, r, s, c, alpha);
    % With the underwood roots, use Equations of 4.50 and 4.53
    [NT, NB, exitflag] = calculate_stages(z, phi, phi_bar, alpha, c, NT_g, NB_g);
    % Find the actual number of stages
    N_theory = NT + NB;
    NF_theory = NB;
    % Find real stages
    N = N_theory * stage_efficiency;
    NF = NF_theory * stage_efficiency;
end

function [phi, phi_bar] = find_underwood_roots(xiD, xiB, r, s, c, alpha)
    % Define equations to solve the two phi equations
    padding = 1e-10;
    sorted_alpha = sort(alpha);
    % Equations 4.49 and 4.52
    objective_rectifying = @(v) sum((alpha .* xiD) ./ (alpha - v)) - r - 1;
    objective_stripping = @(v) sum((alpha .* xiB) ./ (alpha - v)) + s;

    phi = zeros(1, c);
    phi_bar = zeros(1, c);

    % alpha_H > alpha_M > alpha_B > alpha_T > alpha_D > 0
    phi_interval_values = [0 sorted_alpha];
    % Inf > alpha_H > alpha_M > alpha_B > alpha_T > alpha_D
    phi_bar_interval_values = [sorted_alpha max(sorted_alpha)*10];

    opt = optimset('Display', 'none');
    % Exactly c roots for each case...
    for i = 1:c
        % Find the root between the ith and i+1 relative volatilities
        interval_phi = [...
            phi_interval_values(i)+padding phi_interval_values(i+1)-padding ...
        ];
        interval_phi_bar = [...
            phi_bar_interval_values(i)+padding phi_bar_interval_values(i+1)-padding ...
        ];
        try
            phi(i) = fzero(objective_rectifying, interval_phi, opt);
            phi_bar(i) = fzero(objective_stripping, interval_phi_bar, opt);
```

```
        catch ME
        end
    end
end

function [NT, NB, exitflag] = calculate_stages(z, phi, phi_bar, alpha, c, NT_g, NB_g)
    x0 = [z(1:c-1) NT_g NB_g];
    opts = optimoptions('fsolve', ...
        'Algorithm', 'levenberg-marquardt', ...
        'Display', 'none', ... % iter-detailed
        'FunctionTolerance', 1e-9, ...
        'FiniteDifferenceType','central', ...
        'StepTolerance', 1e-15);
    % lb = [zeros(1,c-1) 0 0];
    % ub = [ones(1,c-1), 100 100];
     %[x_sol,~,exitflag,~] = fmincon(@(x) sqrt(sum(underwood_equation(x, phi, phi_bar,
alpha, c).^2)), x0,[],[],[],[],lb,ub);
     [x_sol,~,exitflag,~] = fsolve(@(x) underwood_equation(x, phi, phi_bar, alpha, c),
x0, opts);
    if exitflag == 1
        NT = x_sol(c);
        NB = x_sol(c+1);
    else
        NT = 1e6;
        NB = 1e6;
    end
end

function y = underwood_equation(x, phi, phi_bar, alpha, c)
    partial_xifs = x(1:c-1); % Pick out c-1 components
    xif = [partial_xifs 1-sum(partial_xifs)];
    nt = x(c);
    nb = x(c+1);

    % Iterate over every unique combination of phi and phi_bar values,
    % which ends up being c-1 equations for each set
    y = zeros(1,c-1);
    sum_term = @(v) sum((alpha .* xif) ./ (alpha - v));

    idx = 1;
    for k = 1:(c-1)
        j = k + 1;
        phi_k = phi(k);
        phi_j = phi(j);
        phi_bar_k = phi_bar(k);
        phi_bar_j = phi_bar(j);

        % Equation 4.50
        y(idx) = (phi_k / phi_j)^nt - sum_term(phi_j) / sum_term(phi_k);

        % Increament the index
        idx = idx + 1;
```

```
        % Equation 4.53
        y(idx) = (phi_bar_k / phi_bar_j)^nb - sum_term(phi_bar_k) / sum_term(phi_bar_j);

        % Increament the index
        idx = idx + 1;
    end
end

function [N, NF, r, s] = solve_stages_fug(B, D, z, xD, xB, r, s, rmin, stage_efficiency,
alpha, LK_idx, HK_idx, method)
    alpha_LK_HK = alpha(LK_idx) / alpha(HK_idx);
    % Fenske correlation
    Nmin = log((xD(LK_idx) / xD(HK_idx)) / (xB(LK_idx) / xB(HK_idx))) / log(alpha_LK_HK);
    % Get theoretical stages with Gilland correlation
    N_theory = 0.75 * ((1 - ((r - rmin) / (r + 1))^0.5688) + Nmin) / (1 - 0.75*(1 - (r -
rmin)/(r + 1)^0.5688));
    % Two estimations exist, asymmetric and asymmetric
    if method == "CS"
       NF_theory = optimum_feed_stage(Nmin, N_theory, xD, z, alpha_LK_HK, LK_idx, HK_idx);
    elseif method == "KR"
        NF_theory = asymmetric_feed_stage(B, D, xB, xD, z, N_theory, LK_idx, HK_idx);
    end
    % Apply stage efficiencies
    N = N_theory * stage_efficiency;
    NF = NF_theory * stage_efficiency;
end

function NF_theory = optimum_feed_stage(Nmin, N_theory, xD, z, alpha_LK_HK, LK_idx, HK_idx)
    NFmin = log((xD(LK_idx) / xD(HK_idx)) / (z(LK_idx) / z(HK_idx))) / log(alpha_LK_HK);
    NF_theory = NFmin * N_theory/Nmin;
end

function NF_theory = asymmetric_feed_stage(B, D, xB, xD, z, N_theory, LK_idx, HK_idx)
    opts = optimoptions('fsolve', 'Display', 'off');
      objective = @(n) kirkbride_equation(n, B, D, xB(LK_idx), xD(HK_idx), z(LK_idx),
z(HK_idx), N_theory);
    sol = fsolve(objective, [6, 6], opts);
    NB = sol(2) + 1;
    NF_theory = NB;
end

function f = kirkbride_equation(n, B, D, x_LK_B, x_HK_D, z_LK, z_HK, N)
    Nrect = n(1);
    Nstrip = n(2);
    f(1) = N - Nrect - Nstrip - 1;
    f(2) = ((z_HK / z_LK) * (x_LK_B / x_HK_D)^2 * (B / D))^(0.206) - (Nrect / Nstrip);
end
```

F.5: Heat Exchanger Network

```
function [area, duty, log_mean_delta_T, T_cross_out] = calculate_heat_exchange_cross(F_feed, x_feed, F_cross, x_cross, T_feed_in, T_feed_out, T_cross_in, heat_transfer_coeff, constants)
   T_cross_out = fzero(@(t) energy_balance(t, F_feed, x_feed, F_cross, x_cross, T_feed_in, T_cross_in, constants), [T_feed_in, T_cross_in]);

    % With energy balance satisfied, we solve the heat exchanger
    % requirements, which should be just below the maximum load.
    duty = sum(F_feed * x_feed .* integrated_dipper(T_feed_in, T_feed_out, constants));
        [log_mean_delta_T, area] = calculate_heat_exchanger_specs(duty, T_cross_in, T_cross_out, T_feed_in, T_feed_out, heat_transfer_coeff);
end

function f = energy_balance(T_cross_out, F_feed, x_feed, F_cross, x_cross, T_feed_in, T_cross_in, constants)

    % Finds the total duty to produce this temperature change at constant
    % pressure in J/mol
     Q1 = sum(F_feed * x_feed .* integrated_dipper(T_feed_in, T_feed_out, true(1,5), constants));
    Q2 = sum(F_cross * x_cross .* integrated_dipper(T_cross_in, T_cross_out, true(1,5), constants));

    % This energy is then adiabatically transfered to another product
    % stream, so Q1 = Q2 or Q1 - Q2 = 0
    f(1) = Q1 - Q2;

    % There exists a second equation for the two temperature unknowns and
    % that is the constraint that the temperatures on any side cannot
    % exceed the 120 temperature difference. To maximize the heat transfer,
    % we should set this to be the difference!
    f(2) = 120 - calculate_log_mean_T(T_feed_in, T_feed_out, T_cross_out, T_cross_in);
end

function [log_mean_delta_T, area] = calculate_heat_exchanger_specs(duty, T_working_in, T_working_out, T_product_in, T_product_out, heat_transfer_coeff)
    % Find heat exchanger area
    log_mean_delta_T = calculate_log_mean_T(T_product_in, T_product_out, T_working_out, T_working_in);
    area = abs(duty / (heat_transfer_coeff * log_mean_delta_T));

         function [area, duty, flow_working, log_mean_delta_T, cost_per_mol] = calculate_latent_heat_exchange_fluid(F_product, x_product, P_product, T_product_in, T_product_out, working_fluid, mode, heat_transfer_coeff, mask, constants)
              [T_working_in, T_working_out, energy_per_mole, cost_per_mol] = calculate_working_fuild_requirements(P_product, working_fluid, mode, constants);

    % Finds the total duty to produce this temperature change at constant
    % pressure in J/mol
    duty = sum(F_product .* x_product .* latent_heat(T_working_in, constants, mask));
```

```
        [log_mean_delta_T, area] = calculate_heat_exchanger_specs(duty, T_working_in,
T_working_out, T_product_in, T_product_out, heat_transfer_coeff);
    flow_working = duty / energy_per_mole;
end

function log_mean_delta_T = calculate_log_mean_T(T_product_in, T_product_out,
T_working_out, T_working_in)

    delta_T_1 = T_product_in - T_working_in;
    delta_T_2 = T_product_out - T_working_out;

    log_mean_delta_T = (delta_T_1 - delta_T_2) / log(delta_T_1 / delta_T_2);
    if isnan(log_mean_delta_T)
        log_mean_delta_T = T_working_in - T_product_out;
    end
end

function [area, duty, T_working_in, flow_working, log_mean_delta_T, cost_per_mol] =
calculate_sensible_heat_exchange_fluid(F_product, x_product, P_product, T_product_in,
T_product_out, working_fluid, mode, heat_transfer_coeff, mask, constants)

    % Finds the total duty to produce this temperature change at constant
    % pressure in J/mol
      duty = abs(sum(F_product .* x_product(mask) .* integrated_dipper(T_product_in,
T_product_out, mask, constants)));

              [T_working_in, T_working_out, energy_per_mole, cost_per_mol] =
calculate_working_fuild_requirements(P_product, working_fluid, mode, constants);

        [log_mean_delta_T, area] = calculate_heat_exchanger_specs(duty, T_working_in,
T_working_out, T_product_in, T_product_out, heat_transfer_coeff);
    flow_working = duty / energy_per_mole;
end

function [T_working_in, T_working_out, energy_per_mole, cost_per_mol] =
calculate_working_fuild_requirements(P_product, working_fluid, mode, constants)
    switch working_fluid
        case "water"
            if mode == "heating"
                T_working_in = 35 + 273.15;
                T_working_out = 45 + 273.15;
            elseif mode == "cooling"
                T_working_in = 45 + 273.15;
                T_working_out = 35 + 273.15;
            end
            cost_per_mol = 2 * 1e-9; % $/J

            % Calculate the sensible heat change for water
            R = 8.314; % J/(mol K)
            C1 = 8.712;
            C2 = 1.25e-3;
            C3 = -0.18e-6;
```

```
            % J/mol
                    energy_per_mole  =  (1/6)*R*(T_working_in  -  T_working_out)*(6*C1  +
3*C2*(T_working_in + T_working_out) + 2*C3*(T_working_in^2 + T_working_in*T_working_out
+ T_working_out^2));

        case "steam"
            % Assume for now that we equalize the pressures. It eliminates
            % a degree of freedom while being truthful because of material
            % constraits.
            P_working = P_product;

            % For now, convert Pa (gauge) -> psia
            P_working = P_working / constants.units.pressure.psi + 14.6959;

            % To use the methane fuel that we already use for the furnaces,
            % just use our fuel cost while converting $/GJ -> $/(10^6 Btu)
            % ~ 1.055
              C = (31 * log(P_working) - 214) * constants.econ.electricity_cost + 4.83
* 1.62;
            % Convert $/tonne -> %/mol
            cost_per_mol = C * 10^(-9) * 18.01528;

            % Find operating temperature of steam since P is specified
            T_working_in = 4770 / (15.5 - log(P_working));
            T_working_out = T_working_in; % cause of phase change only

            % Convert from kJ/kg -> J/mol
            energy_per_mole = (2069*((647.1 - T_working_in)/209.4)^0.356)*18.01528;
    end
end

constants = get_constants();

working_fluid = "steam"; % or steam for the moment
mode = "cooling";
U = 820; % heat transfer coefficient W/m^2
P_working = 26 * constants.units.pressure.bar;

x_product = [0.0701, 0.0154, 0.5023, 0.4106, 0.0016];
F_product = 200; % mol/s, divide your hysys by 3.6 to plug in here

T_product_in = 596.6 + 273.15;
T_product_out = T_product_in - 120;

%      [T_working_in,      T_working_out,      energy_per_mole,      cost_per_mol]      =
calculate_working_fuild_requirements(P_working, working_fluid, constants)
%    [log_mean_delta_T,    area]    =    calculate_heat_exchanger_specs(Q,    T_working_in,
T_working_out, T_product_in, T_product_out, U)
% flow_working = Q / energy_per_mole

% For the case you want to find the heat change based only on latent heat
% change
```

```
[area, duty, T_working_in, flow_working, log_mean_delta_T, cost_per_mol] =
calculate_sensible_heat_exchange_fluid(F_product, x_product, P_working, T_product_in,
T_product_out, working_fluid, mode, U, true(1,5), constants);

% For the case you want to find the heat change based only on sensible heat
% change
%  [area, duty, T_working_in, flow_working, log_mean_delta_T, cost_per_mol] =
calculate_sensible_heat_exchange_fluid(F_product, x_product, P_working, T_product_in,
T_product_out, working_fluid, U, logical([1,1,1,1,1]), constants)

UA = area * U * (constants.units.time.hour / constants.units.energy.kJ);
duty = duty * (constants.units.time.hour / constants.units.energy.kJ);

fprintf("The working fluid should be flowing at %0.2f kgmol/hr " +...
    "at a temperature of %0.2f C and %0.2f kPa.\nThe heat exchanger needs to" + ...
    "have a UA of %0.2e kJ/(C-h) with duty %0.2e (kJ/hr) and LMTD of %0.2f\n", ...
    flow_working, ...
    T_working_in - 273.15, ...
    P_working / constants.units.pressure.kPa, ...
    UA, ...
    duty, ...
    log_mean_delta_T);

% For the case that you already have a required duty and you want to find
% the heat exchanger specs
```

F.6: Plant Economic and Plant

```
function [isbl, opex] = equipment_costing(mole_streams, plant_parameters, specs,
constants)
    % --- ISBL Calculations ---

    % Running total of all the Scope 1 and Scope 2 carbon sources
    duty_plant_total = 0;

    % -- Compressors --
    % Recycle stream compressor
    % bhp
    duty_compressor = plant_parameters.compressor.duty;
     isbl.compressor = IC_compressor(duty_compressor / constants.units.power.bhp, 1.0,
constants);
    opex.compressor = power_as_electricity(duty_compressor, constants);
    duty_plant_total = duty_plant_total + duty_compressor;

    % -- Heat Exchangers Utilities --
    % This is an approximation
    Fd_heat_exchanger = 0.85; % U-tubes
    Fm_heat_exchanger = 4.25; % Monel-Monel, avoid hydrogen damage
    IC_utilities = 0;
    for k = 1:length(plant_parameters.heat_exchangers)
        heat_exchanger = plant_parameters.heat_exchangers(k);
       ic_heatexchanger = IC_heatexchanger(heat_exchanger.pressure, heat_exchanger.area,
Fd_heat_exchanger, Fm_heat_exchanger, constants);
```

```
        IC_utilities = IC_utilities + ic_heatexchanger;
    end
    isbl.utilities = IC_utilities;
    % -- Furnace + CO2 Cost --

    % Heater material parameters
    Fd_heater = 1.0;
    Fm_heater = 0.35;
    % cost correlation, we calculate for two heaters
    duty_furnace = plant_parameters.furnace.duty;
    isbl.furnace = IC_furnace(specs.reactor_pressure, duty_furnace, Fd_heater, Fm_heater,
constants);
    opex.furnace = power_as_fuel(duty_furnace, constants);
    duty_plant_total = duty_plant_total + duty_furnace;

    % -- Reactors --
    Fm_reactors = 3.89; % clad Monel
    IC_reactors = 0;
    for reactor_idx = 1:length(plant_parameters.reactors)
        reactor = plant_parameters.reactors(reactor_idx);
            ic_vessel  =  IC_vessel(specs.reactor_pressure,  reactor.volume,  Fm_reactors,
constants);
        IC_reactors = IC_reactors + ic_vessel;
    end

    % -- Seperation System --
    method = "real";
    IC_seperations_cost = 0;
    OPEX_seperations_cost = 0;

    if method == "real"

        % Stablizer and Scrubber Wmin Costs


        % Flashs
        Fm_vessel_flash = 1;
        for flash_idx = 1:length(plant_parameters.flashs)
            flash = plant_parameters.flashs(flash_idx);
          ic_flash = IC_vessel(flash.pressure, flash.volume, Fm_vessel_flash, constants);
            IC_seperations_cost = IC_seperations_cost + ic_flash;
        end

        % Distallation Columns
        Fm_heat_exchanger = 1;
        Fm_vessel_columns = 1;
        Fm_internals_columns = 1;
        Ft_columns = 0;

        for column_idx = 1:length(plant_parameters.columns)
            column = plant_parameters.columns(column_idx);
            Ptot = specs.column_pressures(column_idx);
```

```
            spacing = specs.column_spacing(column_idx);

            % Pick the large diameter
            H = column.height;
            D = max(column.diameter_bottom, column.diameter_top);

            ic_vessel = IC_vessel_DH(Ptot, D, H, Fm_vessel_columns, constants);
        ic_internals = IC_tower_internals(spacing, H, D, Ft_columns, Fm_internals_columns,
constants);
                    ic_condensor = IC_heatexchanger(Ptot, column.condenser.area, 0.85,
Fm_heat_exchanger, constants);
         ic_reboiler = IC_heatexchanger(Ptot, column.reboiler.area, 1, Fm_heat_exchanger,
constants);

            % $/yr
           opex_condensor = column.condenser.cost_per_mol * column.condenser.flow_working
* constants.units.time.plant_yr;
             opex_reboiler = column.reboiler.cost_per_mol * column.reboiler.flow_working
* constants.units.time.plant_yr;

               IC_seperations_cost = IC_seperations_cost + ic_vessel + ic_internals +
ic_condensor + ic_reboiler;
          OPEX_seperations_cost = OPEX_seperations_cost + opex_condensor + opex_reboiler;
                      duty_plant_total = duty_plant_total + column.condenser.duty +
column.reboiler.duty;
        end

        isbl.seperation = IC_seperations_cost;
        opex.seperation = OPEX_seperations_cost;

    elseif method == "sub"
            [Q_sep,isbl.seperation,opex.seperation] = subsitute_sep_cost(mole_streams,
constants);
        duty_plant_total = duty_plant_total + Q_sep;
    end

    price_per_GJ = 1.56;
    opex.co2 = duty_plant_total * 1e-9 * constants.units.time.plant_yr .* price_per_GJ;
end

function ic = IC_vessel(P, V, Fm, constants)
    V_ft3 = V .* 35.314; % V converted to ft^3
    [D, H] = DH_from_ratio(V_ft3, constants);
    Fc = Fm + Fp_vessel(P, constants);
     ic = (constants.econ.CEPCI2025/113.7) .* 101.9 .* (D.^1.066) .* (H.^0.82) .*(2.18
+ Fc);
end

function ic = IC_vessel_DH(P, D, H, Fm, constants)
    D_ft3 = D / constants.units.length.ft;
    H_ft3 = H / constants.units.length.ft;
    Fc = Fm + Fp_vessel(P, constants);
```

```
          ic   =   (constants.econ.CEPCI2025/113.7)   .*   101.9   .*   (D_ft3.^1.066)   .*
(H_ft3.^0.82) .*(2.18 + Fc);
end

function ic = IC_heatexchanger(P, A, Fd, Fm, constants)
    Fc = (Fd + Fp_heat_exchanger(P, constants)) * Fm;
    ic = (constants.econ.CEPCI2025/113.7) .* 101.9 .* A.^0.65 .* Fc;
end

function ic = IC_furnace(P, Q, Fd, Fm, constants)
    Q_Btue6hr_gas = Q * 3.412*10^-6;
    Fc = Fd + Fm + Fp_furnace(P, constants);
     ic = (constants.econ.CEPCI2025/113.7) .* 5.52e3 .* (Q_Btue6hr_gas.^0.85) .* (1.27
+ Fc);
end

function ic = IC_compressor(bhp, Fd, constants)
    Fc = Fd;
    ic = (constants.econ.CEPCI2025/113.7).*517.5.*(bhp^0.82).*(2.11 + Fc);
end

function ic = IC_tower_internals(spacing, H, D, Ft, Fm, constants)
    switch spacing
        case 24
            Fs = 1;
        case 18
            Fs = 1.4;
        case 12
            Fs = 2.2;
        otherwise
            fprintf("Unspecified spacing!\n")
            Fs = 1;
    end
    Fc = Ft + Fm + Fs;
    ic = (constants.econ.CEPCI2025/113.7) .* 4.7 .* D .^ 1.55 .* H .* Fc;
end

function fp = Fp_vessel(P, constants)
    fp = interp1(constants.Fp_vessel.Pa, constants.Fp_vessel.Fp, P);
end

function fp = Fp_heat_exchanger(P, constants)
    fp = interp1(constants.Fp_heatexchanger.Pa, constants.Fp_heatexchanger.Fp, P);
end

function fp = Fp_furnace(P, constants)
    fp = interp1(constants.Fp_furnace.Pa, constants.Fp_furnace.Fp, P);
end

function cost = power_as_electricity(Q, constants)
    % Gives the cost of the specified duty (in W) if its work required by using
    % electricity utilities in $/yr
```

```
   cost = constants.econ.electricity_cost * Q * 2.7778e-7 * constants.units.time.plant_yr;
end

function cost = power_as_fuel(Q, constants)
    % Gives the cost of the specified duty if its used as a fuel
    cost = constants.econ.fuelcost .* 1e-9 .* Q * constants.units.time.plant_yr;
end

function [Q,isbl,opex] = subsitute_sep_cost(mole_streams, constants)
     Wmin = calculate_wmin(mole_streams.TotalReactorEffluent, [mole_streams.RecycleGas,
mole_streams.RecycleToluene, mole_streams.ProductBenzene, mole_streams.ProductDiphenyl]);
    Wreal = 50 * Wmin;
    C = 1;
    isbl = Wreal * constants.units.time.plant_yr * constants.econ.fuelcost; % $/y
    opex = Wreal * C * 1e9;
    Q = Wreal;
end

function [reactantCosts, productRevenue] = flow_costs(specs, constants, FT, FG, PG, PD, PB)
    % T in K, Pressure in atm
    % feedCost: cost of raw material in column 1, cost of waste water
    % processing in column 2

    % kJ/mol |1 GJ/10^6 kJ| = GJ/mol
    delH_D = 0.0062285; %GJ/mol
    delH_H2 = 0.0005716; %GJ/mol
    delH_M = 0.0008903; %GJ/mol
    pD_fuel = constants.econ.fuelcost * delH_D;
    pH2_fuel = constants.econ.fuelcost * delH_H2;
    pM_fuel = constants.econ.fuelcost * delH_M;
    % pD_fuel = 0.026471125000000005; % $/mol
    % pH2_fuel = 0.0012146500000000003; % $/mol
    % pM_fuel = 0.003783775; % $/mol

    % correlation uses pressure in bar
    % Pressure_bar = specs.P_reactor .* 1.01325;
    % uses values of 0.05$/kWh for electricity
    % and $1.62/MM Btu for coal
    % pH2O_MT = (31*log(Pressure_bar * 14.5) - 214)*0.05 + 4.82*1.62;

    % Get molar basis prices of feeds
    % $/MT |1 MT/10^6 g|MM g/mol| = $/mol
    pT = constants.econ.toluenecost*92.14/10^6; % $/mol
    pB = constants.econ.benzeneprice*78.11/10^6; % $/mol
    pMH_M = constants.econ.hydromethcost/10^6; % $/g

    % Find the feed gas stream mass flow
    H2_FG_m = (0.95*FG)*2.016; % g/s
    M_FG_m = (0.05*FG)*16.04; % g/s

    % purge
    priceH2_purge = pH2_fuel*PG*specs.yPH;
```

```
    priceM_purge = pM_fuel*PG*(1 - specs.yPH);

    % pH2O = pH2O_MT*18*power(10,-6); % $/mol

    % Waste water cost
    % from Ulrich and Vasudevan , 2007
    % q = TE .* 18.01 ./ (1000*1000); %m^3/s
    % a = 0.001 + (2e-4).* q.^(-0.6);
    % b = 0.01;
    % pwasteH2O = q.*(a*econ.CEPCI2025 + b*econ.fuelcost)*8400*3600;

    % CALCULATING REVENUES in $/yr
    main_product = pT*FT*constants.units.time.plant_yr;
    fuel = pMH_M*(H2_FG_m + M_FG_m)*constants.units.time.plant_yr;

    reactantCosts = main_product + fuel;
            productRevenue    =    (pD_fuel*PD    +    pB*PB    +    priceH2_purge    +
priceM_purge)*constants.units.time.plant_yr;
end

function econ_out = plant_economics(specs, plant_parameters, constants)

    mole_streams = plant_parameters.mole_streams;

    % Equipment Costing
    [isbl, opex] = equipment_costing(mole_streams, plant_parameters, specs, constants);

    ISBL = sum(table2array(struct2table(isbl)));
    OPEX = sum(table2array(struct2table(opex)));

    econ_out.isbl = isbl;
    econ_out.isbl_total = ISBL;

    econ_out.opex = opex;
    econ_out.opex_total = OPEX;

    % Flow costs
    FT = mole_streams.FeedToluene.totalmolarflowrate;
    FG = mole_streams.FeedGas.totalmolarflowrate;
    PD = mole_streams.ProductDiphenyl.totalmolarflowrate;
    PB = mole_streams.ProductBenzene.totalmolarflowrate;
    PG = mole_streams.PurgeGas.totalmolarflowrate;

    [reactantCosts, productRevenue] = flow_costs(specs, constants, FT, FG, PG, PD, PB);

    AGS_cost = constants.econ.AGS_rate .* productRevenue;

    econ_out.co2mass = plant_parameters.furnace.co2_mass;
    econ_out.reactants_costs = reactantCosts;
    econ_out.product_revenue = productRevenue;
    econ_out.opex_ags = AGS_cost;
```

```
    % Fixed capital investment
    FCI = 2.28 .* ISBL;
    econ_out.fci = FCI;

    % Working capital = 2 months of feedstock cost
    WC = (2/12) .* reactantCosts;
    SU = 0.1 .* FCI; % Startup cost = 10% of FCI

    econ_out.tci = FCI + WC + SU;
    econ_out.wc = WC;
    econ_out.su = SU;

    COM = opex.co2 + OPEX + reactantCosts + AGS_cost;
    salvage = FCI * constants.econ.salvage_rate;
    costbasis = FCI + SU - salvage;

    % Ensure integer indexing for depreciation and cash flow schedules
    construction_years = round(constants.econ.construction);  % <-- Fix applied here
    depreciation = zeros(constants.econ.plant_life, 1);
   depreciation(construction_years+1 : construction_years+constants.econ.dep_sched) = ...
        costbasis / constants.econ.dep_sched;

    % Initialize cash flow array
    CF = zeros(constants.econ.plant_life, 1);

    % During construction phase
    for i = 2:construction_years+1
        CF(i) = - FCI / construction_years;
    end

    % Last year of construction includes WC and SU
    CF(construction_years+1) = CF(construction_years+1) - WC - SU;

    % Operational phase
    for i = (construction_years+2):constants.econ.plant_life-1
            CF(i) = CF(i) + (productRevenue - COM - depreciation(i)) * (1 -
constants.econ.tax_rate) + depreciation(i);
    end

    % Final year: include salvage and return of WC
        CF(end) = (productRevenue - COM - depreciation(end) + salvage) * (1 -
constants.econ.tax_rate) + depreciation(end) + WC;

    % Discounted cash flows
    DCF = zeros(constants.econ.plant_life, 1);
    for k = 1:constants.econ.plant_life
        DCF(k) = CF(k) / ((1 + constants.econ.enterprise_rate)^(k - 1));
    end

    % Generate the cumulative discounted cash flows, NPV, and EP
    econ_out.cdcf = cumsum(DCF)';
    econ_out.npv = econ_out.cdcf(end) + DCF(end);
```

```
    econ_out.ep = productRevenue - reactantCosts;
end
```

F.7: Tornado Plot

```
clear;
clc;
close all;

% Load constants and define model setup
constants = get_constants();
method = "cold-shot";

if method == "single"
    load('sol_single_reactor.mat', 'sol');
    specs = initial_spec_single_reactor(constants);
    specs.reactor_design_fun = @reactor_design_single;
    optimized_spec = objective_spec_one_reactor(sol, specs);
elseif method == "cold-shot"
    load('sol_cold_shot.mat', 'sol');
    specs = initial_spec_two_reactor_cold_shot(constants);
    specs.reactor_design_fun = @reactor_design_cold_shoot;
    optimized_spec = objective_spec_cold_shot(sol, specs);
end

% Run base case
[econ_optimized,  plant_parameters_optimized,  ~]  =  solve_HDA_plant(optimized_spec,
constants);
base_npv = econ_optimized.npv / 1e6; % Convert base NPV to millions

% Define low and high values for each parameter
params = {
    'benzene',      700,    1300;
    'toluene',      270,    390;
    'hydrogen',     1260,   1820;
    'fuel',         2.125,  8.5;
    'construction', 2.0,    5.0;
    'tax',          0.15,   0.35
};

param_labels = {
    'Benzene Price',
    'Toluene Price',
    'Hydrogen Price',
    'Fuel Price',
    'Construction Time',
    'Tax Rate'
};

num_params = size(params, 1);
npv_low = zeros(num_params, 1);
```

```
npv_high = zeros(num_params, 1);

% Sensitivity analysis (low/high only)
for i = 1:num_params
    param_name = params{i, 1};
    low_val = params{i, 2};
    high_val = params{i, 3};
    for j = 1:2
        constants_temp = get_constants();  % Reset constants
        val = low_val;
        if j == 2
            val = high_val;
        end
        % Apply parameter change
        switch param_name
            case 'benzene'
                constants_temp.econ.benzeneprice = val;
            case 'toluene'
                constants_temp.econ.toluenecost = val;
            case 'hydrogen'
                constants_temp.econ.hydromethcost = val;
            case 'fuel'
                constants_temp.econ.fuelcost = val;
            case 'construction'
                constants_temp.econ.construction = val;
            case 'tax'
                constants_temp.econ.tax_rate = val;
        end

        % Solve model with modified constants
        [econ_temp, ~, ~] = solve_HDA_plant(optimized_spec, constants_temp);

        % Store NPV results in millions
        if j == 1
            npv_low(i) = econ_temp.npv / 1e6;  % Convert to millions
        else
            npv_high(i) = econ_temp.npv / 1e6; % Convert to millions
        end
    end
end

% Sort the NPVs based on the absolute difference from the base case
[~, idx] = sort(abs(npv_high - npv_low), 'ascend');
npv_low = npv_low(idx);
npv_high = npv_high(idx);
param_labels_sorted = param_labels(idx);

% Create the tornado plot
fig = figure(1);
hold on;

% Plot low and high NPV bars
```

```
h1 = barh(npv_low, 'FaceColor', '#fc8d62');  % Low values in orange
h2 = barh(npv_high, 'FaceColor', "#8da0cb");  % High values in blue

% Set the baseline to the base case NPV
set(h1, 'BaseValue', base_npv);
set(h2, 'BaseValue', base_npv);

% Customize the plot
ax = gca;
ax.FontSize = 14;
ax.XLim = [-10, 200];  % Set the X-axis limits to ensure it fits the range
ax.LineWidth = 1.5;
xlabel('NPV [$MM]');

% Add y-axis labels with some spread
yticks(1:num_params);  % Set tick positions
yticklabels(param_labels_sorted);  % Set tick labels


% Add labels to the bars
for i = 1:length(param_labels_sorted)
    % Low value labels - place at the very end of the low bar with custom offset
            text(npv_low(i),   i,   sprintf('%.0fM',   npv_low(i)),   'FontSize',   12,
'HorizontalAlignment', 'right', 'VerticalAlignment', 'baseline');

    % High value labels - place at the very end of the high bar with custom offset
           text(npv_high(i),   i,   sprintf('%.0fM',   npv_high(i)),   'FontSize',   12,
'HorizontalAlignment', 'left', 'VerticalAlignment', 'baseline');

    % Base NPV label - place at the center of the bar with slight offset for separation
    %text(base_npv, i, sprintf('%.0fM', base_npv), 'FontSize', 12, 'HorizontalAlignment',
'center', 'VerticalAlignment', 'middle');
end
% Draw a vertical line to mark base NPV
xline(base_npv, '--k', 'Base Case', ...
    'FontSize', 12, ...
    'LabelOrientation', 'horizontal', ...
    'LabelVerticalAlignment', 'top', ...
    'LabelHorizontalAlignment', 'center');
hold off;
```

F.8: Monte Carlo Simulation

```
clear;
clc;
close all;

% Load constants and define model setup
constants = get_constants();
method = "single";  % Hardcoded for now

if method == "single"
    load('sol_single_reactor.mat', 'sol');
```

```
    specs = initial_spec_single_reactor(constants);
    specs.reactor_design_fun = @reactor_design_single;
    optimized_spec = objective_spec_one_reactor(sol, specs, constants);
elseif method == "cold-shot"
    load('sol_cold_shot.mat', 'sol');
    specs = initial_spec_two_reactor_cold_shot(constants);
    specs.reactor_design_fun = @reactor_design_cold_shoot;
    optimized_spec = objective_spec_cold_shot(sol, specs);
end

% Get base NPV
[econ_optimized, ~, ~] = solve_HDA_plant(optimized_spec, constants);
base_npv = econ_optimized.npv / 1e6;  % in millions

% Simulation parameters
N = 10000;  % Large number for HPC
npv_results = NaN(N, 1);  % Preallocate

% Define bounds for triangular distributions
benzene_min = 700; benzene_mode = 1000; benzene_max = 1300;
toluene_min = 270; toluene_mode = 300; toluene_max = 390;
construction_min = 2; construction_mode = 3.0; construction_max = 5.0;
%conversion_min = 0.6; conversion_mode = optimized_spec.x; conversion_max = 0.9;

% Open a parallel pool (let's match # cpus you request)
% parpool('local', str2double(getenv('SLURM_CPUS_PER_TASK')));

parfor k = 1:N
    try
        constants_sim = get_constants();

        % Sample uncertain parameters
        benzene_price = triangular_sample(benzene_min, benzene_mode, benzene_max, 1);
        toluene_price = triangular_sample(toluene_min, toluene_mode, toluene_max, 1);
            construction_time  =  triangular_sample(construction_min,  construction_mode,
construction_max, 1);
      %conversion = triangular_sample(conversion_min, conversion_mode, conversion_max, 1);

        % Update constants with sampled values
        constants_sim.econ.benzeneprice = benzene_price;
        constants_sim.econ.toluenecost = toluene_price;
        constants_sim.econ.construction = construction_time;
        optimized_spec_local = optimized_spec;
        %optimized_spec_local.x = conversion;

        % Solve plant model
        econ_sim = solve_HDA_plant(optimized_spec_local, constants_sim);
        npv_results(k) = econ_sim.npv / 1e6;
    catch
        npv_results(k) = NaN;
    end
end
```

```
delete(gcp('nocreate'));

% Clean and save results
npv_results = npv_results(~isnan(npv_results) & npv_results > 0);
writematrix(npv_results, "monte_results.csv");

fig1 = figure;
hold on;
xline(base_npv, 'LineWidth', 2, 'Color', 'black','LineStyle', '--')
histogram(npv_results, 80, 'FaceColor', '#66c2a5');
xlabel('NPV [$MM]');
ylabel('Frequency');
hold off;
savefig(fig1, "histogram.fig");

% --- CDF Plot with Shaded 95% Prediction Interval and Annotations ---
fig2 = figure;
sorted_npv = sort(npv_results);
cdf = linspace(0, 1, length(sorted_npv));

% Interpolate to smooth
npv_interp = linspace(min(sorted_npv), max(sorted_npv), 500); % 500 smooth points
cdf_interp = interp1(sorted_npv, cdf, npv_interp, 'pchip');

% First, plot the shaded 95% interval
P2_5 = prctile(npv_results, 2.5);
P97_5 = prctile(npv_results, 97.5);

% Create shaded area between 2.5% and 97.5% bounds
x_shade = [P2_5, P97_5, P97_5, P2_5];
y_shade = [0, 0, 1, 1];
fill(x_shade, y_shade, [0.9 0.9 0.9], 'EdgeColor', 'none', 'FaceAlpha', 0.5); % Light
gray shade

hold on;

% Plot the empirical CDF
plot(npv_interp, cdf_interp, 'b-', 'LineWidth', 2, 'DisplayName', 'Empirical CDF');

% Find cumulative probability of the base case NPV
[~, idx_base] = min(abs(sorted_npv - base_npv));
base_cdf_value = cdf(idx_base);

% Plot base NPV case
scatter(base_npv, base_cdf_value, 'red', 'filled')

hold off;
xlabel('NPV [$MM]');
ylabel('Cumulative Probability');
legend('Location', 'southeast');
grid on;
```

```
savefig(fig2, "CDF.fig");

writematrix(x_shade, "quantiles.csv");
```

Appendix G: Team Member Work Statements

Team Member Work Statement

Name of team member here Lauren Bajo-Smith

State here what you contributed to the design project and report

ASPEN HYSYS – w/ Ashley
Economic Design Decisions + Calculation
Sensitivity Analysis, majority of Code
written report
+
appendices

Name of team member here
Ashley Uriostegui

State here what you contributed to the design project and report

ASPEN HYSYS – w/ Lauren
HAZOP analysis
Safety summary and tables in appendix
Safety hazards part of report and pie charts

Name of team member here
Forrest Ramirez

State here what you contributed to the design project and report

MATLAB conceptual design, including reactor modeling, Level 1-3 molar flowrates, implementation of economic/npv calculations, and optiminization algorithm. Key figure plots. Typst/LaTex formatting.

Print Name and Sign: Lauren Bajo-Smith Date: 3-12-25

Print Name and Sign Ashley Uriostegui Date: 3-12-25

Print Name and Sign Forrest Ramirez Date: 3-12-25

Rating of Team Members for Design Project

Please rate each group member's contributions in the categories below:
1-2 - unsatisfactory, 3 - acceptable/adequate, 4 – very good, 5 - excellent
Each member fills out one form and signs the bottom.

Name :	1) Ashley	2) Lauren	3) Forrest
Quality of work presented	5	5	5
Quantity of work performed	5	5	5
Effort	5	5	5
Punctuality (meetings and deadlines)	5	5	5
Knowledge of design methods	4	4	4
Class attendance	5	5	5
Communication	5	5	5

Do you feel that each member of the group deserves the same grade? If not, who does not and why?

Yes! Everyone worked very hard. Amazing Team!

It's important to note that differences in performance will not necessarily affect individual grades; however, large discrepancies may result in differences in grades.

Additional comments:

Print Name and Sign: Forrest Ramirez FAR Date: 3-12-25

Rating of Team Members for Design Project

Please rate each group member's contributions in the categories below:
1-2 - unsatisfactory, 3 - acceptable/adequate, 4 – very good, 5 - excellent
Each member fills out one form and signs the bottom.

Name : Lauren	1) Lauren	2) Forrest	3) Ashley
Quality of work presented	5	5	5
Quantity of work performed	5	5	5
Effort	5	5	5
Punctuality (meetings and deadlines)	5	5	5
Knowledge of design methods	4	4	4
Class attendance	5	5	5
Communication	5	5	5

Do you feel that each member of the group deserves the same grade? If not, who does not and why?

Yes, solid team! I think we work well together.

It's important to note that differences in performance will not necessarily affect individual grades; however, large discrepancies may result in differences in grades.

Additional comments:

Print Name and Sign Lauren Bajo Smith Date: 3-12-25

Rating of Team Members for Design Project

Please rate each group member's contributions in the categories below:
1-2 - unsatisfactory, 3 - acceptable/adequate, 4 – very good, 5 - excellent
Each member fills out one form and signs the bottom.

Name : Ashley	1) Ashley	2) Lauren	3) Forrest
Quality of work presented	5	5	5
Quantity of work performed	5	5	5
Effort	5	5	5
Punctuality (meetings and deadlines)	5	5	5
Knowledge of design methods	4	4	4
Class attendance	5	5	5
Communication	5	5	5

Do you feel that each member of the group deserves the same grade? If not, who does not and why?

Yes, everyone deserves the same grade
Everyone contributed in their own way

Great team!

It's important to note that differences in performance will not necessarily affect individual grades; however, large discrepancies may result in differences in grades.

Additional comments:

Print Name and Sign: Ashley Uriostegui Ashley Uriostegui Date: 3-12-25